Compact Calculus - Calculus for the practitioners

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Prerequisites: *set theory, algebra, geometry, basic trigonometry*

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Preface

Purpose of the textbook

I probably made a lot of mathematicians angry by writing this book. But it misses their points. This textbook is designed for the practitioners of calculus: people that are interested in fields of sciences including biology, chemistry, and physics. Therefore, expect what you aren't expected, and don't expect what you traditionally expect.

The contents of this textbook are entirely self-contained and organized into four parts (volumes)[1]: dynamical systems and ordinary differential equations, signal analysis, real and complex analysis, and vector calculus with partial differential equations. However, the topics deviates greatly from the traditional way calculus is taught. I took a great care of how the material is arranged in better needs practitioners, which means it may deviate from a standard calculus curriculum. To maintain focus within the core chapters, the auxilaries topics typically covered in traditional courses have been moved to the appendices.

Even though the book is very self-contained, it still doesn't serve as a standalone book. Sure, the story that I'm about to tell is quite fruitful,

[1]The exact number of parts may change as more material is added during the writing process.

and you can enjoy it from the beginning to end. But, it lacks one thing: problem solving experiences. As of the revision today (February 5, 2025), there are not that many exercises in the chapters. So, you might have to find some online. I've scattered plenty of them throughout the chapters.

Layout of the book

Layout of Part I

The first part of the book concerns differential equations: a powerful tool that can be used to describe how a system changes over time. The contents are laid out as follows:

- Chapters 1 and 2 develops the basic building blocks of calculus: derivatives and integrals, from the ground up
- Chapter 3 takes the reader through derivatives and integrals of basic functions, e.g., exponentials, and logarithms.
- Chapter 4 takes a detour from the main objective a bit, and looks at how calculus could be applied to solve various geometrical problems.
- Chapter 5 studies the interaction between calculus and various trigonometric functions. It also applies calculus to describe some physical systems, specifically oscillatory ones. From this chapter, most of the foundations of calculus is already laid down, ready to be used.
- Finally, chapter 6 uses all the knowledge developed in earlier chapters to explore many systems that exist in nature, including biological, chemical, and physical systems.

Acknowledgement

Book I

Differential equations

Part I

The foundation of calculus: differential equations

CHAPTER 1

Writing down nature: derivatives

Prerequisites: *graphs, functions, kinematic variables*

1.1 Invitation to calculus from a practitioner

Calculus is the language of change, whether it's changes in weather patterns, movement of an object, evolution of biological systems, or even changes in your relationship! Although the last one might be diabolical to think about for some of you, calculus provides a mathematical framework to describe and predict these systems. Therefore, I've structured this volume around the concept of dynamical systems: systems that evolve through time, whether they are biological, chemical, or physical.

As said in the preface, I have thrown most unrelated contents into the appendices. But as much as I'd love to do that, some purely mathematical contents and exercises are still essential for understanding and solving dynamical systems. While I've kept those in, I tried to make them as brief as possible for it to not disrupt your entertainment.

The problem of dynamical systems are separated into two parts:

1. Describing the system, and
2. Solving for the system's future.

Calculus is also neatly separated into two parts which align with these two exact problems: differential calculus and integral calculus. Differential calculus uses differential equations to describe a system; those equations are then solved using integral calculus to predict the system's future. This chapter is devoted to the first part of calculus: differential calculus.

At the end of this chapter, you should develop the intuition that

1. Derivative measures the rate of change of a function w.r.t. a variable.
2. Derivatives can be thought of the slope of a graph.
3. The universe is described in the language of differential equations

1.1.1 The notation of differential calculus

If a is a variable, then $\mathrm{d}a$ is a very small quantity of a, known as an **infinitesimal**. E.g., if $\mathbf{x}$ is displacement, then $\mathrm{d}\mathbf{x}$ is a very small displacement. If t is time, then $\mathrm{d}t$, a very short time. It's very similar to how we usually write a difference in a variable x, Δx. Replacing the Δ with d conveys that "it is a very small difference in x."

Infinitesimals can also be divided to yield a ratio between infinitesimals. E.g., the ratio between some small distance and some short time, $\dfrac{\mathrm{d}\mathbf{x}}{\mathrm{d}t}$ is the speed $\mathbf{v}$.

1.2 Speed and instantaneous rate of change

Our first dynamical system that I'd like to discuss is the car; specifically the **speedometer**, which displays the speed of the car at an instance in time. How does it do that?

If we're traveling at a speed $5\,\mathrm{m\,s^{-1}}$, when *exactly* are we traveling at $5\,\mathrm{m\,s^{-1}}$? You could say $\mathbf{v} = 5\,\mathrm{m}$ *at the moment of measurement*. But what exactly is *at the moment of measurement?* To measure something is to observe, which is to use a measurement tool to *capture* its current state. So, measurement in this case, is analogous to capturing the picture of a car at that moment. But if we say, *at the moment of measurment,* it's like saying "Oh, I can find the speed of the car by just taking a picture of it." But that's illegal! To calculate speed, we have to compare two points in space through time. Or, the rate of change of distance through time:

$$\mathbf{v} = \frac{\mathbf{s}_2 - \mathbf{s}_1}{t_2 - t_1}. \tag{1.1}$$

While it may seem like cameras grab snapshots in an instant, they actually need time to take in light to construct an image: they need some *exposure time* Δt.

To get a "not blurred" image of a moving object, we reduce the exposure time. If the exposure time is too long, the object will be smeared out. This effect is known as motion blur, which is normally undesired. But motion blur actually helps out a lot with measuring velocity. In which the velocity is just

$$\mathbf{v} = \frac{\text{Distance between the smear and the main object}}{\text{Exposure time}}. \tag{1.2}$$

As illustrated in fig. 1.1, the motion blur clearly shows us the change in position of the car over the exposure time.

Fig. 1.1 | Calculating the velocity of a car from motion blur.

But what will happen with shorter exposure time? Does the motion blur disappear? No! The blur is still there, but it's just smaller. Typically, 12 ms exposure time is short enough to create a "focused image". But it's just an illusion that came from the limitation of the screen's ability to reproduce such little blur. If our camera and screen is good enough, we can *always* calculate the velocity of the car from the blur. The smaller the exposure time, the more detailed the image is, and the closer you'll get to the exact **v** at that moment in time. Finally, if we let the exposure time become infinitesimally short, we can say the **v** we got is *the velocity at that exact point*. Or as we call it, the **instantaneous velocity**. By using the said calculus notation in section 1.1.1, we can just write this as

$$\mathbf{v} = \frac{\mathrm{d}\mathbf{x}}{\mathrm{d}t}, \tag{1.3}$$

and here it is ladies and gentlemen, the **derivative**: the measure of rate of change.

From here on out, I shall use *derivative* and *rate of change* interchangeably. So, every time you see *derivative*, think *rate of change*.

1.3 An attempt to define derivatives

Mathematically, a **derivative** is a measure of a function's rate of change with respect to a variable. In the previous example, $\mathbf{x}$ is a function that's dependent on time, and its derivative w.r.t. time, $\mathbf{v}$, is measuring the rate of change of function $\mathbf{x}(t)$ through time $\mathrm{d}t$.

To find an explicit expression for derivative, let's say we have two points in spacetime $(\mathbf{x}_1, t_1)$ and $(\mathbf{x}_2, t_2)$. The change in $\mathbf{x}$ is $\mathbf{x}_2 - \mathbf{x}_1$, and the change in time is $t_2 - t_1$. The derivative of position w.r.t. time is then just

$$\mathbf{v} = \frac{\mathrm{d}\mathbf{x}}{\mathrm{d}t} = \frac{\mathbf{x}_2 - \mathbf{x}_1}{t_2 - t_1}. \tag{1.4}$$

Let $t_2 = t + h$. Then, $\mathbf{x}_2 = \mathbf{x}(t+h)$ and $\mathbf{x}_1 = \mathbf{x}(t)$. The time difference used to calculate $\mathbf{v}$ must be miniscule: infinitesimally small. I have to introduce the notion of limits, which is just a fancier way of saying "very close to, but not"

$$\begin{aligned}\mathbf{v} = \frac{\mathrm{d}\mathbf{x}}{\mathrm{d}t} &= \lim_{h\to 0} \frac{\mathbf{x}(t+h) - \mathbf{x}(t)}{(t+h) - t} &(1.5)\\ &= \lim_{h\to 0} \frac{\mathbf{x}(t+h) - \mathbf{x}(t)}{h}. &(1.6)\end{aligned}$$

The equation above is what we generally refer to as the *definition of derivative*. Then, we just extend this relation to any function $f(x)$, which requires just a substitution of variables. And then we get:

Definition 1.3.1: *Naive definition of derivative*

A derivative of a function $f(x)$ w.r.t. a variable x is the rate of change of $f(x)$ w.r.t. x, and it is written as

$$\frac{\mathrm{d}f(x)}{\mathrm{d}x} \quad \text{or} \quad \frac{\mathrm{d}}{\mathrm{d}x}f(x), \tag{1.7}$$

where

$$\frac{\mathrm{d}f(x)}{\mathrm{d}x} = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}. \tag{1.8}$$

Equation (1.8) directly reads:

The derivative of $f(x)$ with respect to x is $\dfrac{f(x+h) - f(x)}{h}$ where h is very close to 0, but h is strictly not 0.

On notation: So far, we've been using the Leibniz's notation for derivatives, and it has the property that derivatives behave exactly like fractions, and you can cancel terms.

$$\frac{\mathrm{d}a}{\mathrm{d}b} \times \frac{\mathrm{d}b}{\mathrm{d}c} = \frac{\mathrm{d}a}{\mathrm{d}c}. \tag{1.9}$$

But, this isn't the only accepted notation. Multiple great mathematicians have come up with their own, and some are better than others in certain cases. I'd introduce other notations later on if necessary.

1.4 The geometrical interpretation of the derivative

One way to interpret derivatives is by using slope. Notice that eq. (1.4) looks a lot like the slope equation

$$m = \frac{y_2 - y_1}{x_2 - x_1}. \tag{1.10}$$

We've also seen that eq. (1.4) is analogous to the definition of derivative eq. (1.8). So is the derivative just the slope of a line? If it is, then what is the y-axis and the x-axis of a graph?

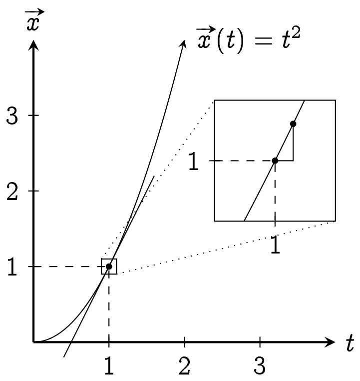


FIG. 1.2 | POSITION VS. TIME GRAPH WHERE $\mathbf{x}(t) = t^2$.

We can compare eq. (1.10) to eq. (1.4): the y-axis should be the position $\mathbf{x}$, and the x-axis, the time t. If we draw that out, we'll get the x-t graph, which can encode the exact trajectory of an object. An example of which is shown in fig. 1.2.

But what does this have to do with derivatives? Equation (1.10) only works for straight line! Well, here's the beauty of it. If you zoom into any points on a curve, eventually, it will look like a line. And thus, *the derivative zooms into the curve at some point, and chooses two very close points on the curve and calculate its slope. In which, that slope represents the rate of change of the function at that point.*

1.5 Evaluation of derivatives: method of increments

The derivative's definition can be used to directly evaluate derivatives. This is called the **method of increment**. E.g., in fig. 1.2, let's evaluate the velocity at time $t = 1\,\text{s}$ where the position function, $\mathbf{x}(t) = t^2$. $t = 1\,\text{s}$. We start from eq. (1.8):

$$\begin{aligned}
\mathbf{v}(1) = \frac{\mathbf{x_2} - \mathbf{x_1}}{t_2 - t_1} &= \lim_{h\to 0} \frac{\mathbf{x}(1+h) - \mathbf{x}(1)}{(1+h) - 1} \\
&= \lim_{h\to 0} \frac{(1+h)^2 - 1}{h} \\
&= \lim_{h\to 0} \frac{2h + h^2}{h} \\
&= \lim_{h\to 0} 2 + h\,.
\end{aligned}$$

Since h is very close to zero, we approximate $2 + h$ as 2. Imagine comparing 2 to 10^{-10}. The 10^{-10} wouldn't make a noticeable difference, and we can ignore it. Therefore, $\mathbf{v}(t = 1\,\mathrm{s}) = 2\,\mathrm{m\,s^{-1}}$.

Now, try evaluating $\mathbf{v}(t = 3\,\mathrm{s})$ for $\mathbf{x}(t) = t^3$. You should get $81\,\mathrm{m\,s^{-1}}$. As a hint, you can also ignore h^2 because if $h < 1$, then $h^2 < h$.

1.6 Higher order derivatives

In kinematics, there are a whole set of quantities that can describe an object's trajectory, e.g., the acceleration, which is defined to be the rate of change of velocity w.r.t. time:

$$\mathbf{a}(t) = \frac{\mathrm{d}\mathbf{v}(t)}{\mathrm{d}t}.$$

But the $\mathbf{v}$ is also the rate of change of position w.r.t. time. Thus,

$$\mathbf{a}(t) = \frac{\mathrm{d}\mathbf{v}(t)}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\mathrm{d}\mathbf{r}(t)}{\mathrm{d}t}\right) = \frac{\mathrm{d}^2\mathbf{r}(t)}{\mathrm{d}t^2},$$

where $\mathbf{r}$ is any position vector.

The $\mathrm{d}^2\mathbf{r}$ and $\mathrm{d}t^2$ is just a matter of symbolic manipulation and should only be interpreted as just a shorthand. $\mathbf{a}$ is called the **second order derivative** of $\mathbf{r}$ because you've differentiated $\mathbf{r}$ twice. **Higher order derivatives** of position w.r.t. time is listed in table 1.1.

Order	Name
1	Velocity/Speed
2	Acceleration
3	Jerk
4	Snap/Jounce
5	Crackle
6	Pop

TABLE 1.1 | HIGHER ORDER DERIVATIVES OF POSITION W.R.T. TIME

1.7 Expressing nature: basic differential equations

The dynamics of a physical system is universally described by the famous Newton's second law $\mathbf{F} = m\mathbf{a}(t)$, which in derivatives form becomes:

$$\mathbf{F} = m\frac{\mathrm{d}^2\mathbf{r}(t)}{\mathrm{d}t^2}. \tag{1.11}$$

Let's try to describe a simple system with this. In fig. 1.3, a ball is dropped from height h. The Earth's gravity pulls the ball with force mg where m is the mass of the ball, and $\mathbf{g}$, the acceleration from Earth's gravity. Newton's second law tells us that

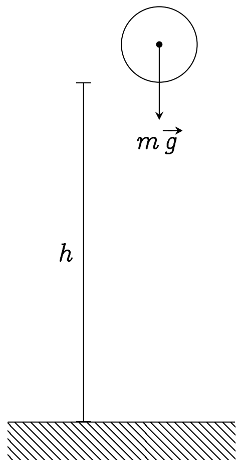


Fig. 1.3 | A BALL DROPPED FROM HEIGHT h

$$\mathbf{F} = m\frac{\mathrm{d}^2\mathbf{r}(t)}{\mathrm{d}t^2} \tag{1.12}$$

$$m\mathbf{g} = m\frac{\mathrm{d}^2\mathbf{r}(t)}{\mathrm{d}t^2} \quad \text{or,} \quad \mathbf{g} = \frac{\mathrm{d}^2\mathbf{r}(t)}{\mathrm{d}t^2}, \tag{1.13}$$

which is sometimes called the **equation of motion**, which packs every information about this system you'd ever want. It directly reads as:

The acceleration of the ball is equals to g.

If you've stayed for this long, congratulations! You now have the power to describe every physical systems with mathematical equations using derivatives: the measure of the rate of change. But it might not be as useful as yet, just as a hammer may seem useless if used to paint, derivatives falls apart when you ask about the future of the system. E.g., how long the ball takes to reach the ground? Or, what's the position of the ball at a certain time? That's the job of the integral to solve, and we'll do so in the next chapter.

1.8 Conclusion for Chapter 1

1. The concept of approaching can be used to bypass dividing by zero.
2. Derivatives are rate of change of a function w.r.t. a variable which can be evaluated by the method of increments.
3. Derivatives can be thought as the slope of a graph, or the tangent to a curve.
4. Physical systems can be described by differential equations of different forms. One of them is the Newton's formulation stated in eq. (1.11)

Remarks 1.8.1

1. In section 1.2, we zoomed in on the graph to approximate the function as a line. Actually, this is quite literally the whole idea of derivatives. If we dig in further in calculus, sometimes the rate of change analogy doesn't even make sense. However, saying that the derivative tries to approximate every function as a line works in all scenario. Though, it's quite abstracted away from the world.

CHAPTER 2

Integrals and antiderivative

Prerequisites: *chapter 1, sigma summation notation*

Terminologies: Every line, including straight, is a **curve**. You might see other textbooks use the term "*area under the graph*" to refer to integrals. But, a **graph** is *a diagram consisting of a line or lines, showing how two or more sets of numbers are related to each other* [5], not the curve itself. Therefore, I'll refer to a curve as any line that connects two points, whether straight or not. Now let's start.

2.1 Invitation to an impossible mission

2.1.1 The mindset of integral calculus

Literally everything that involves a quantity changing is written in the language of derivatives. When derivatives are used in equations to describe a system, it becomes *a differential equation*. Its solution contains the future of the entire system in question. However, it's quite hard to solve; some even impossible.

In classical physics, a **state** represents the configuration that the system *at one point in time*. To predict the system, we need to know the initial state of a system. Classical physics says that if you know the rules that the system plays by (In this case, $\mathbf{F} = m\dfrac{\mathrm{d}^2\mathbf{r}}{\mathrm{d}t^2}$), and the initial condition, we can always determine the state of the system at any point in time. That is, classical physics guarantees that there is always a function, which takes in time as input, and outputs the state of a system. And in order to predict the system, we must know that function. But what may that function be?

Consider the example from chapter 1. The equation of motion of the ball dropped from a height h is a second-order differential equation

$$g = \frac{\mathrm{d}^2\mathbf{r}(t)}{\mathrm{d}t^2}, \tag{2.1}$$

with initial condition being $\mathbf{r} = h$. The function that describes the state is $\mathbf{r}(t)$, which outputs the position of the ball at a certain time t. You can see that this function is the derivative sign. To solve the differential equation for this function, we have to isolate it out, and undo the derivative sign. But how?

It seems impossible at first. We only know that $\mathbf{r}(t)$ must satisfy eq. (2.1), i.e., the second derivative of $\mathbf{r}(t)$ is g. It's like we have to search through a gigantic pool of functions that mathematics have to offer to find a single function $\mathbf{r}(t)$ that satisfies eq. (2.1). It'd be like finding a needle in the haystack!

Of course, this is a textbook, there must be a solution. If there is a will, there is a way. You might have to reverse engineer derivatives, which might look tedious at first. But well, you might find something interesting along the way.

2.1.2 Notation of integral calculus

The $\int$, a.k.a. the **integral**[1], means to sum. This integral symbol is basically sigma summation symbol, but for infinitesimals. Therefore, we have bounds called **integral bounds**. E.g., if we sum a lot of little time step $\mathrm{d}t$ together from t_A to t_B we get $t_B - t_A$, the total time step. Thus,

$$\int_{t_A}^{t_B} \mathrm{d}t = t_B - t_A. \tag{2.2}$$

2.2 Finding a function in the haystack

Equation (2.1) has a second order derivative, let's go slowly and undo one derivative at a time. We'll undo the derivative of the R.H.S. to get the velocity first. Then, we'll undo the derivative again to get the position.

2.2.1 Step one: the velocity function from acceleration

One good strategy for solving any kind of equation is **separation of variables**. We isolate the variable we're interested in solving, which is $\mathbf{r}$, and move everything else to the other side. Here, write $\dfrac{\mathrm{d}^2\mathbf{r}(t)}{\mathrm{d}t^2}$ as $\dfrac{\mathrm{d}\mathbf{v}(t)}{\mathrm{d}t}$. Then, isolate $\mathbf{v}$ on one side and move $\mathbf{t}$ to the other.

$$\mathbf{g} = \frac{\mathrm{d}\mathbf{v}(t)}{\mathrm{d}t} \tag{2.3}$$

$$\mathrm{d}\mathbf{v}(t) = \mathbf{g}\,\mathrm{d}t. \tag{2.4}$$

This equation reads

> *A small change in velocity* $\mathrm{d}\mathbf{v}$ *is product of* $\mathbf{g}$ *and a small time interval* $\mathrm{d}t$.

[1]Which also looks like a beansprout

To find the total change in velocity, we sum up a lot of small changes in velocity $\mathrm{d}\mathbf{v}(t)$, which is equal to $\mathbf{g}\,\mathrm{d}t$. Because $\mathrm{d}v$ is directly proportional to $\mathrm{d}t$, the total change in $\mathbf{v}$ is simply $\mathbf{g}t$. However, *changes* doesn't say anything about the initial condition, so we add a term C to compensate. Therefore,

$$\mathbf{v}(t) = \mathbf{g}t + C. \tag{2.5}$$

To find what C is, just plug in the initial condition. If $\mathbf{v}(t = 0) = \mathbf{v}_0$, then

$$\mathbf{v}(t = 0) = v_0 = \mathbf{g} \times 0 + C \tag{2.6}$$

$$v_0 = C. \tag{2.7}$$

Thus,

$$\mathbf{v}(t) = \mathbf{g}t + \mathbf{v}_0. \tag{2.8}$$

We can also express these ideas symbolically using the integral symbol (section 2.1.2) as

$$\mathbf{g} = \frac{\mathrm{d}\mathbf{v}(t)}{\mathrm{d}t} \tag{2.9}$$

$$\mathrm{d}\mathbf{v} = \mathbf{g}\,\mathrm{d}t \tag{2.10}$$

$$\int \mathrm{d}\mathbf{v} = \int \mathbf{g}\,\mathrm{d}t \tag{2.11}$$

$$\mathbf{v} = \mathbf{g}t + \mathbf{v}_0. \tag{2.12}$$

2.2.2 Step two: the position function from velocity

Rewrite $\mathbf{v}$ as $\frac{\mathrm{d}\mathbf{r}}{\mathrm{d}t}$, then do separation of variables.

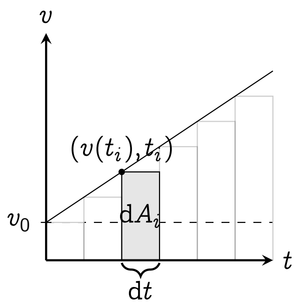
$$\mathbf{v} = \frac{\mathrm{d}\mathbf{r}}{\mathrm{d}t} = \mathbf{g}t + \mathbf{v}_0 \tag{2.13}$$

$$\mathrm{d}\mathbf{r} = \mathrm{d}t(\mathbf{g}t + \mathbf{v}_0), \tag{2.14}$$

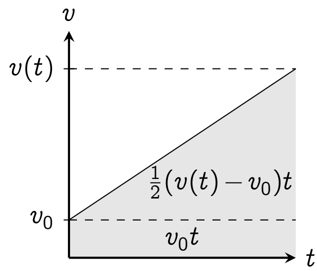
which reads,

> *A small change in position* $\mathrm{d}\mathbf{r}$ *is the product of* $\mathbf{g}t + \mathbf{v}_0$ *and a small time interval* $\mathrm{d}t$.

If we want to find the total change in position $\mathbf{r}$, we just have to sum all $\mathrm{d}\mathbf{r}$'s. This time, it's not as obvious, because there's $\mathbf{g}t + \mathbf{v}_0$, which is also changing with time. So at each point in time, the rate of change of position is different.



(A) A SUBDIVIDED GRAPH



(B) A GRAPH SHOWING THE AREA OF THE GRAY TRAPEZOID BROKEN INTO TWO PARTS

FIG. 2.1 | A v-t GRAPH OF A BALL DROPPED FROM A BUILDING

A good strategy in math if you don't know what to do is to just graph the function. The graph of eq. (2.13) is shown in fig. 2.1a. A small time interval represents a little step in the t axis. The curve shown in the graph represents $\mathbf{g}t + \mathbf{v}_0$. Therefore, $\mathrm{d}t(\mathbf{g}t + \mathbf{v}_0)$ would just represent an area of a little rectangle $\mathrm{d}A$ as shown in the figure.

The total change in position is the sum all those rectangles. When $\mathrm{d}t \to 0$, the sum of all $\mathrm{d}t(\mathbf{g}t + \mathbf{v}_0)$ approaches the area under the graph, which can be calculated geometrically as shown in fig. 2.1b. Thus,

$$\mathbf{r} = \frac{1}{2}(\mathbf{v}(t) - \mathbf{t}_0)t + \mathbf{v}_0 t + C \tag{2.15}$$

$$= \frac{1}{2}(\mathbf{g}t - \mathbf{t}_0)t + \mathbf{v}_0 t + C \tag{2.16}$$

$$= \frac{1}{2}\mathbf{g} \times t^2 + \mathbf{v}_0 t + C \tag{2.17}$$

When $t = 0$, $\mathbf{r} = \mathbf{r}_0$. Therefore,

$$\mathbf{r}_0 = \frac{1}{2}\mathbf{g} \times 0^2 + \mathbf{v}_0 \times 0 + C$$

$$\mathbf{r}_0 = C.$$

Therefore,

$$\mathbf{r}(t) = \frac{1}{2}\mathbf{g}t^2 + \mathbf{v}_0 t + \mathbf{r}_0. \tag{2.18}$$

To model the trajectory of the ball, we set

1. $\mathbf{v}_0 = 0$ (object is dropped and starts at zero speed)
2. $\mathbf{r}_0 = 0$ (convenient initial condition placement)

thus we get,

$$\mathbf{r} = \frac{1}{2}\mathbf{g}t^2 \implies t = \sqrt{\frac{2\mathbf{r}}{\mathbf{g}}} \tag{2.19}$$

Since the ground is at $\mathbf{r} = h$, the time that the ball hits the ground is then $\sqrt{2h/\mathbf{g}}$.

2.2.3 Conclusion: area under the curve and antiderivative

The form of the differential equation that we've solved is in the form $\frac{\mathrm{d}x}{\mathrm{d}t} = f(x)$. We have to undo the derivative, and the simplest way is to do separation of variables and turn the equation into

$$\mathrm{d}x = f(x)\,\mathrm{d}t, \tag{2.20}$$

which reads,

> A small change in x, i.e., $\mathrm{d}x$ is represented by the area of a rectangle width $\mathrm{d}t$ and height $f(x)$.

And, the total change x is represented by the sum all those little rectangles, which is the area under the curve $f(x)$. Then, we add a constant C to compensate for the initial condition. In symbolic form introduced in section 2.1.2, it's just

$$\int \mathrm{d}x = \int g(x)\,\mathrm{d}t \tag{2.21}$$

$$x = \int g(x)\,\mathrm{d}t. \tag{2.22}$$

For now, we could say that integration is the reverse of derivatives. But to clearly see how this is linked for every function, we must study the fundamental theorem of calculus, which is the bridge between integration and differentiation.

Before we go there, let me clarify some terminologies. An **integral** refers to the area under the curve evaluated between two points. We say that an integral must have an **integral bound**. If we want to find the area under a function $f(x)$ from $x = a$ to $x = b$, we write it as

$$A = \int_a^b f(x)\,\mathrm{d}x. \tag{2.23}$$

This just reads

> The area A under the curve $f(x)$ from $x = a$ to $x = b$ is equal to the sum of the area of many thin stripes width $\mathrm{d}x$ height $f(x)$ that lies between $x = a$ and $x = b$.

The **antiderivative** however, refers to the function which takes in a value a, and output the integral of $f(x)$, evaluated from 0 to a. Therefore, if $A(a)$ is the antiderivative of $f(x)$, then

$$A(a) = \int_0^a f(x)\,\mathrm{d}x. \tag{2.24}$$

It also implies that if the area between a and b can be evaluated by

$$\int_a^b f(x)\,\mathrm{d}x = A(b) - A(a). \tag{2.25}$$

2.3 The fundamental theorem of calculus

The intuition of fundamental theorem of calculus states that derivatives and integrals are essentially inverse of each other. In this section, I'll clarify this fact and make it more rigorous.

Theorem 2.3.1: *The (first) fundamental theorem of calculus*

If $A(x)$ is the antiderivative of $f(x)$, then

$$\frac{\mathrm{d}A(x)}{\mathrm{d}x} = f(x). \tag{2.26}$$

If $A(x)$ is the antiderivative of $f(x)$, then

$$\int_0^x f(x)\,\mathrm{d}x = A(x). \tag{2.27}$$

The *actual* area of one of the stripes (not rectangles) width $\mathrm{d}x$ shown in fig. 2.2, it's obviously $A(x+\mathrm{d}x) - A(x)$. The riemann sum approximation approximates the area by a small rectangle area $f(x)\,\mathrm{d}x$. We can write the relation between the actual and the approximated area as

$$A(x+\mathrm{d}x) - A(x) \approx f(x)\,\mathrm{d}x. \tag{2.28}$$

To turn this into an equality, we add a correction term ε

$$A(x+\mathrm{d}x) - A(x) = f(x)\,\mathrm{d}x + \varepsilon. \tag{2.29}$$

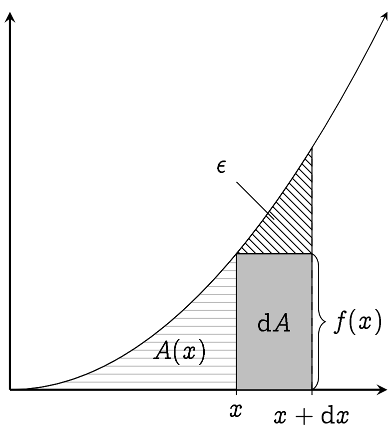


FIG. 2.2 | THE GEOMETRICAL INTERPRETATION OF PART ONE OF THE FUNDAMENTAL THEOREM OF CALCULUS (THEOREM 2.3.1).

If we let $\mathrm{d}x \to 0$, ε is negligible compared to $f(x)\,\mathrm{d}x$; therefore,

$$\lim_{\mathrm{d}x\to 0} A(x+\mathrm{d}x) - A(x) = \lim_{\mathrm{d}x\to 0} f(x)\,\mathrm{d}x. \tag{2.30}$$

Since $f(x)$ is not a variable that's controlled by the limit sign,

$$\lim_{\mathrm{d}x\to 0} A(x+\mathrm{d}x) - A(x) = f(x) \lim_{\mathrm{d}x\to 0} \mathrm{d}x$$
$$\lim_{\mathrm{d}x\to 0} \frac{A(x+\mathrm{d}x) - A(x)}{\mathrm{d}x} = f(x).$$

And here, we see that the L.H.S. is just the derivative of $A(x)$ w.r.t. x, thus

$$\frac{\mathrm{d}A(x)}{\mathrm{d}x} = f(x), \tag{2.31}$$

or "*the rate of change in area is the function itself*". But the area function is given by the integral. This means

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\int f(x)\,\mathrm{d}x\right) = f(x): \tag{2.32}$$

integrals and derivatives are inverses of each other. If we rephrase theorem 2.3.1, we see that "*the integral is the cumulative effect of the function.*"

The second fundamental theorem of calculus,

Theorem 2.3.2: *The second fundamental theorem of calculus (Newton-Leibniz rule)*

If $A(x)$ is the antiderivative of $f(x)$, then the integral of $f(x)$ from $x = a$ to $x = b$ is

$$\int_a^b f(x)\,\mathrm{d}x = \int_0^b f(x)\,\mathrm{d}x - \int_0^a f(x)\,\mathrm{d}x = A(b) - A(a). \tag{2.33}$$

follows as a direct consequence of the geometrical interpretation of integrals: area under the curve.

The fundamental theorem of calculus allows us to evaluate integrals using derivatives. For example, if the derivative of x^2 is $2x$, then we also know that the integral of $2x\,\mathrm{d}x$ is just x^2, which we'll discuss how to do that in the next chapter.

2.4 How to calculate an integral? Riemann sum

In section 2.2, we used geometry to find the area under a curve. However, that is not always possible, e.g., try integrating fig. 2.3 geometrically[2]. It'd be impossible. But we can still approximate its area by slicing the area under the curve into thin rectangular stripes, then summing them. The approximated area is called the **Riemann sum**. Due to its computational cost, you don't really want to use this method. However, to develop a good intuition at the integral, we should still know its symbolic form.

Let there be a function $A(s)$ that represents the actual area of a

[2]This is what you'd get if you solve the simple harmonic oscillator

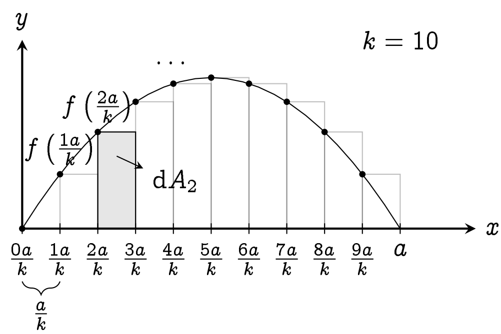


FIG. 2.3 | ILLUSTRATION OF RIEMANN SUM OF A FUNCTION $f(x)$ FROM 0 TO a BY SETTING $k = 10$

function $f(x)$ from 0 to s. The approximated area is then

$$\sum_{i=0}^{k} \mathrm{d}A_i = \sum_{i=0}^{k} \text{width} \times \text{height} \tag{2.34}$$

where k is the amount of subdivisions. From fig. 2.3, the width of each stripe is a / k, and the height of the i'th stripe is $f(ia / k)$. Therefore,

$$\sum_{i=0}^{k} \frac{a}{k} \times f\left(\frac{ia}{k}\right).$$

In fig. 2.3, we $k = 10$. The area of the second rectangle $\mathrm{d}A_2$ is

$$\mathrm{d}A_2 = \frac{a}{k} f\left(\frac{2a}{k}\right).$$

For an arbitrarily finite k, the Riemann sum is just an approximation. If you want to find the *actual* area under the curve, let $k \to \infty$. The limit as $k \to \infty$ is what we actually call the **integral**. Thus, we say

Definition 2.4.1: *Naive definition of integrals*

The (definite) integral, or the area under the curve of $f(x) = y$ from 0 to a, is defined as

$$\int_0^a \mathrm{d}A = \int_0^a f(x)\,\mathrm{d}x = \lim_{k\to\infty} \sum_{i=0}^{k} \frac{a}{k} \times f\left(\frac{ia}{k}\right) = A(a), \tag{2.35}$$

where $\int$ is the integral sign[a]. Here, 0 is the lower bound of integration, and a, the higher bound. The function $A(x)$ is called the **antiderivative**, or the **indefinite integral** of $f(x)$.

[a] Famously known for looking like a beansprout

I shall put these definitions into perspective in the next two examples. It might use a bit of series knowledge. If you don't know, you can simply search up the summation identities that I'll use in *Wikipedia* [6].

Example 2.4.2: *Riemann sum and antiderivative of x^2* Let $f(x) = x^2$, and let $A(a)$ be the antiderivative of $f(x)$, i.e., $A(a)$ is the area under the curve of $f(x)$ from 0 to a. Note that

$$\sum_{i=0}^{k} i^2 = \frac{k(k+1)(2k+1)}{6}, \tag{2.36}$$

The Riemann sum is then

$$\sum_{i=0}^{k} \frac{a}{k} \times f\left(\frac{ia}{k}\right) = \sum_{i=0}^{k} \frac{a}{k} \times \left(\frac{ia}{k}\right)^2 \tag{2.37}$$

$$= \sum_{i=0}^{k} \frac{a^3}{k^3} \times i^2 \tag{2.38}$$

$$= \frac{a^3}{k^3} \times \frac{k(k+1)(2k+1)}{6}. \tag{2.39}$$

To find the antiderivative, let $k \to \infty$. Notice that the $+1$ in the parenthesis are negligible when $k \to \infty$. Therefore, we can write the antiderivative as

$$A(a) = \int_0^a f(x)\,\mathrm{d}x \tag{2.40}$$

$$= \lim_{k\to\infty} \frac{a^3}{k^3} \times \frac{k(k+1)(2k+1)}{6} \tag{2.41}$$

$$= \lim_{k\to\infty} \frac{a^3}{k^3} \times \frac{2k^3}{6} \tag{2.42}$$

$$= \lim_{k\to\infty} \frac{x^3}{3} = \frac{x^3}{3}. \tag{2.43}$$

Example 2.4.3: *Riemann sum and antiderivative of x^3* Let $f(x) = 4x^3$, and let $A(a)$ be the antiderivative of $f(x)$, i.e., $A(a)$ is the area under the curve of $f(x)$ from 0 to a. Note that

$$\sum_{i=0}^{k} i^3 = \left(\frac{k(k+1)}{2}\right)^2, \tag{2.44}$$

The Riemann sum is then

$$\begin{aligned}\sum_{i=0}^{k}\frac{a}{k}\times f\left(\frac{ia}{k}\right) &= \sum_{i=0}^{k}\frac{a}{k}\times 4\left(\frac{ia}{k}\right)^3\\ &= 4\sum_{i=0}^{k}\left(\frac{x}{k}\right)^4 i^3\\ &= 4\left(\frac{x}{k}\right)^4\left(\frac{k(k+1)}{2}\right)^2.\end{aligned}$$

To find the antiderivative, let $k\to\infty$. The $+1$ in $(k+1)$ can be ignored as $k\to\infty$. Therefore,

$$\begin{aligned}A(a) &= \int_0^a f(x)\,\mathrm{d}x\\ &= \lim_{k\to\infty} 4\left(\frac{x}{k}\right)^4\left(\frac{k^2}{2}\right)^2\\ &= \lim_{k\to\infty} x^4 = x^4.\end{aligned}$$

2.5 Basic applications of the integrals

2.5.1 Five equations of linear motion

So far, we have developed an intuition for derivatives, integrals, and their relationship. Let's apply those concepts to derive the famous equations of linear motion:

Theorem 2.5.1: *Motion with uniformed acceleration*

$$v(t) = v_0 + at, \tag{2.45}$$

$$s = \frac{1}{2}(v_0 + v(t))t, \tag{2.46}$$

$$s = v_0(t)t + \frac{1}{2}at^2, \tag{2.47}$$

$$s = v(t)t - \frac{1}{2}at^2, \tag{2.48}$$

$$v(t)^2 = v_0^2 + 2as. \tag{2.49}$$

Here, v_0 represents the initial velocity, $v(t)$, the velocity at any time t, a, the acceleration, and s, the displacement. These equations are derived from the Newton's second law on *constant/uniformed acceleration* motion in one dimension, which we can evaluate using the geometrical interpretation of integrals developed earlier.

Because we've constrained the object to accelerate at a, the force exerted must be ma. Newton's second law $F = m\frac{\mathrm{d}^2x}{\mathrm{d}t^2}$ then simplifies to $a = \frac{\mathrm{d}^2x}{\mathrm{d}t^2}$. The same idea applies, we rewrite the equation in terms of v.

$$a = \frac{\mathrm{d}v}{\mathrm{d}t} \tag{2.50}$$

Which reads, *"the rate of change of v w.r.t. t is a"* or *"the slope of the v-t curve is always equals to a"*. Because a is constant, the v-t curve must be a straight line with slope a, shown in fig. 2.4.

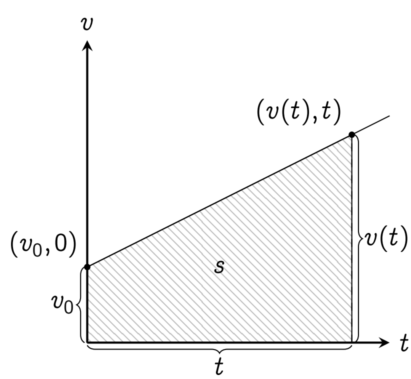


FIG. 2.4 | v-t GRAPH OF AN OBJECT UNDER CONSTANT OR UNIFORMED ACCELERATION

Derivation of eq. (2.45). We take advantage of the linearness of the curve. Just pick two points on fig. 2.4, as already shown, then

$$\begin{aligned} m = a &= \frac{\Delta v}{\Delta t} = \frac{v(t) - v_0}{t - 0} \\ a &= \frac{v(t) - v_0}{t} \\ v(t) &= v_0 + at. \end{aligned}$$

□

Derivation of eq. (2.46). Use the reverse of theorem 2.3.1. We know that[3] $v = \frac{\mathrm{d}}{\mathrm{d}t}s$; therefore,

$$\int v \, \mathrm{d}t = s,$$

which reads "*The displacement s is the area under the curve of a v-t graph*". From fig. 2.4, the area under the curve is a trapezoid with side length v_0, $v(t)$, and width t. Thus,

$$s = \frac{1}{2}(v_0 + v(t))t,$$

which is just eq. (2.46): the area of a trapezoid. □

Derivation of eqs. (2.47) to (2.49). We can arrange eq. (2.45) into three different ways, then plug in eq. (2.46). First, $v(t) = v_0 + at$

$$\begin{aligned} s &= \frac{1}{2}(v_0 + v_0 + at)t \\ &= v_0 t + \frac{1}{2}at^2. \end{aligned}$$

Second, $v_0 = v(t) - at$

$$\begin{aligned} s &= \frac{1}{2}(v(t) - at + v(t))t \\ &= v(t)t - \frac{1}{2}at^2. \end{aligned}$$

Third, $t = \frac{v(t) - v_0}{a}$

$$\begin{aligned} s &= \frac{1}{2}(v(t) + v_0)\frac{(v(t) - v_0)}{a} \\ 2as &= (v(t) + v_0)(v(t) - v_0) \\ 2as &= v(t)^2 - v_0^2 \\ v(t)^2 &= 2as + v_0^2. \end{aligned}$$

□

2.5.2 The area of a circle

The perimeter of the circle is $2\pi r$. For now, we only know how to find areas of polygons, but we want to know the circle's area. So, is there any way to turn a circle into a polygon? From section 2.4, that the riemann sum can approximate areas under the curve using rectangular stripes. It'd

[3]Here, x is replaced with s to represent displacement in one dimension.

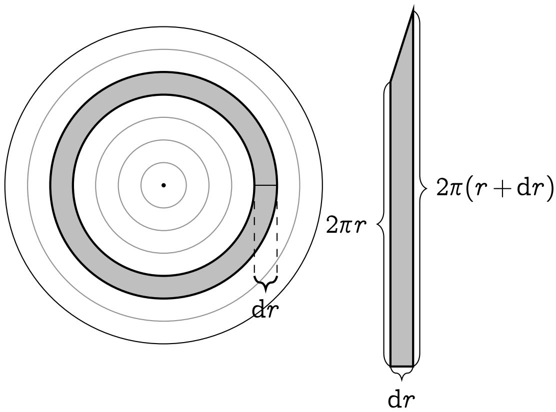


FIG. 2.5 | (LEFT) DISSECTING A CIRCLE RADIUS R RADIALLY INTO RINGS, EACH RING $\mathrm{d}r$ THIN. (RIGHT) STRETCHING A RING INTO A TRAPEZOID (NOT TO SCALE).

be great if this circle can be turned into multiple rectangular stripes on a graph right?

So, we dissect a circle radially into small rings $\mathrm{d}r$ thin, as shown in fig. 2.5. Then, stretch all the rings into a very thin trapezoid-like shape. It might seem impossible at first, but considering that our ring is very thin, it's quite easy to stretch it without breaking. I encourage you to grab a piece of paper, cut a really thin ring and try it out. If you actually do it, it'll look a bit warped. However, the warpedness will go away the thinner you go.

Since we sliced our trapezoid from a circle, for a stripe positioned at r, the inner side will be $2\pi r$ long and the outer side, $2\pi(r + dr)$. The area of the little trapezoid $\mathrm{d}A$ then becomes

$$\begin{aligned}\mathrm{d}A &= \frac{1}{2}\,\mathrm{d}r(2\pi r + 2\pi(r + dr)) \\ &= 2\pi r\,\mathrm{d}r + 2\pi\,\mathrm{d}r^2.\end{aligned}$$

As $\mathrm{d}r \to 0$, $\mathrm{d}r^2$ becomes negligible. Therefore,

$$\mathrm{d}A = 2\pi r\,\mathrm{d}r. \tag{2.51}$$

This equation says that for $\mathrm{d}r \to 0$, the trapezoid becomes a rectangle side length $\mathrm{d}r$ and $2\pi r$. Now, we have to sum it together. If we can put all these rectangles onto a graph, we can easily use the Riemann's sum to evaluate it. A natural way to do this is to put all the rectangles that we got from stretching the rings of the circle onto a graph one by one. The result would look something like fig. 2.6[4].

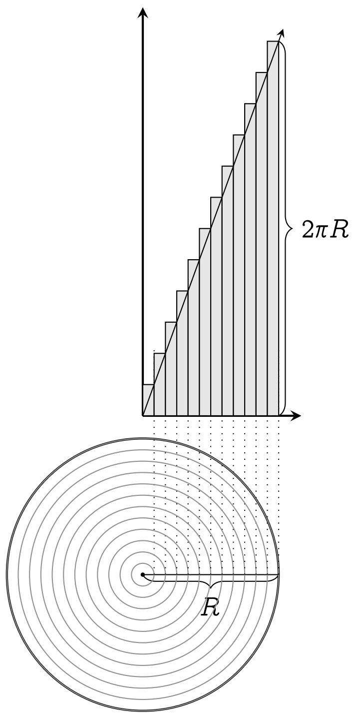


FIG. 2.6 | REARRANGING ALL THE APPROXIMATED RECTANGLES ONTO A GRAPH (NOT TO SCALE).

For every stripe at r, its height is $2\pi r$. If you were to plot the height of all rectangles when $\mathrm{d}r \to 0$, it'll eventually look like a curve that's given by $f(r) = 2\pi r$. We're interested in the area of the circle from 0 to R. Now, it's transformed into the area under the curve of $f(r) = 2\pi r$: a triangle with base R and height $f(R) = 2\pi R$. Thus, the area of a circle becomes

$$A = \frac{1}{2}R(2\pi R) = \pi R^2.$$

And there you go, you've turned essentially turned circle into a triangle and evaluate its area from there. I'd like to end off this chapter by mentioning the spirit of mathematics. Sometimes, you can't solve the

[4]Not to scale

problem directly. Most of the time, you have to re-frame the problem into another more-solvable problem. Problems like these often have the most sublime connections to the foundations of mathematics. This is a common theme in most of mathematics, especailly calculus. So, be sure to keep this in mind while reading through.

2.6 Conclusion for Chapter 2

1. Riemann sum are used to approximate areas under the curve of a function by using little rectangles then summing it.

2. Integrals or anti-derivatives are functions that output the area under the graph of other functions.

3. The limit where the width of the rectangles in the Riemann sum approaches zero, the Riemann sum becomes an integral.

4. Integrals are the cumulative effect of a function.

5. Integrals and derivatives are inverses of each other, and they're related by the fundamental theorem of calculus

6. Integrals can be used in various ways by reframing questions into another simpler question.

CHAPTER 3

Basic derivatives and antiderivatives

This chapter covers the derivatives and integrals of common functions: polynomials, exponential, and logarithms; focusing on their geometrical interpretation. It prepares you for the real deal: differential equations. So, it's better if you can study these contents very thoroughly and get very familiar with doing derivatives and integrals. Most of these rules can be done mentally, and it's advised to practice these skills, so you will have an easier time solving differential equations.

Prerequisites: *binomial theorem (appendix B), basic trigonometry, derivatives, and integrals*

Terminologies: The integral refers to the area under the curve. The antiderivative refers to the function $A(x')$ that outputs the area under the curve from 0 to x'.

3.1 Basic rules

The chain rule The Leibniz' notation treats derivative as fractions; thus, we can cancel terms (eq. (1.9)). We exploit this property to find derivatives of composite functions. Consider two functions $f(x)$ and $g(x)$, the derivative of $f(g(x))$ can be found by a simple substitution. First let $g(x) = u$.

$$\frac{\mathrm{d}}{\mathrm{d}x} f\left(g(x)\right) = \frac{\mathrm{d}f(u)}{\mathrm{d}x}.$$

Let $1 = \dfrac{\mathrm{d}u}{\mathrm{d}u}$, then

$$\frac{\mathrm{d}}{\mathrm{d}x} f(u) = \frac{\mathrm{d}}{\mathrm{d}x} f(u) \times 1 = \frac{\mathrm{d}f(u)}{\mathrm{d}x} \frac{\mathrm{d}u}{\mathrm{d}u}.$$

Perform a change of denominator, then substitute back $u = g(x)$:

$$\frac{\mathrm{d}}{\mathrm{d}x} f\left(g(x)\right) = \frac{\mathrm{d}f(u)}{\mathrm{d}u} \frac{\mathrm{d}g(x)}{\mathrm{d}x}.$$

This is what we call the chain rule, or more generally

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\mathrm{d}y}{\mathrm{d}u} \times \frac{\mathrm{d}u}{\mathrm{d}x}. \tag{3.1}$$

where u is a function of y.

The chain rule holds the intuition of how rate of changes relate to each other. E.g., the cheetah's speed is 10 times the bicycle's speed, which is 4 times the walking speed. The ratio between the cheetah's speed compared to walking speed would obviously be $10 \cdot 4 = 40$:

$$\frac{\mathrm{dCheetah}}{\mathrm{dWalking}} = \frac{\mathrm{dCheetah}}{\mathrm{dBicycle}} \times \frac{\mathrm{dBicycle}}{\mathrm{dWalking}}. \tag{3.2}$$

Integral constant In section 2.2, each time we evaluate the antiderivative, we add an initial condition term, i.e., v_0 and r_0. So for any function $f(x)$,

$$\int f(x)\,\mathrm{d}x = A(x) + C, \tag{3.3}$$

where C is any constant. However, theorem 2.3.2 still holds for integrals with bounds.

Always add an integral constant after evaluating the antiderivative

Integral of the infinitesimal The antiderivative of the small rectangles $\mathrm{d}x$ is just the total rectangle x plus the integral constant C.

$$\int \mathrm{d}x = x + C. \tag{3.4}$$

Rules of equality If two arguments are equal, their derivatives and antiderivatives w.r.t. the same variable must also be equal.

$$\text{If } f = g, \text{ then } \frac{\mathrm{d}f}{\mathrm{d}x} = \frac{\mathrm{d}g}{\mathrm{d}x}, \text{ and } \int f \,\mathrm{d}x = \int g \,\mathrm{d}x + C$$

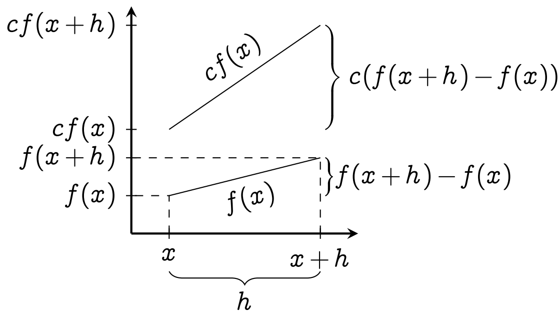
Derivative of a constant A constant doesn't change; thus, the derivative of a constant is zero.

$$\frac{\mathrm{d}}{\mathrm{d}x}(c) = 0. \tag{3.5}$$

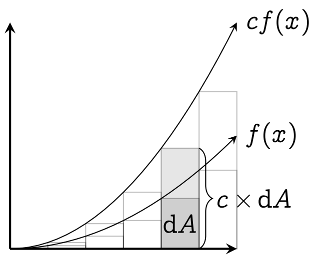
3.2 Linearity of differentiation and integration

A **linear** mathematical entity are defined as anything that is compatible with addition and scaling. E.g., for a function $f(x)$ to be linear, it must obey

1. Additivity: $f(x+y) = f(x) + f(y)$
2. Homogeneity of degree one: $f(ax) = af(x)$ for all constant a



(A) FOR DERIVATIVES



(B) FOR INTEGRALS

FIG. 3.1 | CONSTANT MULTIPLE RULES

The simplest linear function there is, is a line that passes through the origin: $f(x) = ax$, in which

$$f(x+y) = a(x+y) = ax + ay = f(x) + f(y), \tag{3.6}$$

$$f(cx) = a(cx) = c(ax) = cf(x). \tag{3.7}$$

Because this property, we associate a line with being linear. However, linearity doesn't have to always refer to lines.

Both derivatives and integrals are linear; for any function $f(x)$, $g(x)$, and constants a, b, the following rules follow.

The constant multiple rule:

$$\frac{\mathrm{d}}{\mathrm{d}x}(af(x)) = a\frac{\mathrm{d}f(x)}{\mathrm{d}x}, \qquad \text{Illustrated in fig. 3.1a} \tag{3.8}$$

$$\int af(x)\,\mathrm{d}x = a\int f(x)\,\mathrm{d}x. \qquad \text{Illustrated in fig. 3.1b} \tag{3.9}$$

The geometrical interpretation of these two rules are very simple. When a function is multiplied by a constant c, its value is increased by a factor of c everywhere. The slope must also be increased by c, and thus the function's area also.

The sum rule:

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(f(x)+g(x)\right)=\frac{\mathrm{d}}{\mathrm{d}x}f(x)+\frac{\mathrm{d}}{\mathrm{d}x}g(x), \tag{3.10}$$

$$\int f(x)+g(x)\,\mathrm{d}x=\int f(x)\,\mathrm{d}x+\int g(x)\,\mathrm{d}x. \tag{3.11}$$

I.e., adding two functions increases its slope, thus also increasing its area under the graph.

3.3 Derivatives and antiderivatives of polynomials

Now that we've discussed the "trivial rules", we're ready to tackle the easiest family of functions: the **polynomials**. They're in the form

$$f(x)=a_0+a_1x+a_2x^2+a_3x^3+\dots\,. \tag{3.12}$$

Let's focus on the antiderivative first; we'll see later how the antiderivative of polynomials can be found with just a simple substitution trick.

The form mentioned in eq. (3.12) is quite useless if we want to make progress. We can break it down by using the linearity of derivatives:

$$\frac{\mathrm{d}f(x)}{\mathrm{d}x}=\frac{\mathrm{d}}{\mathrm{d}x}(a_0+a_1x^1+a_2x^2+a_3x^3+\dots) \tag{3.13}$$

$$=a_1\frac{\mathrm{d}x^1}{\mathrm{d}x}+a_2\frac{\mathrm{d}x^2}{\mathrm{d}x}+a_3\frac{\mathrm{d}x^3}{\mathrm{d}x}+\dots\,. \tag{3.14}$$

Now we're left with derivatives of *monomials* in the form of x^n. So let's do that instead.

3.3.1 Derivatives of monomials: the power rule

The method of increments allows us to quickly evaluate the derivative of x^n.

$$\frac{\mathrm{d}}{\mathrm{d}x}(x^n) = \lim_{h\to 0} \frac{(x+h)^n - x^n}{h}. \tag{3.15}$$

By using the binomial expansion (appendix B),

$$= \lim_{h\to 0} \frac{\left(\sum_{k=0}^{n} \binom{n}{k} x^{n-k} \cdot h^n\right) - x^n}{h} \tag{3.16}$$

$$= \lim_{h\to 0} \frac{\binom{n}{0} x^n h^0 + \binom{n}{1} x^{n-1} h^1 + \binom{n}{2} x^{n-2} h^2 + \dots \binom{n}{n} + x^0 h^n - x^n}{h} \tag{3.17}$$

$$= \lim_{h\to 0} \frac{x^n + n x^{n-1} h^1 + \binom{n}{2} x^{n-2} h^2 + \cdots + h^n - x^n}{h} \tag{3.18}$$

$$= \lim_{h\to 0} n x^{n-1} + \binom{n}{2} x^{n-2} h + \binom{n}{3} x^{n-3} h^2 + h^{n-1} \tag{3.19}$$

The terms with h vanishes when $h \to 0$; therefore,

$$\frac{\mathrm{d}}{\mathrm{d}x}(x^n) = n x^{n-1}. \tag{3.20}$$

The equation above is what we call the **power rule**. Because the binomial theorem work for all real numbers, this is true for any powers of n.

Now, what is the geometrical interpretation of the power rule? Let's try to answer this question by geometrically interpreting the simplest non-trivial monomial: x^2. This function represents the area of a square sidelength x. Its derivative then represents the ratio between the change in x, and the change of area when the sidelength is increased by $\mathrm{d}x$.

Illustrated in fig. 3.2,

$$\frac{\mathrm{d}}{\mathrm{d}x}(x^2) = \lim_{\mathrm{d}x\to 0} \frac{x\,\mathrm{d}x + x\,\mathrm{d}x + \mathrm{d}x^2}{\mathrm{d}x}. \tag{3.21}$$

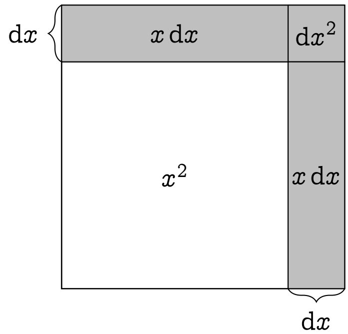


FIG. 3.2 | INTERPRETATION OF $\frac{\mathrm{d}x^2}{\mathrm{d}x}$ ($\mathrm{d}x$ EXAGGERATED)

When $\mathrm{d}x \to 0$, the $\mathrm{d}x^2$ square would just become a single point. Compared to the big $x\,\mathrm{d}x$ on the side, it's negligible; therefore, we can safely ignore it. The equation then becomes

$$\frac{\mathrm{d}}{\mathrm{d}x}(x^2) = \lim_{\mathrm{d}x \to 0} \frac{2x\,\mathrm{d}x}{\mathrm{d}x} = 2x,$$

which is equivalent to the result from the power rule.

Deriving $\frac{\mathrm{d}}{\mathrm{d}x}(x^3)$ geometrically shouldn't be too hard either. Take a cube sidelength x; increase its side length by $\mathrm{d}x$, and compute the ratio between the change in volume and $\mathrm{d}x$. The final answer should be $3x^2$. You could try with more higher order of n, but it'd be very hard to visualize.

Here I leave some exercises which shouldn't be too hard to do. The fourth and fifth one requires you to convert the fractions into negative exponents such as $\frac{1}{x} = x^{-1}$, then use the power rule. The sixth one requires you to convert square roots into fractional exponents. And, the final two would need you to expand the polynomial first, then use the power rule.

1. $\frac{\mathrm{d}}{\mathrm{d}x}(x^2 - 2x + 16)$ $\qquad$ $2x - 2$

2. $\dfrac{\mathrm{d}}{\mathrm{d}x}(x^3 + x^2 + x + 1)$ $\qquad 3x^2 + 2x + 1$

3. $\dfrac{\mathrm{d}}{\mathrm{d}x}(3x^4 + 24x^3 - 2x^2 - 32x + 88)$ $\qquad 12x^3 + 72x^2 - 4x - 32$

4. $\dfrac{\mathrm{d}}{\mathrm{d}x}\left(\dfrac{1}{x^2} + \dfrac{1}{x} + x\right)$ $\qquad -2x^{-3} - 1x^{-2} + 1$

5. $\dfrac{\mathrm{d}}{\mathrm{d}x}\left(\dfrac{3}{x} + x^2 + \dfrac{1}{4}x^{-4}\right)$ $\qquad -3x^{-2} + 2x - x^{-5}$

6. $\dfrac{\mathrm{d}}{\mathrm{d}x}(\sqrt{x} + \sqrt[3]{x})$ $\qquad \dfrac{1}{2}x^{-1/2} + \dfrac{1}{3}x^{-2/3}$

7. $\dfrac{\mathrm{d}}{\mathrm{d}x}((x^2 + 3x)^2)$ $\qquad 4x^3 + 12x^2 + 8x$

8. $\dfrac{\mathrm{d}}{\mathrm{d}x}((x^2 + 1)^2(x - 2))$ $\qquad 5x^4 - 8x^3 + 6x^2 - 8x + 1$

3.3.2 Antiderivatives of polynomials: the reversed power rule

For the antiderivative of x^n, we use the fundamental theorem of calculus (theorem 2.3.1) on the power rule (eq. (3.20)),

$$x^n = \int nx^{n-1}\,\mathrm{d}x.$$

Substitute n with $n + 1$

$$\int (n+1)x^n\,\mathrm{d}x = x^{n+1} + C,$$

and by the linearity of integrations (eq. (3.9)), we get the **reversed power rule**:

$$\int x^n\,\mathrm{d}x = \frac{x^{n+1}}{n+1} + C. \tag{3.22}$$

Notice, this rule does not work for $n = 1$ because we can't divide by zero...*or can we???* (Discussed further in sections 3.5 and 3.7)

In just the same way, it's very important to familiarize yourself to these integrals. Therefore, here I leave some examples for practice. For the forth and fifth one, you also need to turn the fractions into a negative exponent. And the last two, expand the polynomial first.

1. $\displaystyle\int x^2 + 1 \, \mathrm{d}x$ $\qquad \dfrac{x^3}{3} + x + C$
2. $\displaystyle\int 2x^3 + x^2 + 3 \, \mathrm{d}x$ $\qquad \dfrac{x^4}{2} + \dfrac{x^3}{3} + 3x + C$
3. $\displaystyle\int 3x^2 - 5x + 2 \, \mathrm{d}x$ $\qquad x^3 - \dfrac{5x^2}{2} + 2x + C$
4. $\displaystyle\int \sqrt{x} + \frac{1}{3}\sqrt[3]{x} \, \mathrm{d}x$ $\qquad \dfrac{2x^{3/2}}{3} + \dfrac{x^{4/3}}{4}$
5. $\displaystyle\int 3x^{-5/2} + \frac{x^{7/3}}{3}$ $\qquad \dfrac{x^{10/3}}{10} - 2x^{-3/2}$
6. $\displaystyle\int (x^2 + 3x + 1)^2 \, \mathrm{d}x$ $\qquad \dfrac{x^5}{5} + \dfrac{3x^4}{2} + \dfrac{11x^3}{3} + 3x^2 + x$
7. $\displaystyle\int (x + 1)(x - 1)(x + 2) \, \mathrm{d}x$ $\qquad \dfrac{x^4}{4} + \dfrac{2x^3}{3} - \dfrac{x^2}{2} - 2x$

3.3.3 Extending the equations of linear motion

Let's use the power rule to extend the equation of linear motions in section 2.5.1 to include a constant jerk $\jmath$, i.e.,

$$\frac{\mathrm{d}a}{\mathrm{d}t} = \jmath \tag{3.23}$$

Then, we can move the $\mathrm{d}t$ around and integrate both sides by using the reversed power rule.

$$\begin{aligned}
\int \jmath \, \mathrm{d}t &= \int \mathrm{d}a \\
\jmath \int \mathrm{d}t &= \int \mathrm{d}a && \text{Linearity of integrals} \\
\jmath t + a_0 &= a && \text{Integral constant } a_0
\end{aligned}$$

Because a is the derivative of v,

$$\begin{aligned}
\frac{\mathrm{d}v}{\mathrm{d}t} &= jt + a_0 \\
\int \mathrm{d}v &= \int jt + a_0 \,\mathrm{d}t \\
v &= j\int t\,\mathrm{d}t + \int a_0 \,\mathrm{d}t && \text{Linearity of integrals} \\
v &= \frac{1}{2}jt^2 + a_0 t + v_0. && \text{Reversed power rule}
\end{aligned}$$

We add v_0 as an integral constant in a similar manner. Since v is the derivative of r,

$$\begin{aligned}
\frac{\mathrm{d}r}{\mathrm{d}t} &= \frac{1}{2}jt^2 + a_0 t + v_0 \\
\int \mathrm{d}r &= \int \frac{1}{2}jt^2 + a_0 t + v_0 \,\mathrm{d}t \\
r &= \frac{1}{2}j\int t^2\,\mathrm{d}t + a_0\int t\,\mathrm{d}t + v_0\int \mathrm{d}t \\
r &= \frac{1}{6}jt^3 + \frac{1}{2}a_0 t^2 + v_0 t + r_0.
\end{aligned}$$

And there you have it! Further extensions can be made from here: the jerk could be time-dependent. But that's probably enough to illustrate my point. If you'd like to try, extend the equation of motion to have a constant snap: $\frac{\mathrm{d}j}{\mathrm{d}t} = s$. You should get

$$r = \frac{1}{24}st^4 + \frac{1}{6}jt^3 + \frac{1}{2}a_0 t^2 + v_0 t + r_0.$$

3.4 Exponentials and growth

The next function that we're going to discuss is exponentials: the mathematical representation of growth and decay. As an example, let's say there's a magical drop of water that doubles its volume V every hour. I.e., for any time t,

$$V(t+1) = 2V(t). \tag{3.24}$$

t	$V(t)$	$V(t)-V(t-1)$
0	1	
1	2	$2-1=1$
2	4	$4-2=2$
3	8	$8-4=4$
4	16	$16-8=8$
5	32	$32-16=16$
6	64	$64-32=32$
7	128	$128-64=64$
8	256	$256-128=128$

TABLE 3.1 | $V(t)=2^t$ FROM 0 TO 8

If the drop starts at one unit of volume, $V(0)=1$; thus,

$$\begin{aligned}
V(1) &= 2V(0) = 2,\\
V(2) &= 2V(1) = 2(2) = 2^2,\\
V(3) &= 2V(2) = 2(2^2) = 2^3,\\
V(4) &= 2V(3) = 2(2^3) = 2^4,\\
&\vdots
\end{aligned}$$

It's clear that the pattern is $V(t)=2^t$: an exponential function. A natural question to ask is "*what is its rate of change?*". Let's start by plotting the function over time (fig. 3.3). But, exponentials grow too quick to plot! By $V(7)$, we're already in the hundreds. So, it'd probably be better to list the values of each point on a table. Notice that on the right most column of table 3.1, the difference between $V(t)$ and $V(t-1)$ is exactly $V(t-1)$: the function changes as much as its past-self. So does that mean that $\frac{\mathrm{d}}{\mathrm{d}x}(2^x)=2^x$?

Sadly not, but it's very close. See, table 3.1 only shows a *discrete* step where $\mathrm{d}x=1$. You can write it out as

$$\frac{2^{x+1}-2^x}{1}=2^x\left(\frac{2-1}{1}\right)=2^x, \tag{3.25}$$

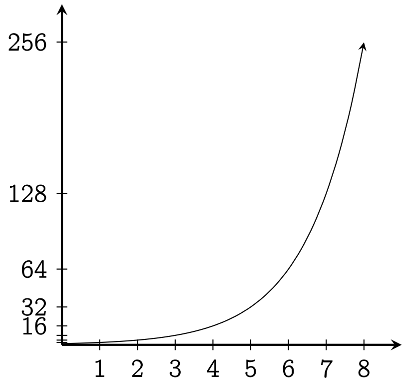


FIG. 3.3 | EXPONENTIAL FUNCTION 2^x PLOTTED FROM 0 TO 8

that is why $V(t) - V(t-1) = V(t-1)$. But we still need the method of increments to calculate the derivative of 2^x:

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}x}(2^x) &= \lim_{h\to 0} \frac{2^{x+h} - 2^x}{\mathrm{d}x} \\ &= 2^x \lim_{h\to 0} \left(\frac{2^h - 1}{h}\right).\end{aligned}$$

Now you could try plugging in a really small value of h, say 0.000001. The term $\frac{2^h - 1}{h}$ will approach 0.69314 .... If you try other bases of exponents, say 3, you might see a pattern emerging.

$$\frac{\mathrm{d}}{\mathrm{d}x}(3^x) = 3^x \lim_{h\to 0} \frac{3^h - 1}{h}. \tag{3.26}$$

The rate of change of an exponential function is always itself times a proportionality constant. For 3^x, it's about 1.09851 .... If we could find a number n where $\frac{n^h - 1}{h} = 0$, we'd have a very pretty function which it is its own derivative. So let's find that!

3.4.1 A function that is its own derivative: Euler's number

Let's set a goal: find the function that is its own derivative. If that function exists, it must satisfy the differential equation

$$\frac{\mathrm{d}f(x)}{\mathrm{d}x} = f(x). \tag{3.27}$$

We've already seen that one possible candidate for this function is the exponential function n^x. But how do we find the number n that satisfies it?

Here is a good time where I introduce a method that's used very often to solve differential equations: **power series expansion**. Most functions have a power series expansion, written as[1]

$$f(x) = a_0 + a_1x^1 + a_2x^2 + a_3x^3 + \dots \,. \tag{3.28}$$

E.g., $\sin(x)$ can be written as[2]

$$\sin(x) = x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \frac{1}{7!}x^7 + \dots \,. \tag{3.29}$$

We can assume that the exponential function can be expanded using a power series; thus,

$$n^x \equiv a_0 + a_1x^1 + a_2x^2 + a_3x^3 + \dots \,. \tag{3.30}$$

To find out each of the leading coefficient, just substitute the expansion into eq. (3.27) and see where mathematics would lead us.

A direct substitution followed by the power rule would give

$$\frac{\mathrm{d}}{\mathrm{d}x}(a_0 + a_1x^1 + a_2x^2 + a_3x^3 + \dots) = a_1 + 2a_2x^1 + 3a_3x^2 + 4a_4x^3 + \dots \,. \tag{3.31}$$

[1] Although the convergence of the series derived is quite questionable; thankfully, the power series of the exponential function converges everywhere.

[2] If you don't believe me, wait for section 5.2.1.

By the eq. (3.27), the R.H.S must also equals n^x; therefore,

$$\begin{aligned} n^x &= a_0 + a_1x^1 + a_2x^2 + a_3x^3 + \ldots \text{ and also,} \\ n^x &= a_1 + 2a_2x^1 + 3a_3x^2 + 4a_4x^3 + \ldots . \end{aligned} \tag{3.32}$$

Since they're equal, their leading coefficient must match, and thus forming a system of equations

$$\begin{aligned} a_0 &= a_1 \\ a_1 &= 2a_2 \\ a_2 &= 3a_3 \\ a_3 &= 4a_4 \\ &\vdots \end{aligned} \tag{3.33}$$

Because n^x is an exponential, n^0 must equal one. Therefore, $x = 0$

$$\begin{aligned} 1 &= a_0 + a_1(0)^1 + a_2(0)^2 + a_3(0)^3 + \ldots \\ a_0 &= 1; \end{aligned}$$

thus, by eq. (3.33), a_1 must also equal 1. Substituting $a_1 = 1$ into eq. (3.33) gives

$$a_2 = \frac{1}{2},\ a_3 = \frac{1}{2\times 3},\ a_4 = \frac{1}{2\times 3\times 4}, \ldots; \tag{3.34}$$

therefore,

$$n^x = 1 + 1 + \frac{1}{2!}x + \frac{1}{3!}x^3 + \frac{1}{4!}x^4 + \ldots$$

Clearly, $a_n = n!$. If we want to find n, just let $x = 1$.

$$n = 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \cdots = \sum_{i=0}^{\infty} \frac{1}{i!}$$

and there it is: the expression for n, which evaluates to 2.71828 … .[3]

The property that the exponentiation of this number is its own derivative makes it a very important tool in mathematics. Even though

[3]Using the continuation that $0! = 1$.

it's intrinsically related to growths, it somehow manages to seek its way to places that are not even remotely related to growths: oscillations, the patterns of prime numbers, wave equations, etc. Don't worry, we shall be seeing it much later on. Because this usefulness, this constant, 2.71828 ... have a name: the **Euler's number**, which is written as e, where[4]

$$\mathrm{e} = 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \cdots = \sum_{i=0}^{\infty} \frac{1}{i!} \tag{3.35}$$

3.4.2 Another interpretation of the Euler's number: continuous bank interests

There are two types of bank interests: simple and compound. Simple interests is the thing that you don't really want: the interest is always the same and doesn't grow with your account. You can calculate it by using

$$n(t) = n_0 + tr \tag{3.36}$$

where $n(t)$ is the total money at time t, n_0 the initial money in your bank account, and r, the interest rate.

Compound interest in the other hand calculates your interest based on how much money you have at that moment:

$$n(t+1) = n(t)r + n(t). \tag{3.37}$$

We can find the expression for $n(t)$ in a similar fashion to what we've done in eq. (3.24). You'll get

$$n(t) = (1+r)^t n_0 \tag{3.38}$$

which is an exponential function.

[4]Not to be confused with the "Euler's constant" which is another constant written γ, and is around 0.57721 ...

Let's say you deposit 100$ into a bank and the bank is offering you two options on **compound interests** rate. 1) Take 100% interest in 1 year, 2) Take 100 / 2% twice a year, or 3) Take 100 / 356% daily. If you take option one, you'd end up with 200$. Option two takes you to 225$, and option three takes you to around 271.447 ... $. You might see a theme here. If you get 100 / n% interest, n times a year, the result keeps getting higher. Is there an upper limit to this?

If we write it in terms of limits as $n \to \infty$ and use eq. (3.38), the compound interests formula,

$$x = \lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n n_0.$$

Where x is the total money after a year. We're interested in the upper limit, so we'll just let n_0 for now. The expression will become

$$x = \lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n \tag{3.39}$$

Technically, we could go in and substitute a very high n, such as 1000000. But I believe you could already see that it would be a nightmare to calculate: exponentiation is not at all an easy task. However, notice that from option three earlier, the total money is 271.447 ... $ which is suspiciously similar to e at 2.71828 ... . If eq. (3.39) equals eq. (3.33), we'd find the upper limit for this problem and solve the mystery.

We can use the binomial theorem on eq. (3.39) and get

$$\begin{aligned}
&\lim_{n\to\infty} \sum_{k=0}^{n} \binom{n}{k} 1^{n-k} \frac{1}{n^k} \\
&= \lim_{n\to\infty} \binom{n}{0}\frac{1}{n^0} + \binom{n}{1}\frac{1}{n^1} + \binom{n}{2}\frac{1}{n^2} + \binom{n}{3}\frac{1}{n^3} + \dots .
\end{aligned}$$

Then, we use the definition of n choose k,

$$= 1 + \lim_{n\to\infty} \frac{n!}{1!(n-1)!}\frac{1}{n^1} + \frac{n!}{2!(n-2)!}\frac{1}{n^2} + \frac{n!}{3!(n-3)!}\frac{1}{n^3} + \dots .$$

Now, we can cancel the $n!$ on the numerator to the denominator and isolate the factorials.

$$= 1 + \lim_{n\to\infty} \frac{n(n-1)!}{(n-1)!}\frac{1}{1!n^1} + \frac{n(n-1)(n-2)!}{(n-2)!}\frac{1}{2!n^2} + ...$$
$$= 1 + \frac{1}{1!} + \lim_{n\to\infty} \frac{n(n-1)}{2!n^2} + \frac{n(n-1)(n-2)}{3!n^3} + \frac{n(n-1)(n-2)(n-3)}{4!n^4} + ...$$

Notice, as $n \to \infty$, the ratio between $n+R$ and n where R is any real numbers would be literally negligible. For every terms in our series, both the numerator and the denominator has the same polynomic degrees. Therefore, all the n's in the series cancel out and we get

$$x = 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + ... \tag{3.40}$$

which is literally eq. (3.33); thus,

$$\mathrm{e} = \lim_{n\to\infty}\left(1 + \frac{1}{n}\right)^n. \tag{3.41}$$

In conclusion, the upper limit that the bank can give you is e, which should make sense geometrically. Because we're gradually turning a discrete interest into a continuous one, e should appear in the limit of the continuous bank interests.

The limit definition of e allows us to express the exponential function e^x in terms of limits.

$$\mathrm{e}^x = \left(\lim_{n\to\infty}\left(1 + \frac{1}{n}\right)^n\right) \tag{3.42}$$
$$= \lim_{n\to\infty}\left(1 + \frac{1}{n}\right)^{nx} \tag{3.43}$$

If $n = u\ /\ x$, then

$$\mathrm{e}^x = \lim_{\frac{u}{x}\to\infty}\left(1 + \frac{u}{h}\right)^h \tag{3.44}$$

For $\dfrac{u}{x} \to \infty$ to be true for all finite x, u must approach ∞; therefore,

$$\mathrm{e}^x = \lim_{u\to\infty}\left(1 + \frac{u}{h}\right)^h \tag{3.45}$$

3.4.3 The antiderivative of exponential functions

It should be trivial that if e^x is the derivative of itself, so is its antiderivative

$$\int \mathrm{e}^x \,\mathrm{d}x = \mathrm{e}^x + C. \tag{3.46}$$

For other bases, we could use the fundamental theorem of calculus (theorem 2.3.1) to find its antiderivative. I.e., if

$$\frac{\mathrm{d}(n^x)}{\mathrm{d}x} = n^x \lim_{h\to 0} \frac{n^h - 1}{h},$$

then

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}x}(n^x) &= n^x \lim_{h\to 0} \frac{n^h - 1}{h} \\
n^x &= \lim_{h\to 0} \frac{n^h - 1}{h} \int n^x \,\mathrm{d}x \\
\int n^x \,\mathrm{d}x &= n^x \lim_{h\to 0} \left(\frac{n^h - 1}{h}\right)^{-1} + C.
\end{aligned}$$

The term in the limit sign still appears here. If we want to uncover the origin of this term, we must discuss the logarithms.

3.5 Logarithms

Logarithms are inverses of exponential, like how roots are inverses of monomials. To illustrate what I mean,

$$\text{For monomials, } \sqrt[a]{x^a} = x, \text{ but with logarithms, } \log_a(a^x) = x.$$

They have the following properties:

$$\log_a(x) + \log_a(y) = \log_a(xy), \tag{3.47}$$

$$\log_a(x) - \log_a(y) = \log_a\left(\frac{x}{y}\right), \tag{3.48}$$

$$\log_a(x) = \frac{\log_b(x)}{\log_b(a)}. \tag{3.49}$$

Since e^x is shown to be a very important function in modelling continuous growth, and is its own derivative, we give its inverse function its own name: the **natural logarithm**, written as $\ln(x)$.

What's the derivative of $\ln(x)$? By the method of increments,

$$\begin{aligned}
\frac{d}{dx}\ln(x) &= \lim_{h\to 0}\frac{\ln(x+h)-\ln(x)}{h} && (3.50)\\
&= \lim_{h\to 0}\frac{\ln(\frac{x+h}{x})}{h} && (3.51)\\
&= \lim_{h\to 0}\frac{\ln(1+\frac{h}{x})}{h} && (3.52)\\
&= \lim_{h\to 0}\ln\left(\left(1+\frac{h}{x}\right)^{\frac{1}{h}}\right). && (3.53)
\end{aligned}$$

This form is very similar to the limit definition of the exponential function e^x, but with some terms swapped around. So let's try to fit this limit to the form of e^x. I shall let $h = \dfrac{1}{n}$. If $h \to 0$, then $n \to \infty$. Therefore,

$$\begin{aligned}
\frac{d}{dx}\ln(x) &= \lim_{n\to\infty}\ln\left(\left(1+\frac{1}{nx}\right)^{n}\right) && (3.54)\\
&= \lim_{n\to\infty}\ln\left(\left(1+\frac{1/x}{n}\right)^{n}\right). && (3.55)
\end{aligned}$$

Recall that

$$e^x = \lim_{n\to\infty}\left(1+\frac{x}{n}\right)^n. \tag{3.56}$$

Thus,

$$\lim_{n\to\infty}\left(1+\frac{1/x}{n}\right)^n = e^{\frac{1}{x}}. \tag{3.57}$$

which fits the form of eq. (3.55). Therefore,

$$\begin{aligned}
\frac{d}{dx}\ln(x) &= \lim_{n\to\infty}\ln(e^{\frac{1}{x}}) && (3.58)\\
&= \lim_{n\to\infty}\frac{1}{x} = \frac{1}{x}: && (3.59)
\end{aligned}$$

the derivative of the natural logarithm of x is the reciprocal of x.

This result can also be interpretted geometrically. If the exponential grows very quickly when x increases, its inverse must grow slow. The slow growing rate is directly captured by the function $\frac{1}{x}$.

Here is the perfect time that I'll give you a glimpse into calculus with many variables: multivariable calculus. I shall present another way to derive the derivative of $\ln(x)$: **implicit differentiation**.

Functions in the form $f(x) = y$ are called **explicit functions**: the relations between the two variables are written explicitly. Functions that can't be written as $f(x) = y$ are called **implicit functions**, e.g., $x^2 + y^2 = r^2$. **Implicit differentiation** concerns the differentiation of implicit functions. In this case, it actually makes our life easier if we turn $\ln(x)$ into an implicit function.

If I let $\ln(x) = y$, I can raise both sides to the power of e and get

$$\mathrm{e}^{\ln(x)} = \mathrm{e}^y. \tag{3.60}$$

Because logarithms are inverses of exponential,

$$x = \mathrm{e}^y. \tag{3.61}$$

By the rule of equality, (section 3.1), we can take the derivative of both sides w.r.t. y instead of x:

$$\frac{\mathrm{d}x}{\mathrm{d}y} = \frac{\mathrm{d}\mathrm{e}^y}{\mathrm{d}y}$$
$$\frac{\mathrm{d}x}{\mathrm{d}y} = \mathrm{e}^y.$$

But we're looking for the derivative of y (which is just $\ln(x)$) w.r.t. x, not the derivative of x w.r.t. y. Here's where Leibniz's notation comes into clutch: we can swap the numerator with the denominator for both sides then, substitute back $y = \ln(x)$:

$$\begin{aligned}\frac{\mathrm{d}\ln(x)}{\mathrm{d}x} &= \frac{1}{\mathrm{e}^{\ln(x)}} \\ \therefore \frac{\mathrm{d}\ln(x)}{\mathrm{d}x} &= \frac{1}{x},\end{aligned}$$

and there: the derivative of the natural logarithm is the reciprocal.

With the power of natural logarithms, we can actually go back at the derivative of n^x and finally uncover the mystery behind the proportionality term that's lingering around. Start with the manipulation of n^x.

$$n^x = \left(\mathrm{e}^{\ln(n)}\right)^x = \mathrm{e}^{x\ln(x)}. \tag{3.62}$$

With the chain rule (section 3.1), let $u = x\ln(n)$:

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}x}(n^x) &= \frac{\mathrm{d}\mathrm{e}^{x\ln(n)}}{\mathrm{d}x} \\ &= \frac{\mathrm{d}\mathrm{e}^u}{\mathrm{d}x} \times \frac{\mathrm{d}u}{\mathrm{d}u} \\ &= \frac{\mathrm{d}\mathrm{e}^u}{\mathrm{d}u} \times \frac{\mathrm{d}u}{\mathrm{d}x} \\ &= \mathrm{e}^{x\ln(n)} \times \frac{\mathrm{d}x\ln(n)}{\mathrm{d}x} \\ &= n^x\ln(n).\end{aligned}$$

And here it is: the mystery proportionality constant. It's just a consequence of the natural logarithm. Thus, one way to define the natural log would be

$$\ln(n) = \lim_{h\to 0}\frac{n^h - 1}{h}. \tag{3.63}$$

The antiderivative of other bases exponents are then given by

$$\int n^x = \frac{1}{\ln(n)}n^x + C. \tag{3.64}$$

3.5.1 The product rule and the quotient rule

Sometimes, we have to multiply the two functions together before taking the derivatives. There are two ways to do this. To keep the spirit of visualization, I shall first introduce the geometrical way, then the analytical way.

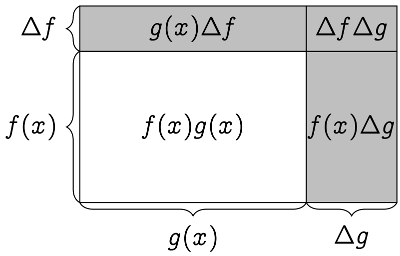


FIG. 3.4 | THE GEOMETRICAL INTERPRETATION OF THE PRODUCT RULE.

The derivative of $f(x)g(x)$ w.r.t. x can be thought of a rectangle with side length that's governed by $f(x)$ and $g(x)$. As shown in fig. 3.4, the area increase on side $f(x)$ is $g(x)\Delta f$ and on $g(x)$, $f(x)\Delta g$. The $\Delta f \Delta g$ part is basically negligible. Therefore,

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}x}\big(f(x)g(x)\big) &= \lim_{h\to 0}\frac{g(x)\Delta f + f(x)\Delta g}{h} \\
&= \lim_{h\to 0}\frac{f(x)(g(x+h)-g(x)) + g(x)(f(x+h)-f(x))}{h} \\
&= f(x)\lim_{h\to 0}\frac{g(x+h)-g(x)}{h} + g(x)\lim_{h\to 0}\frac{f(x+h)-f(x)}{h} \\
&= f(x)\frac{\mathrm{d}g(x)}{\mathrm{d}x} + g(x)\frac{\mathrm{d}f(x)}{\mathrm{d}x}.
\end{aligned}$$

Which is what we call the product rule. Notice the alternation between f and g in the two terms. The derivative of $f(x)$ is multiplied by $g(x)$, and the derivative of $g(x)$ is multiplied by $f(x)$. This is a direct consequence of the diagram: the change in $f(x)$ is multiplied by $g(x)$ to give the area and also the other way around. You could check this with the method of increments, and it would still be true. I encourage you to do it.

To take derivatives of quotients of functions, just plug in $1\,/\,g(x)$

instead of $g(x)$. The final form should be

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{f(x)}{g(x)}\right)=\frac{f(x)\dfrac{\mathrm{d}g(x)}{\mathrm{d}x}-g(x)\dfrac{\mathrm{d}f(x)}{\mathrm{d}x}}{f(x)^2}. \tag{3.65}$$

But how's about the product of multiple functions? We can't really geometrically interpret those anymore. So we have to turn ourselves to the analytical method.

Let's think of this through. We don't know the derivative of products, but we know the derivative of *sums*: the linearity property. So if we can turn products into sum, this problem would be so easy! Gladly, the logarithms can do exactly that. To make our life easier, we shall use the natural logarithm. Because I want to save space, let's write $a(x)b(x)c(x)$ as just abc. Do know that these functions are all dependent on x. Start off with a manipulation of products.

$$\begin{aligned}\frac{\mathrm{d}(abc\dots)}{\mathrm{d}x}&=\frac{\mathrm{d}}{\mathrm{d}x}\left(\mathrm{e}^{\ln(abc\dots)}\right)\\&=\frac{\mathrm{d}}{\mathrm{d}x}\left(\mathrm{e}^{\ln(a)+\ln(b)+\ln(c)+\dots}\right).\end{aligned}$$

Now, let $\ln(a)+\ln(b)+\ln(c)+\cdots=u$ and use the chain rule,

$$\begin{aligned}&=\frac{\mathrm{d}}{\mathrm{d}x}(\mathrm{e}^u)\cdot\frac{\mathrm{d}u}{\mathrm{d}u}\\&=\frac{\mathrm{d}}{\mathrm{d}x}(\mathrm{e}^u)\cdot\frac{\mathrm{d}}{\mathrm{d}x}(\ln(a)+\ln(b)+\ln(c)+\dots)\\&=\mathrm{e}^u\left(\frac{\mathrm{d}\ln(a)}{\mathrm{d}x}+\frac{\mathrm{d}\ln(b)}{\mathrm{d}x}+\frac{\mathrm{d}\ln(c)}{\mathrm{d}x}+\dots\right).\end{aligned}$$

Then use the chain rule again on the terms in the parenthesis

$$\begin{aligned}&=\mathrm{e}^u\left(\frac{\mathrm{d}\ln(a)}{\mathrm{d}x}\frac{\mathrm{d}a}{\mathrm{d}a}+\frac{\mathrm{d}\ln(b)}{\mathrm{d}x}\frac{\mathrm{d}b}{\mathrm{d}b}+\frac{\mathrm{d}\ln(c)}{\mathrm{d}x}\frac{\mathrm{d}c}{\mathrm{d}c}+\dots\right)\\&=(abc\dots)\left(\frac{\mathrm{d}\ln(a)}{\mathrm{d}a}\frac{\mathrm{d}a}{\mathrm{d}x}+\frac{\mathrm{d}\ln(b)}{\mathrm{d}b}\frac{\mathrm{d}b}{\mathrm{d}x}+\frac{\mathrm{d}\ln(c)}{\mathrm{d}c}\frac{\mathrm{d}c}{\mathrm{d}x}+\dots\right)\\&=(abc\dots)\left(\frac{1}{a}\frac{\mathrm{d}a}{\mathrm{d}x}+\frac{1}{b}\frac{\mathrm{d}b}{\mathrm{d}x}+\frac{1}{c}\frac{\mathrm{d}c}{\mathrm{d}x}+\dots\right).\end{aligned}$$

And there we have it: the generalized product rule.

3.5.2 Alternative derivations for the power rule

The power rule can also be derived using the same technique we just used. However, we use a different property of logarithm: $\ln(x^n) = n\ln(x)$.

$$\frac{\mathrm{d}}{\mathrm{d}x}x^n = \frac{\mathrm{d}}{\mathrm{d}x}\mathrm{e}^{n\ln(x)}.$$

Let $u = n\ln(x)$ then use the chain rule

$$\begin{aligned}
&= \frac{\mathrm{d}\mathrm{e}^u}{\mathrm{d}x}\cdot\frac{\mathrm{d}u}{\mathrm{d}u} \\
&= \frac{\mathrm{d}\mathrm{e}^u}{\mathrm{d}u}\cdot\frac{\mathrm{d}u}{\mathrm{d}x} \\
&= \mathrm{e}^u\cdot\frac{\mathrm{d}}{\mathrm{d}x}\big(n\ln(x)\big) \\
&= x^n\cdot n\frac{1}{x} = nx^{n-1}.
\end{aligned}$$

3.6 Implicit differentiation

Implicit differentiation also shows up in problems where two rate of changes are related to each other: the **related rates** problem. Consider a sliding ladder that's 5m long (fig. 3.5). At a certain time t, what's the sliding rate along the x-axis w.r.t. the y-axis?

The problem is asking for $\frac{\mathrm{d}x}{\mathrm{d}y}$. Here, the rate of sliding along the y-axis $\frac{\mathrm{d}y}{\mathrm{d}t}$ and along the x-axis $\frac{\mathrm{d}x}{\mathrm{d}t}$ is related because the ladder length is fixed. If y decreases, x must increase. The Pythagorean theorem says

$$x^2 + y^2 = 5^2. \tag{3.66}$$

We can take the derivative w.r.t. t on both side then use the chain rule

$$\begin{aligned}
\frac{\mathrm{d}x^2}{\mathrm{d}t} + \frac{\mathrm{d}y^2}{\mathrm{d}t} &= \frac{\mathrm{d}(5^2)}{\mathrm{d}t} \\
\frac{\mathrm{d}x^2}{\mathrm{d}x}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{\mathrm{d}y^2}{\mathrm{d}y}\frac{\mathrm{d}y}{\mathrm{d}t} &= 0
\end{aligned}$$

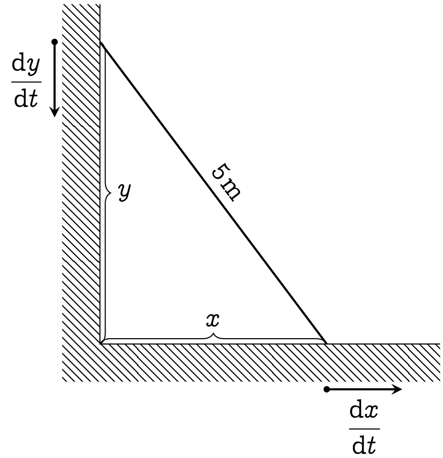


FIG. 3.5 | A LADDER LENGTH 5m SLIDING DOWN A CORNER.

$$2x\frac{\mathrm{d}x}{\mathrm{d}t} + 2y\frac{\mathrm{d}y}{\mathrm{d}t} = 0.$$

We can take advantage of the Leibniz's notation and multiply by $\mathrm{d}t$ on both sides giving

$$2x\,\mathrm{d}x + 2y\,\mathrm{d}y = 0.$$

To find $\frac{\mathrm{d}x}{\mathrm{d}y}$, we just have to isolate the variables,

$$\frac{\mathrm{d}x}{\mathrm{d}y} = -\frac{y}{x}.$$

And this should actually make sense. Because if y is increasing by a bit, x must decrease by some amount, and that amount is $-y\,/\,x$: the higher the y, the larger the rates of sliding.

3.7 Integral of the reciprocal

By the fundamental theorem of calculus (theorem 2.3.1), the antiderivative of $\frac{1}{x}$ is $\ln(x)$. You might believe this, and it's not wrong. But I find it very disturbing and unresolved. It's like taking the result and pointing it back to the origin. It doesn't really make sense. So from this dissatisfaction, I spent a night coming up with a way to derive this using just the reversed power rule. Enjoy the transformation!

$$\begin{aligned}
\int \frac{1}{x}\,\mathrm{d}x &= \int \lim_{h\to 0}\left(\frac{1}{2}x^{-1+h}+\frac{1}{2}x^{-1-h}\right)\mathrm{d}x \\
&= \lim_{h\to 0}\int \left(\frac{1}{2}x^{-1+h}+\frac{1}{2}x^{-1-h}\right)\mathrm{d}x \\
&= \lim_{h\to 0}\int \left(\frac{1}{2}\frac{x^{-1+h+1}}{(-1+h+1)}+\frac{1}{2}\frac{x^{-1-h+1}}{(-1-h+1)}\right)\mathrm{d}x \\
&= \lim_{h\to 0}\left(\frac{1}{2}\frac{x^{h}}{h}-\frac{1}{2}\frac{x^{-h}}{-h}\right) = \lim_{h\to 0}\left(\frac{1}{2}\frac{x^{h}}{h}\frac{x^{h}}{h}-\frac{1}{2}\frac{1}{hx^{h}}\right) \\
&= \lim_{h\to 0}\left(\frac{x^{2h}-1}{2hx^{h}}\right) = \lim_{h\to 0}\left(\frac{\mathrm{e}^{2h\ln(x)}-1}{2h\ln(x)}\cdot\frac{\ln(x)}{x^{h}}\right) \\
&= \lim_{h\to 0}\left(\frac{\mathrm{e}^{2h\ln(x)}}{2h\ln(x)}\right)\cdot\lim_{h\to 0}\left(\frac{\ln(x)}{x^{h}}\right) \\
&= \lim_{h\to 0}\left(\frac{\mathrm{e}^{2h\ln(x)}}{2h\ln(x)}\right)\cdot\ln(x).
\end{aligned} \tag{3.67}$$

Then, we evaluate the limit at the front by letting $u = 2h\ln(x)$. When $h \to 0$, $u \to 0$ as well. Then, use the definition of e from eq. (3.39).

$$\begin{aligned}
\lim_{h\to 0}\left(\frac{\mathrm{e}^{2h\ln(x)}}{2h\ln(x)}\right) &= \lim_{u\to 0}\left(\frac{\mathrm{e}^{u}-1}{u}\right) \\
&= \lim_{u\to 0}\left(\frac{\left(\lim_{n\to\infty}\left(1+\frac{1}{n}\right)^{n}\right)^{u}}{u}\right).
\end{aligned}$$

Change the limits from $n \to \infty$ into $n \to 0$. Notice, $\lim_{n\to 0}\left(1+\frac{1}{n}\right)^{n} = \lim_{n\to 0}(1+n)^{1/n}$. If $n \to 0$ and $u \to 0$, that means $n = u$. Substitute $n = u$ into the limit,

$$= \lim_{u\to 0}\left(\frac{\left(\lim_{u\to 0}(1+u)^{1/u}\right)^{u}-1}{u}\right)$$

$$= \lim_{u\to 0}\left(\frac{1+u-1}{u}\right) = 1.$$

Then, substitute this limit back into eq. (3.67), you'll see that

$$\int \frac{1}{x}\,\mathrm{d}x = \lim_{h\to 0}\left(\frac{\mathrm{e}^{2h\ln(x)}}{2h\ln(x)}\right)\cdot \ln(x) = \ln(x) + C,$$

completing the proof.

3.8 Amalgamation of functions

Technically, I could end the chapter now because we've explored the derivatives and antiderivatives of all non-trigonometric functions: polynomials, exponentials, and logarithms. But I want you to familiarize yourself with evaluating derivatives and antiderivatives of an amalgamation of functions first, whether it be by addition, subtraction, multiplication, division, or jamming one function inside another (composite functions). These techniques will help us solve other harder problems in later chapters. So, this section is written as a compilation of mathematical exercise, ready to be trained on.

The reason that this section is the longest in this chapter is because it contains all solution to the exercises, making it a bit exhaustive even for me to write. So, you don't have to do all the exercises. Just familiarize yourself with it, and you will be mostly fine. However, there are one place that I want you to focus: section 3.8.2, which is on substitution in antiderivatives. It is absolutely necessary, so don't skip over it. You shall see why in later chapters.

Evaluating derivatives is completely different to evaluating antiderivatives. In derivatives, there are strict rules to guide you through. Those rules are gone in antiderivatives. It is an art form; a very difficult one. But it has proven to be too useful to be discarded. So, focus on it. Without further ado, let's crack on.

3.8.1 Technique of differentiation

[5]Put simply, if there's a linear combination of functions, use the linearity of derivatives to break it down into multiple smaller derivatives

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(af(x)+bg(x)\right)=a\frac{\mathrm{d}f(x)}{\mathrm{d}x}+b\frac{\mathrm{d}g(x)}{\mathrm{d}x},$$

if there's product, then use the product rule

$$\frac{\mathrm{d}\left(f_1(x)f_2(x)\dots f_n(x)\right)}{\mathrm{d}x}=f_1f_2\dots f_n\left(\frac{1}{f_1}\frac{\mathrm{d}f_1}{\mathrm{d}x}+\frac{1}{f_2}\frac{\mathrm{d}f_2}{\mathrm{d}x}+\cdots+\frac{1}{f_n}\frac{\mathrm{d}f_n}{\mathrm{d}x}\right),$$

and if there are composite functions $f(g(x))$, and you know the derivative of both $f(x)$ and $g(x)$, just set $u=g(x)$ and use the chain rule.

$$\frac{\mathrm{d}f(g(x))}{\mathrm{d}x}=\frac{\mathrm{d}f(u)}{\mathrm{d}x}=\frac{\mathrm{d}f(u)}{\mathrm{d}u}\times\frac{\mathrm{d}u}{\mathrm{d}x}=\frac{\mathrm{d}f(u)}{\mathrm{d}u}\times\frac{\mathrm{d}g(x)}{\mathrm{d}x}$$

And, implicit derivatives can also help you simplify a whole class of problems if you're stuck. Knowing these, you're more than ready now. Let's get on with solving.

Explicit differentiation

Let's start with some simple composite functions

Example 3.8.1: $\frac{\mathrm{d}}{\mathrm{d}x}\mathrm{e}^{2\sqrt{x}}$ This is a composite function $f(g(x))$ where $f(x)=\mathrm{e}^x$, and $g(x)=2\sqrt{x}$. We know that the derivative of e^x is e^x (section 3.4). And by the linearity of derivatives and the power rule,

$$\frac{\mathrm{d}g(x)}{\mathrm{d}x}=\frac{\mathrm{d}}{\mathrm{d}x}(2\sqrt{x})=2\frac{\mathrm{d}x^{1/2}}{\mathrm{d}x}=2\left(\frac{1}{2}x^{-1/2}\right)=x^{-\frac{1}{2}}=\frac{1}{\sqrt{x}}. \quad (3.68)$$

[5]The rules mentioned here are repeated without numberings for ease of reading

So, setting $g(x) = 2\sqrt{x} = u$, we get

$$\frac{\mathrm{d}e^{2\sqrt{x}}}{\mathrm{d}x} = \frac{\mathrm{d}e^{u}}{\mathrm{d}x} \tag{3.69}$$
$$= \frac{\mathrm{d}e^{u}}{\mathrm{d}u} \times \frac{\mathrm{d}u}{\mathrm{d}x} \tag{3.70}$$
$$= e^{u} \times \frac{\mathrm{d}}{\mathrm{d}x}(2\sqrt{x}) \tag{3.71}$$
$$= e^{2\sqrt{x}}\left(\frac{1}{\sqrt{x}}\right) = \frac{e^{2\sqrt{x}}}{\sqrt{x}}. \tag{3.72}$$

Now let's throw in a bit of products.

Example 3.8.2: $\frac{\mathrm{d}}{\mathrm{d}x}\left(x^3 \ln(x^3)\right)$ By the product rule,

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(x^3 \ln(x^3)\right) = (x^3)\left(\frac{\mathrm{d}\ln(x^3)}{\mathrm{d}x}\right) + \ln(x^3)\left(\frac{\mathrm{d}x^3}{\mathrm{d}x}\right) \tag{3.73}$$

The power rule says that $\frac{\mathrm{d}}{\mathrm{d}x}x^3 = 3x^2$. We then evaluate $\frac{\mathrm{d}\ln(x^3)}{\mathrm{d}x}$ by using the chain rule. Let $x^3 = u$, then

$$\frac{\mathrm{d}\ln(x^3)}{\mathrm{d}x} = \frac{\mathrm{d}\ln(u)}{\mathrm{d}x} \tag{3.74}$$
$$= \frac{\mathrm{d}\ln(u)}{\mathrm{d}u} \times \frac{\mathrm{d}u}{\mathrm{d}x}, \tag{3.75}$$

discussed in section 3.5, $\frac{\mathrm{d}\ln(u)}{\mathrm{d}u} = \frac{1}{u}$,

$$= \frac{1}{u} \times \frac{\mathrm{d}x^3}{\mathrm{d}x} \tag{3.76}$$
$$= \frac{1}{u} \times 3x^2 = \frac{1}{x^3} \times 3x^2 = \frac{3}{x}; \tag{3.77}$$

therefore,

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(x^3 \ln(x^3)\right) = x^3 \times \frac{3}{x} + \ln(x^3) \times 3x^2 \tag{3.78}$$
$$= 3x^2 + 3x^2\ln(x^3) = 3x^2\left(1 + \ln(x^3)\right). \tag{3.79}$$

How's about some purely polynomial derivatives?

Example 3.8.3: $\frac{\mathrm{d}}{\mathrm{d}x}\left((x^2+1)^5(3x^3+4)^9\right)$ This derivative is quite tough to evaluate by expansion. A better way would be to use the product rule and the chain rule in combination. Firstly,

$$\begin{aligned}&\frac{\mathrm{d}}{\mathrm{d}x}\left((x^2+1)^5(3x^3+4)^9\right)\\&=(x^2+1)^5\left(\frac{\mathrm{d}}{\mathrm{d}x}\left((3x^3+4)^9\right)\right)+(3x^3+4)^9\left(\frac{\mathrm{d}}{\mathrm{d}x}\left((x^2+1)^5\right)\right).\end{aligned} \tag{3.80}$$

The derivative of $(3x^3+4)^9$ can be evaluated using the chain rule by setting $u = 3x^3+4$

$$\frac{\mathrm{d}}{\mathrm{d}x}\left((3x^3+4)^9\right) = \frac{\mathrm{d}u^9}{\mathrm{d}x} \tag{3.81}$$

$$= \frac{\mathrm{d}u^9}{\mathrm{d}u} \times \frac{\mathrm{d}u}{\mathrm{d}x} \tag{3.82}$$

By the power rule, $\frac{\mathrm{d}u^9}{\mathrm{d}u} = 9u^8$

$$= 9u^8 \times \frac{\mathrm{d}}{\mathrm{d}x}(3x^3+4) \tag{3.83}$$

Using the power rule, $\frac{\mathrm{d}}{\mathrm{d}x}(3x^3+4) = 9x^2$

$$= 9(3x^3+4)^8 \times 9x^2 = 81x^2(3x^3+4)^8. \tag{3.84}$$

Similarly, use the chain rule to evaluate $(x^2+1)^5$ and set $u = x^2+1$

$$\frac{\mathrm{d}}{\mathrm{d}x}\left((x^2+1)^5\right) = \frac{\mathrm{d}u^5}{\mathrm{d}x} \tag{3.85}$$

$$= \frac{\mathrm{d}u^5}{\mathrm{d}u} \times \frac{\mathrm{d}u}{\mathrm{d}x} \tag{3.86}$$

By the power rule, $\frac{\mathrm{d}u^5}{\mathrm{d}u} = 5u^4$

$$= 5u^4 \times \frac{\mathrm{d}(x^2+1)}{\mathrm{d}x} \tag{3.87}$$

By the power rule, $\dfrac{\mathrm{d}(x^2+1)}{\mathrm{d}x} = 2x$

$$= 5(x^2+1)^4 \times 2x = 10x(x^2+1)^4. \tag{3.88}$$

Therefore,

$$\frac{\mathrm{d}}{\mathrm{d}x}\left((x^2+1)^5(3x^3+4)^9\right) = (x^2+1)^5 \times 81x^3(3x^3+4)^8 \tag{3.89}$$
$$+ (3x^3+4)^9 \times 10x(x^2+1)^4$$
$$= 81x^3(x^2+1)^5(3x^3+4)^8 \tag{3.90}$$
$$+ 10x(x^2+1)^4(3x^3+4)^9.$$

There are certain cases of differentiation where using algebraical rules is better than using the chain rule with the product rule. Here is such example

Example 3.8.4: $\dfrac{\mathrm{d}}{\mathrm{d}x}\left(\ln\left(\sqrt{\dfrac{x+1}{x-1}}\right)\right)$ Breaking this derivative with the chain rule will be a nightmare. As said, let's use the algebraical properties of logarithm to help us with this. Namely, $\ln(\frac{a}{b}) = \ln(a) - \ln(b)$, and $\ln(a^b) = b\ln(a)$.

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\ln(\sqrt{\frac{x+1}{x-1}})\right) = \frac{\mathrm{d}}{\mathrm{d}x}\left(\ln(\sqrt{x+1}) - \ln(\sqrt{x-1})\right) \tag{3.91}$$
$$= \frac{\mathrm{d}}{\mathrm{d}x}\left(\ln\left((x+1)^{1/2}\right) - \ln\left((x-1)^{1/2}\right)\right) \tag{3.92}$$
$$= \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{1}{2}\ln(x+1) - \frac{1}{2}\ln(x-1)\right) \tag{3.93}$$

By the linearity of derivative,

$$= \frac{1}{2}\left(\frac{\mathrm{d}\ln(x+1)}{\mathrm{d}x} - \frac{\mathrm{d}\ln(x-1)}{\mathrm{d}x}\right). \tag{3.94}$$

These smaller derivatives are in the form $\frac{\mathrm{d}\ln(u)}{\mathrm{d}x}$ where in the first one, $u = x + 1$, and in the second one, $u = x - 1$. By the chain rule and the derivative of logarithms (section 3.5),

$$\frac{\mathrm{d}\ln(u)}{\mathrm{d}x} = \frac{\mathrm{d}\ln(u)}{\mathrm{d}u} \times \frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{u} \times \frac{\mathrm{d}u}{\mathrm{d}x}. \tag{3.95}$$

Since in both cases, $\frac{\mathrm{d}u}{\mathrm{d}x} = 1$,

$$\frac{\mathrm{d}\ln(x+1)}{\mathrm{d}x} = \frac{1}{x+1} \quad \text{and} \quad \frac{\mathrm{d}\ln(x-1)}{\mathrm{d}x} = \frac{1}{x-1}; \tag{3.96}$$

thus,

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\ln\left(\sqrt{\frac{x+1}{x-1}}\right)\right) = \frac{1}{2}\left(\frac{1}{x+1} - \frac{1}{x-1}\right) = -\frac{1}{x^2-1}. \tag{3.97}$$

Implicit differentiation

Implicit differentiation can help us with evaluating the derivative of functions in the form $f(x)^{g(x)}$. Let $y = f(x)^{g(x)}$, then using the algebraic properties of logarithms,

$$\ln(y) = \ln\left(f(x)^{g(x)}\right) \tag{3.98}$$

$$\ln(y) = g(x)\ln\big(f(x)\big) \tag{3.99}$$

$$\frac{\mathrm{d}\ln(y)}{\mathrm{d}x} = \frac{\mathrm{d}}{\mathrm{d}x}\big(g(x)\ln\big(f(x)\big)\big). \tag{3.100}$$

Using the chain rule on the L.H.S. and the product rule on the R.H.S.,

$$\frac{\mathrm{d}\ln(y)}{\mathrm{d}y} \times \frac{\mathrm{d}y}{\mathrm{d}x} = g(x)\frac{\mathrm{d}\ln\big(f(x)\big)}{\mathrm{d}x} + \ln\big(f(x)\big)\frac{\mathrm{d}g(x)}{\mathrm{d}x} \tag{3.101}$$

The derivative of the $\ln(x)$ is $\frac{1}{x}$ (section 3.5), and use the chain rule on the R.H.S.

$$\frac{1}{y} \times \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{g(x)}{f(x)}\frac{\mathrm{d}f(x)}{\mathrm{d}x} + \ln\big(f(x)\big)\frac{\mathrm{d}g(x)}{\mathrm{d}x} \tag{3.102}$$

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \left(\frac{g(x)}{f(x)}\frac{\mathrm{d}f(x)}{\mathrm{d}x} + \ln\big(f(x)\big)\frac{\mathrm{d}g(x)}{\mathrm{d}x}\right)y \tag{3.103}$$

Since $y = f(x)^{g(x)}$,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \left(\frac{g(x)}{f(x)}\frac{\mathrm{d}f(x)}{\mathrm{d}x} + \ln\big(f(x)\big)\frac{\mathrm{d}g(x)}{\mathrm{d}x}\right)f(x)^{g(x)} \tag{3.104}$$

Frankly, it can also be done explicitly by rewriting $f(x)^{g(x)}$ as $\mathrm{e}^{g(x)\ln(f(x))}$ and using the chain rule, but it's just neater this way. Let's try some examples. First, with the infamous x^x.

Example 3.8.5: $\dfrac{\mathrm{d}x^x}{\mathrm{d}x}$ This function x^x is in the form $f(x)^{g(x)}$ where $f(x) = x$, and $g(x)$ is also equal to x. Trivially, $\dfrac{\mathrm{d}f(x)}{\mathrm{d}x} = \dfrac{\mathrm{d}g(x)}{\mathrm{d}x} = 1$, so we can use eq. (3.104) right away and get

$$\frac{\mathrm{d}x^x}{\mathrm{d}x} = \left(\frac{x}{x}(1) + \ln(x)(1)\right)x^x = x^x\big(\ln(x) + 1\big). \tag{3.105}$$

Implicit differentiation can also be used if you are too lazy to deal with roots, e.g.,

Example 3.8.6: $\dfrac{\mathrm{d}\sqrt{2\ln(x)^2 - 1}}{\mathrm{d}x}$ It is very possible to go through this and use the chain rule and the power rule. But who wants to deal with derivative of roots? They are confusing. Well, we can actually use implicit differentiation to turn the square root into an integer power. Set $y = \sqrt{2\ln(x)^2 + 1}$, then

$$y = \sqrt{2\ln(x)^2 + 1} \tag{3.106}$$

$$y^2 = 2\ln(x)^2 + 1. \tag{3.107}$$

Take the derivative on both sides w.r.t. x. Use the linearity of derivative to break down the R.H.S., and use the chain rule to take care of the rest

$$\frac{\mathrm{d}y^2}{\mathrm{d}x} = 2\frac{\mathrm{d}\ln(x)^2}{\mathrm{d}x} + \cancelto{0}{\frac{\mathrm{d}}{\mathrm{d}x}(1)} \tag{3.108}$$

$$\frac{\mathrm{d}y^2}{\mathrm{d}y} \times \frac{\mathrm{d}y}{\mathrm{d}x} = 2\frac{\mathrm{d}\ln(x)^2}{\mathrm{d}x} \tag{3.109}$$

$$2y \times \frac{\mathrm{d}y}{\mathrm{d}x} = 2\frac{\mathrm{d}\ln(x)^2}{\mathrm{d}x} \tag{3.110}$$

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{y}2\frac{\mathrm{d}\ln(x)^2}{\mathrm{d}x} \tag{3.111}$$

Use the chain rule to evaluate $\frac{\mathrm{d}\ln(x)^2}{\mathrm{d}x}$. Set $u = \ln(x)$

$$\frac{\mathrm{d}\ln(x)^2}{\mathrm{d}x} = \frac{\mathrm{d}u^2}{\mathrm{d}x} \tag{3.112}$$

$$= \frac{\mathrm{d}u^2}{\mathrm{d}u} \times \frac{\mathrm{d}u}{\mathrm{d}x} \tag{3.113}$$

$$= 2u \times \frac{\mathrm{d}\ln(x)}{\mathrm{d}x} \tag{3.114}$$

$$= \frac{2\ln(x)}{x} \tag{3.115}$$

Therefore,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{y}2\left(\frac{2\ln(x)}{x}\right) \tag{3.116}$$

Substitute in $y = \sqrt{2\ln(x)^2 + 1}$, then

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{4\ln(x)}{x\sqrt{2\ln(x)^2 + 1}}. \tag{3.117}$$

Or maybe, you just have a very wonky equation that you can't even rearrange in terms of $y = f(x)$, e.g.,

Example 3.8.7: *Find* $\frac{\mathrm{d}y}{\mathrm{d}x}$ *given that* $x^y = y^x$ e start from taking the logarithm of both sides

$$\ln(x^y) = \ln(y^x) \tag{3.118}$$

$$y \ln x = x \ln y \tag{3.119}$$

Then, take the derivative w.r.t. x by using the product rule

$$\frac{\mathrm{d}}{\mathrm{d}x}(y \ln x) = \frac{\mathrm{d}}{\mathrm{d}x}(x \ln y) \tag{3.120}$$

$$y\frac{\mathrm{d}\ln x}{\mathrm{d}x} + \ln x\frac{\mathrm{d}y}{\mathrm{d}x} = x\frac{\mathrm{d}\ln y}{\mathrm{d}x} + \ln y\overbrace{\frac{\cancel{\mathrm{d}x}}{\cancel{\mathrm{d}x}}}^{1} \tag{3.121}$$

First, the derivative of $\ln(x)$ is $\frac{1}{x}$, and we can use the chain rule to take care of the rest

$$\frac{y}{x} + \ln x\frac{\mathrm{d}y}{\mathrm{d}x} = x\frac{\mathrm{d}\ln y}{\mathrm{d}y}\frac{\mathrm{d}y}{\mathrm{d}x} + \ln y \tag{3.122}$$

$$\frac{y}{x} + \ln x\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{x}{y}\frac{\mathrm{d}y}{\mathrm{d}x} + \ln y \tag{3.123}$$

$$\frac{x}{y}\frac{\mathrm{d}y}{\mathrm{d}x} + \ln x\frac{\mathrm{d}y}{\mathrm{d}x} = \ln y - \frac{y}{x} \tag{3.124}$$

$$\frac{\mathrm{d}y}{\mathrm{d}x}\left(x - \ln x\right) = \ln y - \frac{y}{x} \tag{3.125}$$

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\ln y - \frac{y}{x}}{x - \ln x}. \tag{3.126}$$

3.8.2 Technique of integration: substitution

Before we finish this chapter off, I'd want to take a look at a very useful technique for dealing with complicated integrals: substitution.

Substitution is a technique to simplify algebraic expressions. I believe you've already seen a multitude of them. Basically, we substitute some expression of x with u, then use everything in our power to change every x to u, including the differential element: from $\mathrm{d}x$ to $\mathrm{d}u$.

u-substitution works when the integrals are of the form

$$\int f\left(g(x)\right) \frac{\mathrm{d}g(x)}{\mathrm{d}x}\,\mathrm{d}x. \tag{3.127}$$

In this notation, it's easy to see that the dx will cancel, leaving us with

$$\int f\left(g(x)\right)\mathrm{d}g(x). \tag{3.128}$$

If $g(x) = u$, then our integral simplifies to

$$\int f(u)\,\mathrm{d}u. \tag{3.129}$$

Let's apply this knowledge to solve some integrals. Starting with the simplest: integrand with polynomials.

Example 3.8.8: $\int x(x^2+1)^{10}\,\mathrm{d}x$ If $f(x) = x^{10}$, and $u = x^2 + 1$, then

$$\int x(x^2+1)^{10}\,\mathrm{d}x = \int u^{10} x\,\mathrm{d}x. \tag{3.130}$$

Because $x = \frac{1}{2}\frac{\mathrm{d}(x^2+1)}{\mathrm{d}x}$,

$$\int u^{10} x\,\mathrm{d}x = \int u^{10} \frac{1}{2}\frac{\mathrm{d}(x^2+1)}{\mathrm{d}x}\,\mathrm{d}x = \frac{1}{2}\int u^{10}\,\mathrm{d}(x^2+1) \tag{3.131}$$

Since $u = x^2 + 1$, our integral simplifies to

$$\frac{1}{2}\int u^{10}\,\mathrm{d}u. \tag{3.132}$$

Using the reversed power rule (eq. (3.20)), we get

$$\frac{1}{2}\frac{u^{11}}{11} = \frac{u^{11}}{22} = \frac{(x^2+1)^{11}}{22} + C. \tag{3.133}$$

Example 3.8.9: $\int x^2 \mathrm{e}^{4x^3+4}\,\mathrm{d}x$ The integrand is a composite function: $f(u) = \mathrm{e}^u$ and $u = 4x^3 + 4$. In which, $\dfrac{\mathrm{d}u}{\mathrm{d}x} = 12x^2$; therefore,

$$\int x^2 \mathrm{e}^{4x^3+4}\,\mathrm{d}x = \frac{1}{12}\int \mathrm{e}^u 12x^2\,\mathrm{d}x \tag{3.134}$$

$$= \frac{1}{8}\int \mathrm{e}^u \frac{\mathrm{d}u}{\mathrm{d}x}\,\mathrm{d}x \tag{3.135}$$

$$= \frac{1}{8}\int \mathrm{e}^u\,\mathrm{d}u = \frac{1}{8}\mathrm{e}^u \tag{3.136}$$

$$= \frac{1}{12}\mathrm{e}^{4x^3+4} + C. \tag{3.137}$$

Notice any patterns? If the integral has a polynomial u with degree n nested inside another function, there must be a term outside with the degree $n-1$ to substitute with. In ex. 3.8.8, $u = x^2 + 1$ and there's an x multiplied outside waiting to be substituted. In ex. 3.8.10, $u = 4x^3 + 4$, and there's an x^2 outside. This pattern is a consequence of the power rule, and is pretty common. So, here are some practice problems with answers and hints on the right.[6] Try not to look at it much.

1. $\int 3x^2\sqrt{x^3+5}\,\mathrm{d}x$ — $\frac{2}{3}(x^3+5)^{\frac{3}{2}} + C$, $u = x^3 + 5$
2. $\int \sqrt{4+3x}\,\mathrm{d}x$ — $\frac{2}{9}(3x+4)^{\frac{3}{2}} + C$, $u = 4 + 3x$
3. $\int \frac{24x}{(4x^2+4)^2}\,\mathrm{d}x$ — $\frac{3}{4x^2+4} + C$, $u = 4x^2 + 4$

Now let's move on from polynomials to other functions that are nested inside another function. For the substitution to work, there must be the function's derivative multiplied, waiting to be substituted. A classic example is:

[6] Practice problems taken from [2].

Example 3.8.10: $\displaystyle\int \frac{\ln(x)}{x}\,\mathrm{d}x$ This integral is not quite a composite function, but rather just a product between a function and its derivative. If $u = \ln(x)$, then $\dfrac{\mathrm{d}u}{\mathrm{d}x} = \dfrac{1}{x}$; therefore,

$$\begin{aligned}
\int \frac{\ln(x)}{x}\,\mathrm{d}x &= \int u\frac{\mathrm{d}u}{\mathrm{d}x}\,\mathrm{d}x && (3.138)\\
&= \int u\,\mathrm{d}u && (3.139)\\
&= \frac{u^2}{2} = \frac{\ln(x)^2}{2} + C. && (3.140)
\end{aligned}$$

Example 3.8.11: $\displaystyle\int xe^{x^2}(e^{x^2}+3)\,\mathrm{d}x$ Let $e^{x^2}+3 = u$, then

$$\begin{aligned}
\frac{\mathrm{d}u}{\mathrm{d}x} &= \frac{\mathrm{d}e^{x^2}+3}{\mathrm{d}x} && (3.141)\\
&= \frac{\mathrm{d}e^{x^2}}{\mathrm{d}x} && (3.142)
\end{aligned}$$

Let $x^2 = v$, then by the chain rule,

$$\begin{aligned}
&= \frac{\mathrm{d}e^{v}}{\mathrm{d}x}\cdot\frac{\mathrm{d}v}{\mathrm{d}v} && (3.143)\\
&= \frac{\mathrm{d}e^{v}}{\mathrm{d}v}\cdot\frac{\mathrm{d}x^2}{\mathrm{d}x} && (3.144)\\
&= 2xe^{v} = 2xe^{x^2}. && (3.145)
\end{aligned}$$

Thus,

$$\begin{aligned}
\int xe^{x^2}(e^{x^2}+3)\,\mathrm{d}x &= \frac{1}{2}\int u\left(2xe^{x^2}\right)\mathrm{d}x && (3.146)\\
&= \frac{1}{2}\int u\frac{\mathrm{d}u}{\mathrm{d}x}\,\mathrm{d}x = \frac{1}{2}\int u\,\mathrm{d}u && (3.147)\\
&= \frac{1}{2}\frac{u^2}{2} = \frac{(e^{x^2}+3)^2}{4} + C. && (3.148)
\end{aligned}$$

Sometimes, they need some *creative* algebraic manipulations, and sometimes even multiple substitution. E.g.,[7]

Example 3.8.12: $\displaystyle\int \frac{1}{x(x^5+1)}\,\mathrm{d}x$ If we want to use power rule substitution, there's only x^5 but no x^4 to substitute with. So, we need to get a bit creative.

First, factor the x^5 out of the inner parenthesis and get

$$\int \frac{1}{x^6\left(1+\frac{1}{x^5}\right)}\,\mathrm{d}x. \tag{3.149}$$

Now, we can let $u = 1 + \dfrac{1}{x^5}$. By the power rule, $\dfrac{\mathrm{d}u}{\mathrm{d}x} = -\dfrac{5}{x^6}$. Our integral then becomes

$$-5\int \frac{1}{u}\left(-\frac{5}{x^6}\right)\mathrm{d}x = -\frac{1}{5}\int \frac{1}{u}\frac{\mathrm{d}u}{\mathrm{d}x}\,\mathrm{d}x \tag{3.150}$$

$$= -\frac{1}{5}\int \frac{1}{u}\,\mathrm{d}u = -5\ln(x) \tag{3.151}$$

$$= -\frac{1}{5}\ln\left(1+\frac{1}{x^5}\right) + C \tag{3.152}$$

In certain problems, there aren't any outside terms to substitute with, so you have to create them. E.g.,

Example 3.8.13: $\displaystyle\int \left(1+\sqrt{x}\right)^4\mathrm{d}x$ We want to get rid of the term inside the parenthesis. So, let $u = 1+\sqrt{x}$, and by the power rule, $\dfrac{\mathrm{d}u}{\mathrm{d}x} = \dfrac{1}{2\sqrt{x}}$. However, there is no $\dfrac{1}{\sqrt{x}}$ outside to substitute with. Let's be adventurous and substitute u in anyway and see what happens.

$$\int (1+\sqrt{x})^4\,\mathrm{d}x = \int u^4\,\mathrm{d}x \tag{3.153}$$

[7] Taken from MIT Integration Bee 2019, I_9 [1]

If $\dfrac{\mathrm{d}u}{\mathrm{d}x} = \dfrac{1}{2\sqrt{x}}$, then $\mathrm{d}x = 2\sqrt{x}\,\mathrm{d}u$,

$$= \int u^4 (2\sqrt{x})\,\mathrm{d}u. \tag{3.154}$$

We're almost there, now we just have to replace the leftover x with u. Since $u = 1 + \sqrt{x}$, $\sqrt{x} = u - 1$; thus,

$$\int u^4(2\sqrt{x})\,\mathrm{d}u = 2\int u^4(u-1)\,\mathrm{d}u \tag{3.155}$$

$$= 2\int u^5 - u^4\,\mathrm{d}u = 2\left(\frac{u^6}{6} - \frac{u^5}{5}\right) \tag{3.156}$$

$$= 2\left(\frac{(1+\sqrt{x})^6}{6} - \frac{(1+\sqrt{x})^5}{5}\right). \tag{3.157}$$

These problems definitely take a while to get used to. So I highly encourage you to practice integrating using a problem set ([1], [2]). The unfortunate thing is that most problem sets include trigonometry. So, you'd have to pick the one without them to practice with. Or, you can just wait until we discuss trigonometry and then practice from there.

3.9 Formula for Chapter 3

3.9.1 Formula for derivatives of functions

1. $f(x) = g(x) \implies \dfrac{\mathrm{d}f(x)}{\mathrm{d}x} = \dfrac{\mathrm{d}g(x)}{\mathrm{d}x}$ (Rules of Equality)
2. For $c \in \mathbb{R}$, $\dfrac{\mathrm{d}(c)}{\mathrm{d}x} = 0$ (Derivative of a constant)
3. $\dfrac{\mathrm{d}x}{\mathrm{d}y} = \dfrac{\mathrm{d}x}{\mathrm{d}u} \times \dfrac{\mathrm{d}u}{\mathrm{d}y}$ (Chain rule)
4. $\dfrac{\mathrm{d}}{\mathrm{d}x}(af(x) + bg(x)) = a\dfrac{\mathrm{d}f(x)}{\mathrm{d}x} + b\dfrac{\mathrm{d}g(x)}{\mathrm{d}x}$ (Linearity of differentiation)
5. $\dfrac{\mathrm{d}}{\mathrm{d}x}(ax^n) = anx^{n-1}$ (Power rule)

6. $\dfrac{\mathrm{d}}{\mathrm{d}x}(n^x) = n^x \ln(n)$, $\dfrac{\mathrm{d}(\mathrm{e}^x)}{\mathrm{d}x} = \mathrm{e}^x$ (Derivative of exponentials)
7. $\dfrac{\mathrm{d}}{\mathrm{d}x}\big(\ln(x)\big) = \dfrac{1}{x}$ (Derivative of natural logarithms)
8. $\dfrac{\mathrm{d}}{\mathrm{d}x}\big(f_1(x)f_2(x)\big) = f_1(x)\dfrac{\mathrm{d}f_2}{\mathrm{d}x} + f_2(x)\dfrac{\mathrm{d}f_1}{\mathrm{d}x}$ (Product rule for two functions)
9. $\dfrac{\mathrm{d}}{\mathrm{d}x}(f_1 f_2 \ldots f_n) = f_1 f_2 \ldots f_n \left(\dfrac{1}{f_1}\dfrac{\mathrm{d}f_1}{\mathrm{d}x} + \dfrac{1}{f_2}\dfrac{\mathrm{d}f_2}{\mathrm{d}x} + \cdots + \dfrac{1}{f_n}\dfrac{\mathrm{d}f_n}{\mathrm{d}x}\right)$ (Generalized product rule)

3.9.2 Formula for antiderivatives of functions

1. $f(x) = g(x) \implies \displaystyle\int f(x)\,\mathrm{d}x = \int g(x)\,\mathrm{d}x$ (Rules of Equality)
2. $\displaystyle\int af(x) + bg(x)\,\mathrm{d}x = a\int f(x)\,\mathrm{d}x + b\int g(x)\,\mathrm{d}x$ (Linearity of integration)
3. $n \neq -1$, $\displaystyle\int ax^n\,\mathrm{d}x = a\frac{x^{n+1}}{n+1} + C$ (Reversed power rule)
4. $\displaystyle\int n^x\,\mathrm{d}x = \frac{1}{\ln(n)}n^x + C$, $\displaystyle\int \mathrm{e}^x\,\mathrm{d}x = \mathrm{e}^x + C$ (Antiderative of exponentials)
5. $\displaystyle\int \frac{1}{x}\,\mathrm{d}x = \ln(x) + C$ (Antiderivative of the reciprocal)
6. $\displaystyle\int f(g(x))\frac{\mathrm{d}g}{\mathrm{d}x}\,\mathrm{d}x = \int f(u)\,\mathrm{d}u; u = g(x)$ (u-Substitution)

3.9.3 Definition for various functions and constants

1. $\mathrm{e}^x = \displaystyle\lim_{k\to 0}\sum_{i\to 0}^{k}\frac{1}{i!} = 1 + \frac{x^1}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots = \lim_{n\to\infty}\left(1 + \frac{x}{n}\right)^n$
2. $\mathrm{e} = \displaystyle\sum_{i=0}^{\infty}\frac{1}{i!} = 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \cdots = \lim_{n\to\infty}\left(1 + \frac{1}{n}\right)^n$
3. $\ln(x) = \displaystyle\lim_{n\to 0}\left(\frac{x^h - 1}{h}\right)$

CHAPTER 4

A study of graphs with calculus

4.1 Invitation: sketching mountain ranges

If I told you to *sketch* out x^2, it'd be easy, right? Just a simple parabola would do. But if I say: plot $x^4 - 2x^3 + x^2 - x + 1$? It isn't so easy now is it. You could substitute various x and plot it into the graph directly. But I'm asking for a rough sketch, not a plot. I just want the general shape and structure of the function: where are the highest and lowest points, the x-intercept, and the y-intercept. We're going to learn how to do exactly that in this chapter.

Imagine a function as a vast mountain range. When sketching this landscape, we're interested in identifying its peaks, valleys, and plateaus. A peak represents the sharp summit of a mountain, where the land falls away in all directions; a valley is a sunken area, nestled lower than the surrounding terrain; and a plateau is a level stretch, a temporary calm before the ascent resumes toward higher peaks or descends into deeper valleys. Standing at a peak, the land around you dips downward. In contrast, in a valley, everything rises above you. On a plateau, the ground is flat under your feet, but when gazed into the distance, one side raises higher, but the other reveals the path down. These important features are all characterized by a flat ground: a place with zero slope. If the mountain range represents function, then all of these points have zero first derivative.

Still, the information about zero first derivative is not sufficient to determine whether the point is a peak, a valley, or a plateau. What helps classify these points is the sign of the second derivative.

The second derivative suggests the concavity of a graph at that point. If the first derivative gives you the slope, the second derivative will give you the rate of change of that slope. If the second derivative is positive, it means that the function is increasing around that point; thus, that point must be a valley. If it's negative, then everything is decreasing around; thus, it's a peak. If it's zero, it's simply a plateau.

Just like a mountain, a function can have many points that have zero derivatives. Functions with more than one peaks exists. The first derivative does not provide us with any information on which one of those peaks are the highest. It only tells you where the peak is. In order to find which one is the highest, we'd just have to directly find the function's value at those points, then compare it.

In mathematics terminology, the points that have zero first derivative is called a **stationary point**: the value of the surrounding function is stationary. The valleys are called, **local minima**; the peaks, **local maxima**, and the plateau, **inflection point**. The word *local* suggests that locally, these are either the maximum or minimum point, but it might not be globally. Like how you can stand on your house and say that "this is the highest point of my house," but you're still not higher than Mount Everest. We call the global maximum or minimum point, the **absolute maximum** and **absolute minimum**.

4.2 Minima and maxima

This section focuses on finding the minima and maxima of functions and sketching it. These methods are applicable for all kinds of function, but we'll only study the important ones: quadratics and cubics.

4.2.1 Shape of quadratics

Let's start off with quadratics: a function with only one peak or valley. For any given quadratic

$$f(x) = ax^2 + bx + c, \tag{4.1}$$

Its derivative can be found using the power rule (eq. (3.20)).

$$\frac{\mathrm{d}}{\mathrm{d}x} f(x) = \frac{\mathrm{d}}{\mathrm{d}x}\left(ax^2 + bx + c\right) \tag{4.2}$$

$$= 2ax + b. \tag{4.3}$$

The peak or valley of $f(x)$ is at $x = x_0$ where $\frac{\mathrm{d}}{\mathrm{d}x} f(x)\Big|_{x=x_0} = 0$. The notation $\frac{\mathrm{d}}{\mathrm{d}x} f(x)\Big|_{x=x_0}$ tells you that this derivative needs to be evaluated at $x = x_0$. For a quadratic, this means

$$2ax_0 + b = 0 \tag{4.4}$$

$$x_0 = -\frac{b}{2a}, \tag{4.5}$$

which matches with the usual result from traditional algebra. The y value of these points can be found by plugging in $x = -\frac{b}{2a}$ into $f(x)$.

$$f\left(-\frac{b}{2a}\right) = a\left(-\frac{b}{2a}\right)^2 + b\left(-\frac{b}{2a}\right) + c \tag{4.6}$$

$$= \frac{b^2}{4a} - \frac{b^2}{2a} + c \tag{4.7}$$

$$= c - \frac{3b^2}{4a} \tag{4.8}$$

Other important points include the y and x intercept. The y-intercept can easily be found by setting $x = 0$:

$$y_0 = a(0)^2 + b(0) + cy_0 = c. \tag{4.9}$$

Sadly, calculus does not offer any tools to find the exact root of an equation; you still have to use the quadratic equation

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}. \tag{4.10}$$

Classically, the direction that the quadratic The second derivative of a quadratic is determined by the sign of a. The same result can be derived from the second derivative.

$$\frac{\mathrm{d}^2}{\mathrm{d}x^2} f(x) = \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\mathrm{d}f(x)}{\mathrm{d}x}\right) \tag{4.11}$$

$$= \frac{\mathrm{d}}{\mathrm{d}x}(2ax + b) \tag{4.12}$$

$$= 2a. \tag{4.13}$$

Since we only care about the sign, the two in $2a$ does not matter. This result tell us that if a is positive, it's a right side up parabola, and the stationary point is a local minimum. If it isn't, then it's upside-down and the stationary point is a local maxima.

A parabola doesn't have any inflection point because it only occurs when the second derivative is zero. However, the only way that $\frac{\mathrm{d}^2}{\mathrm{d}x^2} f(x) = 0$ is for a to be zero. If it happens, then $f(x)$ isn't a parabola anymore, but rather a straight line.

Example 4.2.1: *Sketch the graph of* $f(x) = -2x^2 + x + 6$ Because a is negative, the parabola is upside-down. The vertex is at

$$x = -\frac{b}{2a} = -\frac{1}{2(-2)} = \frac{1}{4}, \tag{4.14}$$

$$y = -2\left(\frac{1}{4}\right)^2 + \left(\frac{1}{4}\right) + 6 = 6.125. \tag{4.15}$$

The y-intercept is trivially 6. The x intercept are the roots of this polynomial, given by

$$x = \frac{-(1) \pm \sqrt{(1)^2 - 4(-2)(6)}}{2(-2)} = 2, -\frac{3}{2}. \tag{4.16}$$

Factorization would also yield the same result.

$$-2x^2 + x + 6 = -(2x^2 - x - 6) \tag{4.17}$$

$$= -(x-2)(2x+3) \rightarrow x = 2, -\frac{3}{2} \tag{4.18}$$

Therefore, the sketched graph would be

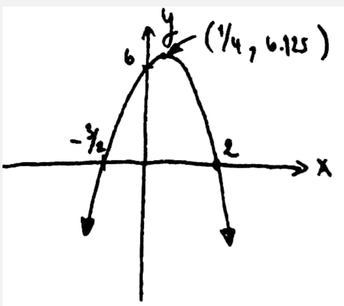


FIG. 4.1 | SKETCH OF THE FUNCTION $-2x^2 + x + 6$.

We can also determine the amount of roots by just looking at the y-value of the vertex point, given by eq. (4.8), and the direction of the parabola. But it's more computationally expensive than the determinant method which says,

> A parabola has no roots if $b^2 - 4ac < 0$, one root if $b^2 - 4ac = 0$, and two roots if $b^2 - 4ac > 0$.

This is a direct result of the quadratic formula: $x = \frac{-b\pm\sqrt{b^2-4ac}}{2a}$ only have real solution if the terms in the root, $b^2 - 4ac$, is positive.

4.2.2 Shape of cubics

Cubics are third order polynomials

$$f(x) = ax^3 + bx^2 + cx + d. \tag{4.19}$$

It's one of the many families of function that can contain more than one stationary point. The x coordinate of the stationary points can be found by using the same method as earlier.

$$\begin{aligned}\frac{\mathrm{d}f(x)}{\mathrm{d}x} &= \frac{\mathrm{d}}{\mathrm{d}x}ax^3 + bx^2 + cx + d && (4.20)\\ &= 3ax^2 + 2bx + c, && (4.21)\end{aligned}$$

in which

$$\begin{aligned}\left.\frac{\mathrm{d}f(x)}{\mathrm{d}x}\right|_{x=x_0} &= 0 = 3ax_0^2 + 2bx_0 + c && (4.22)\\ x_0 &= \frac{-2b \pm \sqrt{(2b)^2 - 4(3a)(c)}}{2(3a)} && (4.23)\\ &= \frac{-2b \pm 2\sqrt{b^2 - 3ac}}{6a} && (4.24)\\ &= \frac{-b \pm \sqrt{b^2 - 3ac}}{3a}. && (4.25)\end{aligned}$$

By the terms under the root, a cubic has no stationary point if $\frac{b^2-3ac}{3a} < 0$, one if $\frac{b^2-3ac}{3a} = 0$, and two if $\frac{b^2-3ac}{3a} > 0$.

The function $f(x)$ intersects the y axis when $x = 0$:

$$f(0) = a(0)^3 + b(0)^2 + c(0) + d = d; \tag{4.26}$$

thus, the y intercept is at $(0, d)$. For the x-intercept, again, there is no tool in calculus that can be used to find the exact root. So, you'd just have to either factor the function, or use the very lengthy cubic formula.

$$x = \sqrt[3]{q + \sqrt{q^2 + (r - p^2)^3}} + \sqrt[3]{q - \sqrt{q^2 + (r - p^2)^3}} + p \tag{4.27}$$

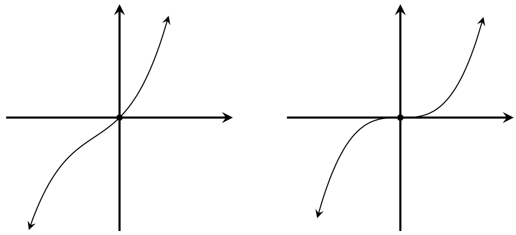
where

$$p = -\frac{b}{3a}, \quad q = p^3 + \frac{bc - 3ad}{6a^2} \quad \text{and} \quad r = \frac{c}{3a}. \tag{4.28}$$

Another topic that I'd like to discuss is the amount of root of cubics. Normally, we would use factorization to determine the amount. From the shape of the cubic, one side shoots of to positive infinity, and the other, negative infinity. And by continuity, it must intersect the x axis somewhere in between and create a root. Therefore, all cubics must have at least one root over the reals. But what if we want to know whether it has a second or third roots or not? We can just look at the amount and the position of the stationary point, given by eq. (4.25).

A key feature of stationary points is that they indicate where a function changes direction. If a function is increasing and then begins to decrease, there must be a stationary point marking this transition. So, if a function lacks a stationary point, and one end tends toward positive infinity while the other falls to negative infinity, we can conclude the function is always either increasing or decreasing. Once the function intersects the x-axis to form a root, it cannot turn back to create another root.

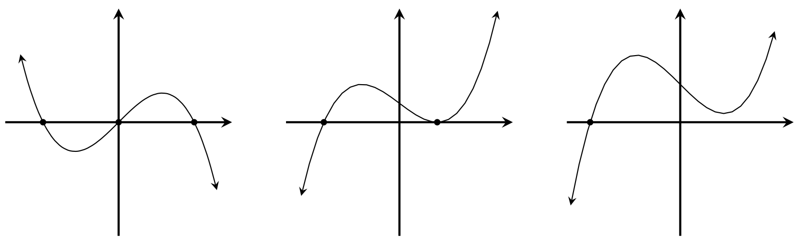
A cubic with zero stationary points ($\frac{b^2-3ac}{3a} < 0$) are such kind of function. One side goes to infinity, and the other, negative infinity. Since it has no stationary point, it must only have one root. One such example is the function $f(x) = x^3 + x^2 + x$, plotted in fig. 4.2a.



(A) $x^3 + x^2 + x$: NO VERTEX **(B)** x^3: ONE VERTEX

FIG. 4.2 | CUBICS WITH ZERO AND ONE VERTEX

Similarly, cubics with only one stationary point ($\frac{b^2-3ac}{3a} = 0$) only have one root. Illustrated in fig. 4.2b, the stationary point must be an inflection point, as it cannot represent a local maximum or minimum. This is due to the behavior of the function, which extends toward both positive and negative infinity. If the stationary point were a local maximum or minimum, it would turn into a parabola, and thus needs another point to reverse its direction again to be a cubic. Hence, cubic functions with only one stationary point have just one root.



(A) $x^3 + x$: THREE ROOTS **(B)** $x^3 - \frac{3}{4}x + \frac{1}{4}$: TWO ROOTS **(C)** $x^3 - x$: ONE ROOT

FIG. 4.3 | TWO VERTICES CUBICS WITH VARYING AMOUNT OF ROOTS

For cubics with two vertices, one of those must be a local maximum, the other, a local minimum. The amount of roots can be directly determined by the position of these points. A cubic will have three roots if the vertices aren't on the same side of the x axis, it must have three roots, illustrated in fig. 4.3a. If one of the vertex is at $y = 0$, then it has two roots;

one of them being the vertex itself, illustrated in fig. 4.3b. If both vertices are on the same side, then it must have only one root; illustrated in fig. 4.3c. It is left as an exercise for the reader to find the position of vertices of the cubics given in fig. 4.3

4.2.3 Optimization problems

The method of finding local maxima, minima, and inflection points are very useful for sketching graphs. Are there any real life applications to this? The answer is in optimization.

Optimization is about finding the best solution for a problem. What's the best time to harvest the crop? What's the best placement for an air conditioner? What's the best way to make profit?, etc. These problems are all about finding the maximum values of some variables. That's exactly what our tools have been able to do: finding the maxima and minima of some functions. Thus, I'd like to discuss some applications of maxima and minima on some physical problems.

Classic maximum area problem. Given a rope length L, what's the maximum rectangular area that this rope can enclose?

To solve this problem, we start by constructing our rectangle. Drawn in fig. 4.4, the perimeter of a rectangle is given by $2(a+b) = L$. Without loss of generality, let side a has length l. Thus,

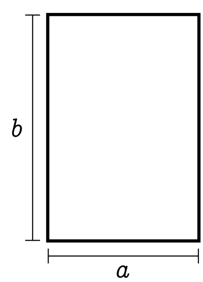


Fig. 4.4 | A rectangle for optimization

$$2(l+b) = L \tag{4.29}$$

$$b = \frac{L}{2} - l \tag{4.30}$$

The area of the rectangle, which is a function of l, $A(l)$ is then

$$A(l) = a \times b = l\left(\frac{L}{2} - l\right) = -l^2 + l\frac{L}{2}. \tag{4.31}$$

Since this function is a parabola, its stationary point must be the local maxima, and must also be the global maxima. This point is where our function is the highest. It can be found by using the derivative w.r.t. l.

$$\frac{\mathrm{d}}{\mathrm{d}l}A(l) = -2l + \frac{L}{2} \tag{4.32}$$

Using the same argument as earlier, we want to find the point l_0 where $\frac{\mathrm{d}}{\mathrm{d}l}A(l)\Big|_{l=l_0} = 0$:

$$0 = -2l_0 + \frac{L}{2} \tag{4.33}$$

$$l_0 = \frac{L}{4}; \tag{4.34}$$

thus, the maximum $A(l)$ is

$$A\left(\frac{L}{4}\right) = -\left(\frac{L}{4}\right)^2 + \left(\frac{L}{4}\right)\frac{L}{2} = \frac{L^2}{16} \tag{4.35}$$

Funnily enough, the maximum area that a perimeter length L can increase is also a parabola $\frac{1}{16}L^2$.

Modified maximum area problem. Given a fence length L. What's the maximum area that I can enclose, given that one of the sides doesn't need any fence?

If our rectangle has sides a and b, and one side doesn't need a fence, without loss of generality, the perimeter is $a + 2b = L$. If $a = l$, then $b = \frac{L-l}{2}$. The area of the rectangle then becomes

$$A(l) = a \times b = l\left(\frac{L-l}{2}\right) = -\frac{1}{2}l^2 + \frac{L}{2}l. \tag{4.36}$$

Using the same argument,

$$\frac{\mathrm{d}}{\mathrm{d}l}A(l) = -l + \frac{L}{2} \tag{4.37}$$

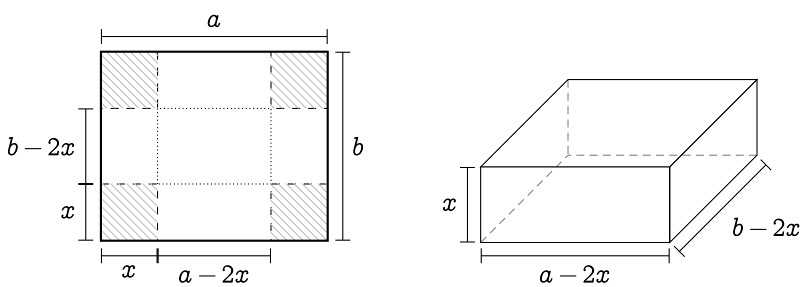
$$\left.\frac{\mathrm{d}}{\mathrm{d}l}A(l)\right|_{l=l_0} = 0 = -l_0 + \frac{L}{2} \tag{4.38}$$

$$l_0 = \frac{L}{2}, \tag{4.39}$$

and

$$A(l_0) = -\frac{1}{2}\left(\frac{L}{2}\right)^2 + \frac{L}{2}\left(\frac{L}{2}\right) = \frac{3}{8}L^2. \tag{4.40}$$

Volume of a box. Let's say you have a paper size $a \times b$, shown in fig. 4.5a. If you want to fold it up into a box without a closing top by cutting a square size x on each corner, shown in fig. 4.5b, find the maximum volume that the box can enclose.



(A) UNFOLDED. THE GRAY PART IS CUT OUT. **(B)** FOLDED

FIG. 4.5 | BOX VOLUME OPTIMIZATION

Shown in fig. 4.5a, if we cut out a square side length x from all corners, then the side length of the final box will be $a-2x$, $b-2x$, and x. The volume $V(x)$, is then

$$V(x) = x(a-2x)(b-2x) = 4x^3 + (-2a-2b)x^2 + (ab)x \tag{4.41}$$

This function is a cubic, and we can use eq. (4.25) to find the x coordinate of the vertex.

$$x_0 = \frac{-(-2a-2b) \pm \sqrt{(-2a-2b)^2 - 3(4)(ab)}}{3(4)} \tag{4.42}$$
$$= \frac{2a+2b \pm \sqrt{4a^2+4b^2-4ab}}{12} \tag{4.43}$$
$$= \frac{a+b \pm \sqrt{a^2+b^2-ab}}{6} \tag{4.44}$$

If $a^2+b^2-ab=0$, there's only one stationary point at $x_0 = \frac{a+b}{6}$, which is an inflection point. Otherwise, it must have two stationary points. Since the leading coefficient of $V(x)$ is positive, $V(x)$ must approach negative infinity when x approaches infinity. Meaning that, the highest local maximum is on the right vertex; thus, we select the negative root:

$$x_0 = \frac{a+b-\sqrt{a^2+b^2-ab}}{6}. \tag{4.45}$$

The full solution is left out due to arithmetical tediousness. It'd be better if you just plug in the numbers.

4.3 Tangent to a curve

In this section, I'll show you how to find a line that's tangent to a curve. But to do it the traditional way would be a bit boring, as it's quite dry and actually doesn't help us with graph sketching at all. So, I shall introduce it via the Newton-Raphson's root finding algorithm; as it's an algorithm that directly takes advantage of the line tangent to a function, and it shows up in many places. So much so that it exists inside the core of the fast square root algorithm which is implemented in every computer.

4.3.1 Newton-Raphson method

Polynomials appears everywhere. Most of the time, you're required to find its root. But it might come as a surprise for you that it is *impossible* to find a general for finding the root of polynomials with degrees higher than four. This is a direct consequence of Galois theory[1], one of the most important theorems in abstract algebra. So most of the time, we resort to root approximation methods.

The root approximation method which is relevant to calculus is the Newton-Raphson's root finding algorithm. I'd try to explain this algorithm has much as I can, but I think it's better if you see it.

The Newton-Raphson algorithm takes advantage of slopes descent. Let's say you want to approximate the root of $f(x)$. You put down a random guess, say x_1. If that point is a root, $f(x_1) = 0$. If it's not, the root is either above ($f(x_1) < 0$) the point or under ($f(x_1) > 0$). What you do next is that you draw a line tangent to the function $f(x)$ at $x = x_1$ and let it cross the x-axis, shown in fig. 4.6. Intuitively, the tangent line will intersect the x-axis at x_2 where x_2 should hopefully be closer to the root. Our new guess will then be $(x_2, f(x_2))$. We then repeat this method until the approximation are satisfactory.

Here, I shall use the Newton-Raphson root finding algorithm as a way to introduce you about how calculus can be further applied to geometry.

The first part of the Newton-Raphson method is to be able to find

[1] The Galois theory is a profound result from abstract algebra. You would usually learn in graduate mathematics, but if you want to just scratch the surface, I found a video by *Math Visualized*, on *Galois Theory Explained Simply* [3], that explains it quite clearly. Still, I need to watch it a couple of times before understanding it.

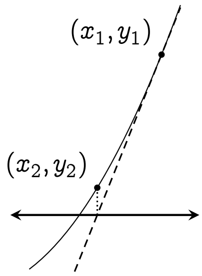


FIG. 4.6 | NEWTON-RAPHSON'S METHOD ILLUSTRATED

the tangent to a curve. Let's first start with a line equation

$$y = mx + c. \tag{4.46}$$

If my first guess is on the point $(x_1, f(x_1))$, the line that passes through that point and tangent to $f(x)$ is

$$y = \left.\frac{\mathrm{d}f(x)}{\mathrm{d}x}\right|_{x=x_1} x + c, \tag{4.47}$$

where $m = \left.\frac{\mathrm{d}f(x)}{\mathrm{d}x}\right|_{x=x_1}$. And c is the intersection with the x axis.

But we don't have to actually find c. For a line tangent to a curve, we just want to move the line from the origin to the tangent point $(x_1, f(x_1))$. To move the line to the right by x_1 units, we substitute x with $x - x_1$, and to shift it up by y_1, substitute y with $y - y_1$. Thus, the line equation turns into

$$y - y_1 = (x - x_1) \left.\frac{\mathrm{d}f(x)}{\mathrm{d}x}\right|_{x=x_1} \tag{4.48}$$

$$y - f(x_1) = (x - x_1) \left.\frac{\mathrm{d}f(x)}{\mathrm{d}x}\right|_{x=x_1}. \tag{4.49}$$

For simplicity of notation, I shall let $\mathrm{D}_x f(x_1)$ represent $\frac{\mathrm{d}f(x)}{\mathrm{d}x}$:

$$y - f(x_1) = (x - x_1)\,\mathrm{D}_x f(x_1). \tag{4.50}$$

Then, Newton-Raphson method tells us that the intersection between the tangent line and the x axis is what our next guess, x_2, should be. It can be easily evaluated by setting $y = 0$.

$$0 - f(x_1) = (x_2 - x_1)\,\mathrm{D}_x f(x_1) \tag{4.51}$$

$$x_2 = x_1 - \frac{f(x_1)}{\mathrm{D}_x f(x_1)} \tag{4.52}$$

Generally, we can repeat this forever. Thus,

$$x_{n+1} = x_n - \frac{f(x_n)}{\mathrm{D}_x f(x_n)} \tag{4.53}$$

a recurrence relation for finding roots of functions. Notice that we don't even need to know the y value for this algorithm to work. And that's the beauty of it.

Let's use this root finding method to approximate $\sqrt{2}$, which is the root of the polynomial $x^2 - 2 = 0$. Since the derivative of $x^2 - 2$ is $2x$, Newton-Raphson method (eq. (4.53)) gives

$$x_{n+1} = x_n - \frac{x_n^2 - 2}{2x_n} \tag{4.54}$$

$$= \frac{x_n^2 + 2}{2x_n}. \tag{4.55}$$

Since $\sqrt{2} \approx 1.414213\ldots$, let's use $x_1 = 1$ as our first guess.

$$x_2 = \frac{x_1^2 + 2}{2x_1} \tag{4.56}$$

$$= \frac{1^2 + 2}{2(1)} = 1.5, \tag{4.57}$$

which is closer to 1.41 than our first guess. Next,

$$x_3 = \frac{x_2^2 + 2}{2x_2} \tag{4.58}$$

$$= \frac{1.5^2 + 2}{2(1.5)} = 1.4166\ldots, \tag{4.59}$$

which is off by just a 0.2% error. By the forth iteration, we get 1.414215 ..., which is off by a mere 0.0002%. And we can stop here.

4.3.2 Numerical version of the Newton-Raphson method

For a function that you can't find a derivative, or you're just too lazy to explicitly find the derivative, you can turn the Newton-Raphson method into a numerical method via the definition of derivativeeq. (4.53).

$$x_{n+1} = x_n - \frac{f(x_n)}{\lim_{h\to 0} \frac{f(x_n+h)-f(x_n)}{h}} \tag{4.60}$$

$$= x_n - \lim_{h\to 0} \frac{f(x_n)h}{f(x_n+h)-f(x_n)}. \tag{4.61}$$

Computationally, we can set h to any arbitrarily small values: 0.01 for example. If you have some experiences with coding, this can be easily implemented by using a `for` loop. Here's some python code that I've written to do just that. Just put in the function in `def f(x):` block, put the amount of iteration in `iterationCount`, your first guess in `firstGuess`, and the value of h in `derivativeStep`. If you have some experiences, I highly encourage you to tinker around.

(The python code is unavailable due to some LaTeXerror. It is still in the source code.)

```
import matplotlib.pyplot as plt
import math

def f(x):  # Insert function here
    return x**3 - 2 * x + 1

iterationCount = 10  # Amount of iteration
firstGuess = -2  # First guess
derivativeStep = 0.01  # For calculating the method of increments
approximationList = [float(firstGuess)]  # List of approximations
```

```python
for i in range(iterationCount):
    diff = (
        f(approximationList[i] + derivativeStep) - f(approximationList[i])
    ) / derivativeStep  # Finding the derivative at a certain point
    nextGuess = (
        approximationList[i] - f(approximationList[i]) / diff
    )  # Newton-Raphson's method
    approximationList.append(nextGuess)  # Add next guess to the list of approximations

print(approximationList)  # Print out the list of approximations
fig, axes = plt.subplots()  # Initiate plot
axes.plot(list(range(iterationCount + 1)), approximationList)  # Plotting
axes.set_xlabel("Iteration")  # Setting the x-axis label
axes.set_ylabel("Guess")  # Setting the y-axis label
plt.show()  # Showing the plot
```

4.4 Normal lines

Alongside the tangent line, there is another line that we're interested in: the normal line, a line that's perpendicular to the tangent. It's actually pretty simple to solve. By analytical geometry, if two lines are perpendicular to each other, the product of their slope must be -1.

If the slope of tangent is m_T and the normal is m_N,

$$m_T \times m_N = -1 \tag{4.62}$$

$$m_N = -\frac{1}{m_T}. \tag{4.63}$$

Since the slope of the tangent at a point $(x_0, f(x_0))$ is $\mathrm{D}_x f(x_0)$,

$$m_N = -\frac{1}{\mathrm{D}_x f(x_0)}. \tag{4.64}$$

The normal line equation can then be formed by translating the line with slope m_N to the point $(x_0, f(x_0))$:

$$y - f(x_0) = -\frac{x - x_0}{\mathrm{D}_x f(x_0)}. \tag{4.65}$$

4.5 Arc length of functions

Before we wrap off this chapter, here's a fun geometrical problem to think about. Given a function $f(x)$, what's the length of the line drawn by the function from a to b? Illustrated in fig. 4.7, the total arc length s can be subdivided into smaller arc lengths $\mathrm{d}s$ that we can sum up. How? Integrals.

The total arc length s of the function $f(x)$ from a to b is just

$$\int_a^b \mathrm{d}s \tag{4.66}$$

By the Pythagorean theorem $\mathrm{d}s = \sqrt{\mathrm{d}x^2 + \mathrm{d}y^2}$. Can we plug this directly into the integral? Of course, we can:

$$s = \int_{x=a}^{x=b} \sqrt{\mathrm{d}x^2 + \mathrm{d}y^2}. \tag{4.67}$$

Factorizing $\mathrm{d}x$ out of the square root would yield an integral

$$\int_{x=a}^{x=b} \sqrt{\mathrm{d}x^2 + \mathrm{d}y^2} = \int_{x=a}^{x=b} \sqrt{1 + \left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2}\,\mathrm{d}x \tag{4.68}$$

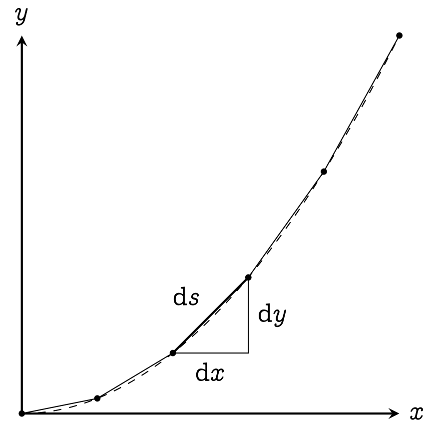


FIG. 4.7 | ARC LENGTH CALCULATION

$$= \int_{x=a}^{x=b} \sqrt{1 + \left(\frac{\mathrm{d}f(x)}{\mathrm{d}x}\right)^2}\, \mathrm{d}x. \tag{4.69}$$

Simply plug in the derivative of $f(x)$ and evaluate the integral, and you would get the arc length.

CHAPTER 5

Calculus, trigonometry, and oscillations

Prerequisites: *basic trigonometry.*

5.1 Invitation: oscillation and trigonometry

Trigonometry is already pretty useful on its own, as it connects the angles of a triangle to its side. But as you study further, trigonometric functions appears in many unexpected places, especially in oscillatory systems. Whether it's a motion of a spring, the synchronized blinks of fireflies, or even dynamics of love; they're somehow all modeled using trigonometric functions. Sure, the usual sine and cosine is already oscillating up and down with a definite pattern. But there are so many oscillatory functions in this world! Why these two? Is it baked into the nature of oscillation? You'll find out in this chapter.

Before moving on, I'd like to point out the whole goal of the rest of the chapters in this volume. After this chapter, the foundation of calculus are in a sense, complete. From here on out, we shall be focusing on various differential equations that describe nature; starting with this chapter. But honestly, I have no idea how I could introduce you to oscillations without studying how trigonometric functions interact with calculus first. So, in the first few sections, it's just going to be mathematics which, I tried to keep it as short but yet as complete as possible.

5.1.1 Newton's fluxion notation

Here's the notation of derivative that will be used throughout the chapter, called the **Newton's fluxion notation** or, **dot notation**. *This notation is used only when the derivative is taken w.r.t. time.* It places a dot over the variables, e.g., the first derivative of position r w.r.t. time is $\dot{r}$.

Higher derivatives are written with more dots, e.g., the second derivative of position r w.r.t. is $\ddot{r}$. The third derivative is $\dddot{r}$, forth derivative, $\ddddot{r}$ and so on.

5.2 Derivative of trigonometric functions

5.2.1 Two fundamental functions: sine and cosine

The derivative of sine is cosine. We'll prove it later, but the geometrical interpretation is that the derivative of sine oscillates along off-sync with the function itself. When $x = 0$, there's a 45° rising slope. At $x = \frac{\pi}{2}$, the slope flattens down to zero. Later at $x = \pi$, the slope is going 45° down, etc. If we plot the slope of the sine function at each point (shown as dots in fig. 5.1), it resembles a cosine curve.

To prove this, let's start with the definition of the derivative given in eq. (1.8),

$$\frac{\mathrm{d}}{\mathrm{d}x}\sin(x) = \lim_{h\to 0}\frac{\sin(x+h)-\sin(x)}{h}. \tag{5.1}$$

Using the sum of sines ($\sin(x+h) = \sin(x)\cos(h) + \cos(x)\sin(h)$),

$$\frac{\mathrm{d}}{\mathrm{d}x}\sin(x) = \lim_{h\to 0}\frac{\sin(x)\cos(h)+\cos(x)\sin(h)-\sin(x)}{h} \tag{5.2}$$

When $h \to 0$, $\cos(h) \to 1$

$$= \lim_{h\to 0}\frac{\sin(x)+\cos(x)\sin(h)-\sin(x)}{h} \tag{5.3}$$

$$= \cos(x)\lim_{h\to 0}\frac{\sin(h)}{h}. \tag{5.4}$$

For small angles, $\sin(h) \approx h$, therefore

$$= \cos(x)\lim_{h\to 0}\frac{h}{h} = \cos(x). \tag{5.5}$$

In conclusion, the derivative of sine is just the cosine just as we predicted with the graph.

Similarly, the derivative of cosine can also be found using the method of increments and using the sum of cosines formula $\cos(x+h) = \cos(x)\cos(h) - \sin(x)\sin(h)$

$$\frac{\mathrm{d}}{\mathrm{d}x}(\cos(x)) = \lim_{h\to 0}\frac{\cos(x+h)-\cos(x)}{h} \tag{5.6}$$

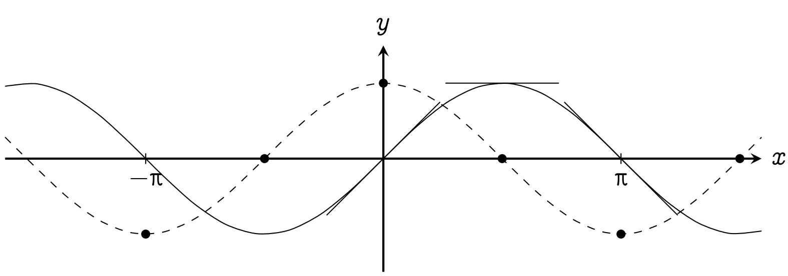


FIG. 5.1 | THE RELATION BETWEEN SINE (IN BLACK) AND COSINE (IN DASHED)

$$= \lim_{h \to 0} \frac{\cos(x)\cos(h) - \sin(x)\sin(h) - \cos(x)}{h} \tag{5.7}$$

When $h \to 0$, $\cos(h) = 1$

$$= \lim_{h \to 0} \frac{\cos(x) - \cos(x) - \sin(x)\sin(h)}{h} \tag{5.8}$$

$$= \lim_{h \to 0} \frac{-\sin(x)\sin(h)}{h} \tag{5.9}$$

For small angles of h, $\sin(h) \approx h$

$$= \lim_{h \to 0} \left(-\sin(x)\right) = -\sin(x) \tag{5.10}$$

I.e., the derivative of cosine is the negative of sine. From those two alone, we can form a set of formulas for differentiating sines and cosines:

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}x}\left(\sin(x)\right) &= \cos(x) \\
\frac{\mathrm{d}^2}{\mathrm{d}x^2}\left(\sin(x)\right) &= \frac{\mathrm{d}}{\mathrm{d}x}\left(\cos(x)\right) = -\sin(x) \\
\frac{\mathrm{d}^3}{\mathrm{d}x^3}\left(\sin(x)\right) &= \frac{\mathrm{d}}{\mathrm{d}x}\left(-\sin(x)\right) = -\cos(x) \\
\frac{\mathrm{d}^4}{\mathrm{d}x^4}\left(\sin(x)\right) &= \frac{\mathrm{d}}{\mathrm{d}x}\left(-\cos(x)\right) = \sin(x)
\end{aligned} \tag{5.11}$$

We can see that the derivative of these two functions cycles in four: $\sin(x), \cos(x), -\sin(x), -\cos(x)$. From these alone, we can use the rules developed in chapter 3, e.g., the chain rule, power rule, etc., in order to find the derivative of other trigonometric functions.

5.2.2 Derivative of other trigonometric functions

Note that you can continue reading the next section straightaway. This section serves as a reference. But for those who are still interested, I'm glad to have you here. This is one of the places that uses the chain rule, product rule, and power rule a lot. It's a good exercise. Without further ado, let's start with the cosecant ($\csc(x) = \frac{1}{\sin(x)}$).

$$\frac{\mathrm{d}}{\mathrm{d}x}\csc(x) = \frac{\mathrm{d}}{\mathrm{d}x}\frac{1}{\sin(x)} \tag{5.12}$$

$$= \frac{\mathrm{d}}{\mathrm{d}x}\frac{1}{u}\cdot\frac{\mathrm{d}u}{\mathrm{d}u} \qquad \text{Chain rule: } \sin(x) = u \tag{5.13}$$

$$= \frac{\mathrm{d}}{\mathrm{d}u}u^{-1}\cdot\frac{\mathrm{d}u}{\mathrm{d}x} \tag{5.14}$$

$$= -1u^{-2}\cdot\frac{\mathrm{d}}{\mathrm{d}x}\sin(x) \qquad \text{Power rule, } u = \sin(x) \tag{5.15}$$

$$= -\csc(x)\cot(x) \qquad \cot(x) = \frac{\cos(x)}{\sin(x)}. \tag{5.16}$$

And the secant function ($\sec(x) = \frac{1}{\cos(x)}$),

$$\frac{\mathrm{d}}{\mathrm{d}x}\sec(x) = \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{1}{\cos(x)}\right) \tag{5.17}$$

$$= \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{1}{u}\right)\cdot\frac{\mathrm{d}u}{\mathrm{d}u} \qquad \text{Chain rule: } \cos(x) = u \tag{5.18}$$

$$= \frac{\mathrm{d}}{\mathrm{d}u}u^{-1}\cdot\frac{\mathrm{d}u}{\mathrm{d}x} \tag{5.19}$$

$$= -1u^{-2}\cdot\frac{\mathrm{d}}{\mathrm{d}x}\cos(x) \qquad \text{Power rule, } u = \cos(x) \tag{5.20}$$

$$= -\frac{1}{\cos^2(x)}(-\sin(x)) \qquad \frac{\mathrm{d}}{\mathrm{d}x}\cos(x) = -\sin(x) \tag{5.21}$$

$$= \sec(x)\tan(x). \tag{5.22}$$

For the tangent function ($\tan(x) = \frac{\sin(x)}{\cos(x)}$), we use the product rule and the results from earlier

$$\frac{\mathrm{d}}{\mathrm{d}x}\tan(x) = \frac{\mathrm{d}}{\mathrm{d}x}\frac{\sin(x)}{\cos(x)} \tag{5.23}$$

$$= \frac{\mathrm{d}}{\mathrm{d}x}\sin(x)\sec(x) \tag{5.24}$$

$$= \sin(x)\frac{\mathrm{d}}{\mathrm{d}x}\sec(x) + \sec(x)\frac{\mathrm{d}}{\mathrm{d}x}\sin(x). \tag{5.25}$$

$$= \sin(x)\sec(x)\tan(x) + \sec(x)\cos(x) \tag{5.26}$$

$$= \sin(x)\frac{1}{\cos(x)}\frac{\sin(x)}{\cos(x)} + \frac{1}{\cos(x)}\cos(x) \tag{5.27}$$

$$= \left(\frac{\sin(x)}{\cos(x)}\right)^2 + 1 = \tan^2(x) + 1 = \sec^2(x) \tag{5.28}$$

The Pythagorean identity is used in the last line to convert $1 + \tan^2(x)$ to

$\sec^2(x)$. Of course, we cannot miss its reciprocal function: $\cot(x) = \frac{1}{\tan(x)}$.

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}x}\cot(x) &= \frac{\mathrm{d}}{\mathrm{d}x}\frac{1}{\tan(x)} && (5.29)\\
&= \frac{\mathrm{d}}{\mathrm{d}x}\frac{\cos(x)}{\sin(x)} && (5.30)\\
&= \frac{\mathrm{d}}{\mathrm{d}x}\cos(x)\csc(x) && (5.31)\\
&= \cos(x)\frac{\mathrm{d}}{\mathrm{d}x}\csc(x) + \csc(x)\frac{\mathrm{d}}{\mathrm{d}x}\cos(x) && (5.32)\\
&= \cos(x)\big(-\csc(x)\cot(x)\big) + \csc(x)\big(-\sin(x)\big) && (5.33)\\
&= -\cos(x)\frac{1}{\sin(x)}\frac{\cos(x)}{\sin(x)} - \frac{1}{\sin(x)}\sin(x) && (5.34)\\
&= -\big(\cot^2(x)+1\big) = -\csc^2(x), && (5.35)
\end{aligned}$$

where we've again used the Pythagorean identity $\cot^2(x) + 1 = -\csc^2(x)$.

Altogether, I've summarized all the derivatives of trigonometric functions into table 5.1.

$f(x)$	$\frac{\mathrm{d}}{\mathrm{d}x}f(x)$
$\sin(x)$	$\cos(x)$
$\cos(x)$	$-\sin(x)$
$\tan(x)$	$\sec^2(x)$
$\csc(x)$	$-\csc(x)\cot(x)$
$\sec(x)$	$\sec(x)\tan(x)$
$\cot(x)$	$-\csc^2(x)$

TABLE 5.1 | THE DERIVATIVES OF TRIGONOMETRIC FUNCTIONS

5.3 Power series of trigonometric functions

The approximation method that I'd be discussing are commonly taught to high schoolers with the name **small angle approximation**, i.e., for $\theta \to 0$,

$$\sin(\theta) \approx \tan(\theta) \approx \theta \quad \text{and} \quad \cos(\theta) \approx 1 - \frac{\theta^2}{2} \tag{5.36}$$

These approximations are very useful for calculating limits, e.g.,[1]

$$\lim_{\theta \to 0} \frac{\sin(n\theta)}{\theta} = \lim_{\theta \to 0} \frac{n\theta}{\theta} = n. \tag{5.37}$$

Here, I want to focus on where these approximations really comes from: the power series development of trigonometric functions.

5.3.1 Power series of sine

Assume that the sine function can be written as a power series

$$\sin(x) = s_0 + s_1 x^1 + s_2 x^2 + s_3 x^3 + ... \tag{5.38}$$

Because $\sin(0) = 0$,

$$\sin(0) = s_0 + s_1 (0)^1 + s_2 (0)^2 + s_3 (0)^3 + ... \tag{5.39}$$

$$s_0 = 0; \tag{5.40}$$

Because sine is an odd function, $\sin(-x) = -\sin(x)$; thus,

$$\begin{aligned} s_1(-x)^1 + s_2(-x)^2 \\ + s_3(-x)^3 + s_4(-x)^4 + ... = & - s_1 x^1 - s_2 x^2 \\ & - s_3 x^3 - s_4 x^4 + ... \end{aligned} \tag{5.41}$$

$$\begin{aligned} - s_1 x^1 + s_2 x^2 \\ - s_3 x^3 + s_4 x^4 - ... = & - s_1 x^1 - s_2 x^2 - \\ & s_3 x^3 - s_4 x^4 - ... \end{aligned} \tag{5.42}$$

[1] Normally these limits are evaluated using the squeeze theorem. However, I don't want the mathematical intricacies to disrupt the flow of our journey right now. The full theorem will be discussed later at the end of the book.

In the L.H.S., the sign alternates between negative and positive. In order for the L.H.S. to match the R.H.S., all the positive terms must vanish, leaving only the matching negative terms; therefore,

$$\sin(x) = s_1 x^1 + s_3 x^3 + s_5 x^5 + \dots . \tag{5.43}$$

This is what we mean by "sine is an odd function": there are only odd powers in its power series.

To find s_1, take the derivative of sine

$$\frac{\mathrm{d}}{\mathrm{d}x}\sin(x) = \cos(x) = s_1 + 3s_3x^2 + 5s_5x^4 + \dots , \tag{5.44}$$

then substitute $x = 0$

$$\cos(0) = 1 = s_1 + 3s_3(0)^2 + 5s_5(0)^4 + \dots \tag{5.45}$$

$$1 = s_1. \tag{5.46}$$

We can then take the derivative of sine again to get a recurrence relation.

$$\frac{\mathrm{d}^2}{\mathrm{d}x^2}\sin(x) = 3\cdot 2s_3x + 5\cdot 4s_5x^3 + 7\cdot 6s_7x^5 + \dots \tag{5.47}$$

$$-\sin(x) = \frac{3!}{1!}s_3x + \frac{5!}{3!}s_5x^3 + \frac{7!}{5!}s_7x^5 + \dots \tag{5.48}$$

$$-s_1x - s_3x^3 - s_5x_5 - \dots = \frac{3!}{1!}s_3x + \frac{5!}{3!}s_5x^3 + \frac{7!}{5!}s_7x^5 + \dots \tag{5.49}$$

By matching terms with the same order,

$$\begin{aligned} -s_1 &= \frac{3!}{1!}s_3x, \\ -s_3 &= \frac{5!}{3!}s_5x, \\ -s_5 &= \frac{7!}{5!}s_7x. \\ &\vdots \end{aligned} \tag{5.50}$$

From $s_1 = 1$, we get that $s_3 = -\frac{1}{3!}$, $s_5 = -\frac{1}{5!}$, $s_7 = -\frac{1}{7!}$ etc. The negative sign alternates every term. We can then summarize the whole thing as

$$\sin(x) = \sum_{n=0}^{\infty} \frac{(-1)^{2n+1}}{(2n+1)!}x^{2n+1} \tag{5.51}$$

$$= x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \frac{1}{7!}x^7 + \dots, \tag{5.52}$$

which is the polynomial expansion of the sine function.

5.3.2 Power series of cosine

The power series of cosine can be developed using the same method as discussed earlier. But notice that the derivative of sine is cosine. Hence, we can just take the derivative of the sine's power series once to get the cosine's power series.

$$\frac{\mathrm{d}}{\mathrm{d}x}\sin(x) = \frac{\mathrm{d}}{\mathrm{d}x}\left(x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \frac{1}{7!}x^7\right) \tag{5.53}$$

$$\cos(x) = 1 - \frac{1}{2!}x^2 + \frac{1}{4!}x^4 - \frac{1}{6!}x^6 + \dots = \sum_{n=0}^{\infty}\frac{(-1)^{2n}}{(2n)!}x^{2n} \tag{5.54}$$

5.3.3 Power series of other trigonometric functions

The same method could be used to find the power series expansion of secant, cosecant, tangent, etc. However, their power series doesn't actually represent the function because the other functions are all ratios of sine and cosines. To see what I mean, consider the behavior of the tangent function.

The tangent function is defined as $\tan\theta = \frac{\sin\theta}{\cos\theta}$. From this definition, $\tan\theta$ approaches infinity whenever $\cos\theta$ approaches zero, which is at $n\pi$ where $n \in \mathbb{Z}$[2]. As seen, there are infinitely many places in which the function approaches infinity. However, polynomials only have two infinities: at $x \to \infty$, and $x \to -\infty$; therefore, any polynomial expansion can't possibly represent the tangent function. The same logic also works for other trigonometric functions.

[2] $\mathbb{Z}$ is the set of integers

5.3.4 Approximation of trigonometric functions

One way a function can be approximated is by truncating its power series. To see why this is viable, consider the limit

$$\lim_{x\to 0}\sin(x)=\lim_{x\to 0}\left(x-\frac{1}{3!}x^3+\frac{1}{5!}x^5-\frac{1}{7!}x^7+\ldots\right)\tag{5.55}$$

When $x\to 0$, all the higher order degrees term vanishes; thus,

$$\lim_{x\to 0}\sin(x)=x.\tag{5.56}$$

It tells us that when $x\approx 0$, the sine function behaves like a linear function. Illustrated in fig. 5.2,

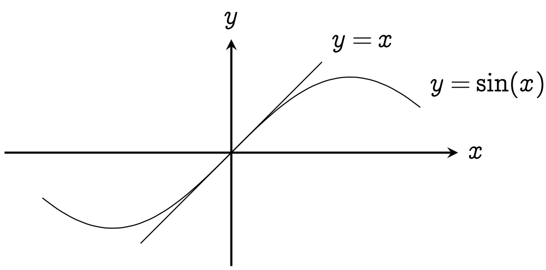


FIG. 5.2 | COMPARISON OF $y=\sin(x)$ AND $y=x$ NEAR $x=0$

When x gets larger, the higher order terms in the power series contributes more and more, so we need to include them. An excellent approximation for the sine function when $x\in[-\pi,\pi]$ is a truncation at the fifth term:

$$\sin(x)\approx x-\frac{1}{3!}x^3+\frac{1}{5!}x^5-\frac{1}{7!}x^7+\frac{1}{9!}x^9,\tag{5.57}$$

plotted in fig. 5.3. It is extremely accurate in that range. For cosine,

$$\lim_{x\to 0}\cos(x)=\lim_{x\to 0}\left(1-\frac{x^2}{2!}+\frac{x^4}{4!}-\ldots\right)\approx 1-\frac{x^2}{2!}\approx 1.\tag{5.58}$$

An acceptable approximation around zero is 1 or $1-\frac{x^2}{2}$. Similarly, an excellent approximation when $x\in[-\pi,\pi]$ is also a truncation at the fourth

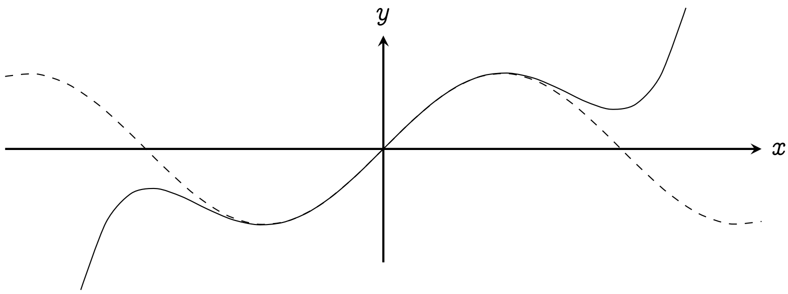


Fig. 5.3 | An approximation of sine by truncating its power series at the fifth term.

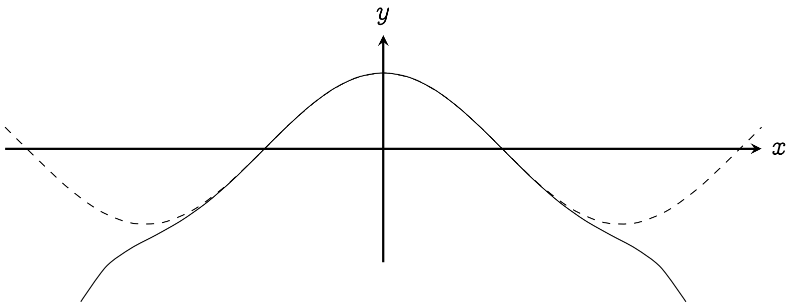


Fig. 5.4 | An approximation of cosine by truncating its power series at the fourth term

term:

$$\cos(x) \approx 1 - \frac{1}{2!}x^2 + \frac{1}{4!}x^4 - \frac{1}{6!}x^6 \tag{5.59}$$

plotted in fig. 5.4

5.4 The harmonic oscillator

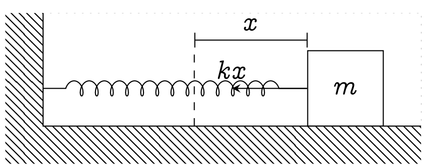


Fig. 5.5 | Setup of the harmonic oscillator

Finally, we're ready to tackle our first oscillatory system: the harmonic oscillator. Illustrated in fig. 5.5, consider a mass m connected to a spring. The point that the spring is neither stretched nor compressed is called the **equilibrium position**, drawn as the dashed line. The force that the spring acts on the mass depends on the displacement from the equilibrium position x:

$$F = -kx, \tag{5.60}$$

where k is the stiffness of the spring. Higher k represents a stiffer spring, and lower k represents a looser spring. Newton's second law reads

$$-kx = m\frac{\mathrm{d}^2 x}{\mathrm{d}t^2} \tag{5.61}$$

$$-\frac{k}{m}x = \frac{\mathrm{d}^2 x}{\mathrm{d}t^2}. \tag{5.62}$$

We're interested in the solution of eq. (5.62) given an initial condition that at time $t = 0$, the mass is distance A away from the equilibrium position, and it's released with zero initial velocity, i.e.,

$$x(t=0) = A \quad \text{and} \quad \dot{x}(t=0) = 0. \tag{5.63}$$

5.4.1 Your first trigonometric substitution

The equation that we got, eq. (5.62), is an inseparable second order differential equation. It is extremely hard to work with. But by exploiting the chain rule, we can reduce it into a first order differential equation.

$$\frac{\mathrm{d}^2 x}{\mathrm{d}t^2} = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\mathrm{d}x}{\mathrm{d}t}\right) = \frac{\mathrm{d}\dot{x}}{\mathrm{d}t} = \frac{\mathrm{d}\dot{x}}{\mathrm{d}x}\frac{\mathrm{d}x}{\mathrm{d}t} = \dot{x}\frac{\mathrm{d}\dot{x}}{\mathrm{d}x}. \tag{5.64}$$

Equation (5.62) then becomes

$$-\frac{k}{m}x = \dot{x}\frac{\mathrm{d}\dot{x}}{\mathrm{d}x} \tag{5.65}$$

$$-\frac{k}{m}x\,\mathrm{d}x = \dot{x}\,\mathrm{d}\dot{x} \tag{5.66}$$

$$-\frac{k}{m}\int x\,\mathrm{d}x = \int \dot{x}\,\mathrm{d}\dot{x} \tag{5.67}$$

$$-\frac{k}{m}\frac{x^2}{2} = \frac{\dot{x}^2}{2} + C \tag{5.68}$$

$$-\frac{k}{m}x^2 = \dot{x}^2 + C. \tag{5.69}$$

To find C, substitute in the initial condition eq. (5.63).

$$-\frac{k}{m}(A)^2 = (0)^2 + C \tag{5.70}$$

$$C = -\frac{k}{m}A^2. \tag{5.71}$$

Thus,

$$-\frac{k}{m}x^2 = \dot{x}^2 - \frac{k}{m}A^2 \tag{5.72}$$

$$\dot{x}^2 = \frac{k}{m}\left(A^2 - x^2\right) \tag{5.73}$$

$$\dot{x} = \sqrt{\frac{k}{m}}\sqrt{A^2 - x^2}. \tag{5.74}$$

Because I'm too lazy to write square roots, let's use $\omega = \sqrt{k\,/\,m}$; thus,

$$\dot{x} = \omega\sqrt{A^2 - x^2} \tag{5.75}$$

$$\frac{\mathrm{d}x}{\mathrm{d}t} = \omega\sqrt{A^2 - x^2} \tag{5.76}$$

$$\omega\,\mathrm{d}t = \frac{1}{\sqrt{A^2 - x^2}}\,\mathrm{d}x \tag{5.77}$$

$$\omega\int \mathrm{d}t = \int \frac{1}{\sqrt{A^2 - x^2}}\,\mathrm{d}x. \tag{5.78}$$

The L.H.S. directly evaluates to ωt. But on the R.H.S., power rule substitution wouldn't work because there's a polynomial with degree two the root, but no terms with lower degree outside. If $u = A^2 - x^2$, we'd be going in loops. So how do we evaluate this?

The answer is, you can't do anything about it if we're going to use the methods so far. It's time for a new tool: **trigonometric substitution**.

We want to take advantage of the Pythagorean identity $1 - \sin^2\theta = \cos^2\theta$ to bring $A^2 - x^2$ out of the root. Substitute in $x = A\sin\theta$ and see what happens.

$$\int \frac{1}{\sqrt{A^2 - x^2}}\,\mathrm{d}x = \int \frac{1}{\sqrt{A^2 - A^2\sin^2\theta}}\,\mathrm{d}x \tag{5.79}$$

$$= \frac{1}{A}\int \frac{1}{\sqrt{1-\sin^2\theta}}\,\mathrm{d}x \tag{5.80}$$

$$= \frac{1}{A}\int \frac{1}{\sqrt{\cos^2\theta}}\,\mathrm{d}x = \frac{1}{A}\int \frac{1}{\cos\theta}\,\mathrm{d}x \tag{5.81}$$

Taking care of the $\mathrm{d}x$; if $x = A\sin\theta$, then

$$\frac{\mathrm{d}x}{\mathrm{d}\theta} = A\frac{\mathrm{d}\sin\theta}{\mathrm{d}\theta} \tag{5.82}$$

$$\frac{\mathrm{d}x}{\mathrm{d}\theta} = A\cos\theta \tag{5.83}$$

$$\mathrm{d}x = A\cos\theta\,\mathrm{d}\theta. \tag{5.84}$$

Substituting back into the integral gives

$$\frac{1}{A}\int \frac{1}{\cos\theta}\left(A\cos\theta\,\mathrm{d}\theta\right) = \int \mathrm{d}\theta = \theta + C. \tag{5.85}$$

Because $x = A\sin\theta$, $\theta = \arcsin\left(x\,/\,A\right)$

$$\int \frac{1}{\sqrt{A^2 - x^2}}\,\mathrm{d}x = \arcsin\left(\frac{x}{A}\right) + C. \tag{5.86}$$

Thus, eq. (5.78) becomes

$$\omega t = \arcsin\left(\frac{x}{A}\right) + C. \tag{5.87}$$

Then, substitute in the initial condition (eq. (5.63)) to find C:

$$\omega(0) = \arcsin\left(\frac{A}{A}\right) + C \tag{5.88}$$

$$C = \frac{\pi}{2}. \tag{5.89}$$

Thus,

$$\omega t = \arcsin\left(\frac{x}{A}\right) + \frac{\pi}{2} \tag{5.90}$$

$$\arcsin\left(\frac{x}{A}\right) = \omega t + \frac{\pi}{2} \tag{5.91}$$

$$x = A\sin\left(\omega t + \frac{\pi}{2}\right). \tag{5.92}$$

Since $\sin(\theta + \frac{\pi}{2}) = \cos(\theta)$,

$$x(t) = A\cos(\omega t) \tag{5.93}$$

which is the solution of the differential equation. It represents where the mass is relative to the equilibrium point over time. Taking the derivative of this once,

$$\dot{x}(t) = A\frac{\mathrm{d}\cos(\omega t)}{\mathrm{d}t} \tag{5.94}$$

Let $u = \omega t$. By the chain rule,

$$= A\frac{\mathrm{d}\cos(u)}{\mathrm{d}u} \times \frac{\mathrm{d}u}{\mathrm{d}t} \tag{5.95}$$

$$= -A\sin(u)\frac{\mathrm{d}}{\mathrm{d}t}(\omega t) \tag{5.96}$$

$$= -\omega A\sin(\omega t), \tag{5.97}$$

we get the velocity as a function w.r.t. time. Similarly, taking the derivative of velocity again gives the acceleration

$$\ddot{x} = \frac{\mathrm{d}\dot{x}}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}\big(-\omega A\sin(\omega t)\big) = -\omega^2 A\sin(\omega t). \tag{5.98}$$

5.4.2 The power series method

Perhaps a faster method to solve eq. (5.62) is to just assume that the solution can be expanded in terms of power series

$$x(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3 + \dots. \tag{5.99}$$

Consequently,

$$\dot{x}(t) = a_1 + 2a_2 t + 3a_3 t^2 + 4a^4 t_3 + \dots \tag{5.100}$$

$$\ddot{x}(t) = 2a_2 + 3 \cdot 2a_3 t + 4 \cdot 3a_4 t^2 + 5 \cdot 4a_5 t_3 + \dots \tag{5.101}$$

The initial condition given in eq. (5.63) ($x(t = 0) = A$, $\dot{x}(t = 0) = 0$) can then be used to find a_0 and a_1.

$$x(t = 0) = a_0 + a_1(0) + a_2(0)^2 + a_3(0)^3 + \dots \tag{5.102}$$

$$a_0 = A \tag{5.103}$$

and from eq. (5.100),

$$\dot{x}(t = 0) = a_1 + 2a_2(0) + 3a_3(0)^2 + 4a_4(0)^3 + \dots \tag{5.104}$$

$$a_1 = 0 \tag{5.105}$$

The rest of the terms can be found by using the differential equation itself. First, rearrange the equation into

$$\frac{\mathrm{d}^2 x}{\mathrm{d}t^2} + \omega^2 x = 0. \tag{5.106}$$

Then, substitute in eqs. (5.99) and (5.101)

$$0 = \frac{\mathrm{d}^2 x}{\mathrm{d}t^2} + \omega^2 x \tag{5.107}$$

$$0 = \left(2a_2 + 3 \cdot 2a_3 t + 4 \cdot 3a_4 t^2 + 5 \cdot 4a_5 t^3 + 6 \cdot 5a_6 t^4 + 6 \cdot 7a_7 t^5 + \dots\right) + \left(\omega^2 A + a_2\omega^2 t^2 + a_3\omega^2 t^3 + a_4\omega^2 t^4 + a_5\omega^2 t^5 \dots\right)$$

$$\begin{aligned} 0 = {} & \left(2a_2 + \omega^2 A\right) \\ & + (3 \cdot 2a_3)t \\ & + (4 \cdot 3a_4 + \omega^2 a_2)t^2 \\ & + (5 \cdot 4a_5 + \omega^2 a_3)t^3 \\ & + (6 \cdot 5a_6 + \omega^2 a_4)t^4 \\ & + (7 \cdot 6a_7 + \omega^2 a_5)t^5 + \dots \end{aligned} \tag{5.108}$$

For the R.H.S. to be 0, all terms must vanish; thus, we get an infinite series of equations to be evaluated

$$\begin{aligned}
0 &= \left(2a_2 + \omega^2 A\right) \\
0 &= (3 \cdot 2a_3) \\
0 &= (4 \cdot 3a_4 + \omega^2 a_2) \\
0 &= (5 \cdot 4a_5 + \omega^2 a_3) \\
0 &= (6 \cdot 5a_6 + \omega^2 a_4) \\
0 &= (7 \cdot 6a_7 + \omega^2 a_5) \\
&\vdots
\end{aligned} \tag{5.109}$$

Starting from the second one: $0 = 3 \cdot 2a_3$ means that $a_3 = 0$. This eliminates every term that includes a_3. The fourth equation says $5 \cdot 4a_5 + \omega^2 a_3 = 0$, thus $a_5 = 0$. This pattern continues, eliminating every odd terms in the series. We're then left with

$$\begin{aligned}
0 &= 2a_2 + \omega^2 A \\
0 &= 4 \cdot 3a_4 + \omega^2 a_2 \\
0 &= 6 \cdot 5a_6 + \omega^2 a_4 \\
0 &= 8 \cdot 7a_8 + \omega^2 a_6 \\
&\vdots
\end{aligned} \tag{5.110}$$

Starting from the first one, we get $a_2 = -\frac{\omega^2}{2}A$. The second equation gives $a_4 = \frac{\omega^4}{4!}A$. The third gives $a_6 = \frac{\omega^6}{6!}A$. This pattern of $a_{2n} = \frac{\omega^{2n}}{2n}A$ continues on forever. The power series expansion for $x(t)$ gives

$$x(t) = a_0 + \cancelto{0}{a_1} + a_2 t^2 + \cancelto{0}{a_3} t^3 + a_4 t^4 + \cancelto{0}{a_5} t^5 + a_6 t^6 + \ldots \tag{5.111}$$

$$= A + A\frac{\omega^2}{2!}t^2 + A\frac{\omega^4}{4!}t^4 + A\frac{\omega^6}{6!}t^6 + \ldots \tag{5.112}$$

$$= A\left(\frac{(\omega t)^2}{2!} + \frac{(\omega t)^4}{4!} + \frac{(\omega t)^6}{6!} + \ldots\right) \tag{5.113}$$

Looks familiar yet? *This*, the result that we got is exactly the power series expansion for cosine (eq. (5.54)); thus,

$$x(t) = A\cos(\omega t). \tag{5.114}$$

5.4.3 Physical aspects of the harmonic oscillator

Now that we have solved the harmonic oscillator, let's bring our attention to the qualitative description of the system.

The mass is released with zero velocity at $x = A$. As the spring tries to restore its equilibrium, it accelerates the mass towards the equilibrium position. The further the mass is from the equilibrium, the stronger the spring pulls. Initially, the spring pulls very hard on the mass, causing it to accumulate much velocity. But as it approaches the equilibrium position, the spring's force diminishes and there is barely any force left to stop the mass, so it overshoots the other side, compressing the spring. This oscillatory cycle continues on forever. Intuitively, we can predict that the mass will have the highest velocity and lowest acceleration as it directly passes through the equilibrium; and, the lowest velocity and the highest acceleration when the spring is fully stretched or compressed. But does the mathematics agree with this?

In fig. 5.6, the position function (eq. (5.93)) is plotted in thick lines, and the velocity function (eq. (5.97)) are plotted in dashed lines.

The furthest point that a mass could be from a spring is at $x = \pm A$. This is called the oscillation **amplitude**. Mathematically, this represents the local maxima and minima of the function, which can be obtained by taking the first derivative and setting it to zero (section 4.2). The derivative of the position w.r.t. time is just the velocity (eq. (5.97)), and it reaches zero whenever $\sin(\omega t)$ is zero

$$\sin(\omega t) = 0 \tag{5.115}$$

$$\omega t = \arcsin(0) + 2n\pi \qquad n \in \mathbb{Z} \tag{5.116}$$

$$t = \frac{(2n+1)\pi}{\omega}. \tag{5.117}$$

Another way to derive this fact is to just notice that the range of the cosine function only extends from -1 to 1; therefore, the highest that $A\cos(\omega t)$ can go is just A. So, another equation that we can form is

$$\cos(\omega t) = 1 \tag{5.118}$$

$$\omega t = \arccos(1) + 2n\pi \qquad n \in \mathbb{Z} \tag{5.119}$$

$$t = \frac{(2n+1)\pi}{\omega}, \tag{5.120}$$

which has the same solution as $\sin(\omega t) = 0$.

The time that the mass goes through the equilibrium position can be found by just setting $A\cos(\omega t)$ to zero

$$A\cos(\omega t) = 0 \tag{5.121}$$

$$\omega t = \arccos(0) + 2n\pi \qquad n \in \mathbb{Z} \tag{5.122}$$

$$t = \frac{2n\pi}{\omega} \tag{5.123}$$

Since the velocity function (eq. (5.97)) is a sine function, it must oscillate π radians off-phase relative to the position function. Therefore,

1. Whenever the mass is furthest away from the equilibrium, the velocity is zero

2. Whenever the mass is directly passing through the equilibrium, the velocity reaches its maximum at $\pm\omega A$

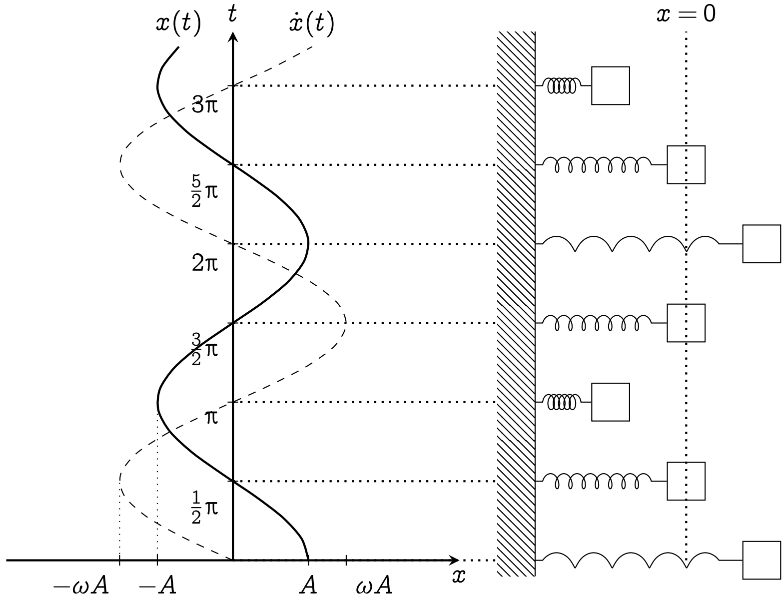


FIG. 5.6 | THE TRAJECTORY OF A MASS IN THE HARMONIC OSCILLATOR SYSTEM. (LEFT) THE POSITION IS PLOTTED IN THICK LINES, AND THE VELOCITY, IN DOTTED LINES. (RIGHT) DEPICTION OF AN OSCILLATING MASS, ACCORDING TO THE SOLUTION OF THE DIFFERENTIAL EQUATION.

The first fact can be easily verified by setting the velocity function, $-\omega A \sin(\omega t)$ equals to zero

$$-\omega A \sin(\omega t) = 0 \tag{5.124}$$

$$\omega t = \arcsin(0) + 2n\pi \tag{5.125}$$

$$t = \frac{2n\pi + 1}{\omega}, \tag{5.126}$$

which directly matches with the point of furthest position (eq. (5.117)). The second one can be verified by taking the derivative of velocity, which is just the acceleration, found in eq. (5.98). Setting that equals to zero yields

$$-\omega^2 A \cos(\omega t) = 0 \tag{5.127}$$

$$\omega t = \arccos(0) + 2n\pi \tag{5.128}$$

$$t = \frac{2n\pi}{\omega}, \tag{5.129}$$

which matches our result from eq. (5.123).

The acceleration function is again, a cosine function that's in phase with the position. Therefore, the local minima and maxima from the position function must match. We can quickly conclude that the acceleration is at its highest at $t = \frac{2n\pi}{\omega}$, and is zero at $\frac{(2n+1)\pi}{\omega}$.

You would think that we have completed the physical aspects of the harmonic oscillator. But there is something I left over: the ω term. Since the beginning, I have assigned $\omega = \sqrt{\frac{k}{m}}$ as a notational trick for convenience. Is there any real physical meaning behind ω?

If you don't know much about units, it's fine if you skip these paragraphs straightaway. But one clue that suggests the physical meaning of ω is its unit. Since $\omega = \sqrt{\frac{k}{m}}$, we have to analyze the units of its components separately. k is a spring stiffness constant which is got from

$$F = -kx \quad \text{or} \quad k = -\frac{F}{x}. \tag{5.130}$$

F is the unit of force (Newton N), which is just $\mathrm{kg}\frac{\mathrm{m}}{\mathrm{s}^2}$), and x, the units of distance (Meters m); therefore, k is the unit of

$$\frac{\mathrm{kg}\frac{\mathrm{m}}{\mathrm{s}^2}}{\mathrm{m}} = \frac{\mathrm{kg}}{\mathrm{s}^2}. \tag{5.131}$$

And since m has the unit of kilogram (kg), ω which is $\sqrt{\frac{k}{m}}$ has the unit of

$$\sqrt{\frac{\frac{\mathrm{kg}}{\mathrm{s}^2}}{\mathrm{kg}}} = \sqrt{\frac{1}{\mathrm{s}^2}} = \frac{1}{\mathrm{s}} \tag{5.132}$$

Funnily enough, this is the unit of Hertz, ($\frac{1}{\mathrm{s}} = \mathrm{Hz}$) which describes frequencies. But it cannot possibly be just a unit of frequency because the solution of the differential equation says so.

The solution of our equation is $x(t) = A\cos(\omega t)$. If ω is really a unit of Hertz, then ωt would have no units, which contradicts that the cosine function either accepts radians (rad), or degrees (deg). But these are units that are dimensionless, meaning that we can just multiply it into ω; therefore, ω must have the unit of $\mathrm{rad\,s^{-1}}$, which is the unit of angular frequencies.

The conclusion that ω is the angular frequencies can also be directly deducted from the solution of the equation. From both of eq. (5.117) and eq. (5.123), the time between each maximum and minima point is divided by ω. If ω is very high, then the time between the maximum and minimum is decreased. If it's low, then the time is increased. Therefore, ω is directly associated to the natural frequency of the system. The more ω is, the more frequent the oscillator oscillates.

Another way ω can be interpreted is by looking at the expression of ω itself: $\sqrt{\frac{k}{m}}$. The spring stiffness constant k is on the top, so it means the stiffer the spring, the higher omega is. A stiffer spring would allow the spring to pull the mass faster; thus, increasing the frequencies. The mass m is on the bottom: the larger the mass, the lower the omega. That just means a more massive block is harder for the spring to move; therefore, the mass will travel slower, decreasing the frequencies.

5.4.4 Ansatz method, and solution for other differential equations

There's actually a method that I've been hiding all along. In reality, the differential equation that describes the harmonic oscillator is "*too easy*" to be integrated by hand. Why integrate when you can just *guess* the solution?

The **ansatz method**[3] is a guessing method that's commonly taught in MIT (Massachusetts Institute of Technology). Let's go back to the harmonic oscillator differential equation

$$\frac{\mathrm{d}^2 x}{\mathrm{d}t^2} = -\omega^2 x. \tag{5.133}$$

It's asking

> What function, when differentiated twice is the negative of itself times a certain constant?

This should already ring a bell that it's obviously either the cosine or sine function. Which one is it?

To deal with this, notice that the differential equation is linear. That is, if $f(t)$ and $g(t)$ are solutions, then $Af(t) + Bg(t)$ where A and B are constants is a solution also. The proof is as follows. First, $f(t)$ and $g(t)$ are solutions mean

$$\frac{\mathrm{d}^2 f(t)}{\mathrm{d}t^2} + \omega^2 f(t) = 0 \quad \text{and} \quad \frac{\mathrm{d}^2 g(t)}{\mathrm{d}t^2} + \omega^2 g(t) = 0. \tag{5.134}$$

Thus,

$$\frac{\mathrm{d}^2\big(Af(t) + Bg(t)\big)}{\mathrm{d}t^2} + \omega^2\big(Af(t) + Bg(t)\big) = 0 \tag{5.135}$$

$$A\left(\frac{\mathrm{d}^2 f(t)}{\mathrm{d}t^2} + \omega^2 f(t)\right) + B\left(\frac{\mathrm{d}^2 g(t)}{\mathrm{d}t^2} + \omega^2 g(t)\right) = 0 \tag{5.136}$$

$$0 + 0 = 0, \tag{5.137}$$

which is true. Since both sines and cosines represents solution of the differential equation, the actual solution could contain both of them, so we shall guess that

$$x(t) = A\cos(\omega t) + B\sin(\omega t) \tag{5.138}$$

[3] Borrowed from the German noun, *ansetzen*, which means to estimate.

Indeed, it does satisfy the differential equation. Using the chain rule twice,

$$\frac{\mathrm{d}^2x(t)}{\mathrm{d}t^2} = \frac{\mathrm{d}^2}{\mathrm{d}t^2}\big(A\cos(\omega t) + B\sin(\omega t)\big) \tag{5.139}$$

$$= -\omega^2 A\cos(\omega t) - \omega^2 B\sin(\omega t) \tag{5.140}$$

$$= -\omega^2\big(A\cos(\omega t) + B\sin(\omega t)\big) \tag{5.141}$$

$$\frac{\mathrm{d}^2x(t)}{\mathrm{d}t^2} = -\omega^2 x(t). \tag{5.142}$$

Similarly, this is just a guess. To actually find what A and B is, we have to plug in the initial condition.

Here is where it gets quite interesting: by the simplicity of this method, we are not restricted to the initial condition that $x(t = 0) = A$ and $v(t = 0) = 0$. So, let's try solving for the case where $x(t = 0) = X$ and $v(t = 0) = V$.

Frankly, there's nothing restricting us in the first place, but I decided not to show it using the direct integration method because the process is too tedious. I prefer this method to analyze other conditions. So, let's start solving!

Let's use the ansatz $x(t) = A\cos(\omega t) + B\sin(\omega t)$. Now, let's substitute our initial condition into $x(t)$. At $t = 0$, $x(t = 0) = X$

$$x(0) = A\cos\big(\omega(0)\big) + B\sin\big(\omega(0)\big) \tag{5.143}$$

$$X = A\cancelto{1}{\cos(0)} + B\cancelto{0}{\sin(0)} \tag{5.144}$$

$$X = A \tag{5.145}$$

Thus, $x(t) = X\cos(\omega t) + B\sin(\omega t)$. To find B, we just have to differentiate this expression once and substitute in $v(t = 0) = V$

$$v(t = 0) = -\omega X\sin\big(\omega(0)\big) + \omega B\cos\big(\omega(0)\big) \tag{5.146}$$

$$\frac{V}{\omega} = -\cancelto{0}{X\sin(0)} + B\cancelto{1}{\cos(0)} \tag{5.147}$$

$$\frac{V}{\omega} = B \tag{5.148}$$

Therefore,

$$x(t) = X\cos(\omega t) + \frac{V}{\omega}\sin(\omega t). \tag{5.149}$$

In this form, it's quite hard to see what is the amplitude and the point of maximum and minimum velocity is. To make it easier for us to interpret, we can combine the cosine and sine term using the **harmonic addition theorem**

$$a\cos\theta + b\sin\theta = c\cos(\theta + \varphi) \tag{5.150}$$

where

$$c = \operatorname{sgn}(a)\sqrt{a^2+b^2} \quad \text{and} \quad \varphi = \arctan\left(-\frac{b}{a}\right). \tag{5.151}$$

Equation (5.149) then becomes

$$\sqrt{X^2 + \frac{V^2}{\omega^2}}\cos\left(\omega t + \arctan\left(-\frac{V}{\omega X}\right)\right) \tag{5.152}$$

In this form, it's easy to see that the amplitude: the point that the mass is furthest away from the spring, is $x = \sqrt{X^2 + \frac{V^2}{\omega^2}}$. But what is that $\arctan\left(-\frac{V}{\omega X}\right)$ doing there? What does it represent? We shall discuss this in detail in **??**

5.5 Trigonometric substitutions

As seen earlier, trigonometric substitution can be used in evaluating integrals which has terms of $\sqrt{A^2 - x^2}$ by substituting $x = \cos\theta$, and taking advantage of the Pythagorean identity. But it doesn't stop at just $\sin^2\theta + \cos^2\theta = 1$, we can use other Pythagorean identities to evaluate integrals which contains $\sqrt{A^2 + x^2}$ and $\sqrt{x^2 - A^2}$ also. We shall explore integral of these forms and solve some integrals in this section.

Honestly, when I was writing this section, I feel bored. But that picture when I first saw these integrals four years ago, caught me:

> *"How in the world is it possible? A simple substitution could do such wonderful things! It just destroyed this integrals involving square roots and turned it into a much more managable form. Amazing!"*

The magic is in the interplay between two seemingly unrelated things: square roots and trigonometric functions. So, I try to find examples that would hopefully let you appreciate these things more. I hope you find the excitement in it, and let's crack on.

Integrals containing $\sqrt{A^2 - x^2}$

Let the integral be described by

$$\int f\left(\sqrt{A^2 - x^2}, x\right) \mathrm{d}x. \tag{5.153}$$

where f is a function of $\sqrt{A^2 - x^2}$ and x. As said earlier, substitute $x = a \sin\theta$; thus,

$$\frac{\mathrm{d}x}{\mathrm{d}\theta} = \frac{\mathrm{d}}{\mathrm{d}\theta}(A \sin\theta) \tag{5.154}$$

$$\mathrm{d}x = A\cos(\theta)\,\mathrm{d}\theta. \tag{5.155}$$

Using $1 - \cos^2\theta = \sin^2\theta$, our integral becomes

$$A \int f\left(\sqrt{A^2 - A^2\cos^2\theta}, A\sin\theta\right)\cos\theta\,\mathrm{d}\theta$$

$$= A \int f\left(A\sqrt{1 - \cos^2\theta}, A\sin\theta\right)\cos\theta\,\mathrm{d}\theta \tag{5.156}$$

$$= A \int f\left(A\sqrt{\sin^2\theta}, A\sin\theta\right)\cos\theta\,\mathrm{d}\theta \tag{5.157}$$

$$= A \int f\left(A\sin\theta, \sin\theta\right)\cos\theta\,\mathrm{d}\theta \tag{5.158}$$

Now let's see some magic: two magic tricks with circles.

The next two examples evaluates the perimeter and area of a circle by using trigonometric substitution. The equation of a circle radius R centered at the origin is given by $x^2+y^2=R^2$, which is in an implicit form. To integrate this, we have to turn it into an explicit form

$$y = f(x) = \sqrt{R^2 - x^2}. \tag{5.159}$$

Since $f(x)$ contains a square root, the function can only output a positive value; therefore, it only represents half a circle. After calculating the perimeter and area covered by $f(x)$, we have to double that to get the actual perimeter and area of a circle.

Example 5.5.1: *Perimeter of a circle* This is an excuse for me to revisit the arc length formula in section 4.5. The perimeter of a circle is essentially double of the arc length of $\sqrt{R^2-x^2}$ from $-R$ to R which from eq. (4.69), is given by

$$\int_{-R}^{R} \sqrt{1 + \left(\frac{\mathrm{d}\sqrt{R^2-x^2}}{\mathrm{d}x}\right)^2}\,\mathrm{d}x. \tag{5.160}$$

The derivative inside can be evaluated using the chain rule:

$$\begin{aligned}
\frac{\mathrm{d}\sqrt{R^2-x^2}}{\mathrm{d}x} &= \frac{\mathrm{d}\sqrt{u}}{\mathrm{d}u} \times \frac{\mathrm{d}u}{\mathrm{d}x} && u = R^2 - x^2 && (5.161)\\
&= \frac{\mathrm{d}u^{\frac{1}{2}}}{\mathrm{d}u} \times \frac{\mathrm{d}R^2 - x^2}{\mathrm{d}x} && && (5.162)\\
&= \frac{1}{2\sqrt{u}}(-2x) && \text{Power rule} && (5.163)\\
&= -\frac{x}{\sqrt{R^2-x^2}} && u = R^2 - x^2 && (5.164)
\end{aligned}$$

Our integral then becomes

$$\int_{-R}^{R} \sqrt{1 + \left(-\frac{x}{\sqrt{R^2-x^2}}\right)^2}\,\mathrm{d}x = \int_{-R}^{R} \sqrt{1 + \frac{x^2}{R^2-x^2}}\,\mathrm{d}x \tag{5.165}$$

$$= \int_{-R}^{R} \sqrt{\frac{R^2 - x^2 + x^2}{R^2 - x^2}}\, \mathrm{d}x \tag{5.166}$$

$$= R \int_{-R}^{R} \frac{1}{\sqrt{R^2 - x^2}}\, \mathrm{d}x. \tag{5.167}$$

This form can be easily evaluated by substituting $x = R\sin\theta$, $\mathrm{d}x = R\cos\theta\,\mathrm{d}x$. The bounds are then changed by $\theta = \arcsin\left(\frac{x}{R}\right)$ to

$$\begin{aligned} x = -R \to \theta = \arcsin\left(\frac{-R}{R}\right) &= -\frac{\pi}{2}, \\ x = R \to \theta = \arcsin\left(\frac{R}{R}\right) &= \frac{\pi}{2}. \end{aligned} \tag{5.168}$$

Our integral then becomes

$$R \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{\sqrt{R^2 - R^2 \sin^2\theta}} R\cos\theta\,\mathrm{d}\theta$$

$$= R \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{R\sqrt{1 - \sin^2\theta}} R\cos\theta\,\mathrm{d}\theta \tag{5.169}$$

$$= R \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\cos\theta}{\cos\theta}\,\mathrm{d}\theta = R \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \mathrm{d}\theta \tag{5.170}$$

$$= R(\theta|_{-\frac{\pi}{2}}^{\frac{\pi}{2}} = R\left(\frac{\pi}{2} - \left(-\frac{\pi}{2}\right)\right) = R\pi; \tag{5.171}$$

therefore, the perimeter of a whole circle radius R is $2\pi R$.

Example 5.5.2: *Area of a circle* The area of a circle radius R is double the integral of $\sqrt{R^2 - x^2}$:

$$\int_{-R}^{R} \sqrt{R^2 - x^2}\,\mathrm{d}x. \tag{5.172}$$

Perform a trigonometric substitution similar to the example above: $x = R\sin\theta$, $\mathrm{d}x = R\cos\theta\,\mathrm{d}\theta$, $x = -R \to \theta = -\frac{\pi}{2}$, and $x = R \to \theta = \frac{\pi}{2}$.

Our integral then becomes

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sqrt{R^2 - R^2\sin^2\theta} R\cos\theta\,\mathrm{d}\theta = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} R^2\sqrt{1 - \sin^2\theta}\cos\theta\,\mathrm{d}\theta$$

$$= R^2 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^2\theta \, \mathrm{d}\theta. \tag{5.173}$$

To evaluate this, we have to use the trigonometric identity $\cos(2\theta) = 2\cos^2\theta - 1$. Rearranging the identity gives $\cos^2\theta = \frac{\cos(2\theta)+1}{2}$; therefore, our integral becomes

$$R^2 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\cos(2\theta)+1}{2} \, \mathrm{d}\theta = \frac{R^2}{2}\left[\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos(2\theta)\,\mathrm{d}\theta + \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \mathrm{d}\theta\right]. \tag{5.174}$$

The second integral simply evaluates to $\frac{\pi}{2} - \left(-\frac{\pi}{2}\right)$, which is just π. The first integral requires another substitution.

Integrals containing $\sqrt{x^2 - A^2}$

Let the integral be described by

$$\int f\left(\sqrt{x^2 - A^2}, x\right) \mathrm{d}x. \tag{5.175}$$

We want to take advantage of $\sec^2\theta - 1 = \tan^2\theta$, so substitute $x = A\sec\theta$. From table 5.1,

$$\frac{\mathrm{d}x}{\mathrm{d}\theta} = \frac{\mathrm{d}}{\mathrm{d}\theta}(A\sec\theta) \tag{5.176}$$

$$\mathrm{d}x = A\sec\theta\tan\theta\,\mathrm{d}\theta. \tag{5.177}$$

Our integral then becomes

$$\int f\left(\sqrt{A^2\sec^2\theta - A^2}, A\sec\theta\right) A\sec\theta\tan\theta\,\mathrm{d}\theta$$

$$= A\int f\left(A\sqrt{1-\sec^2\theta}, A\sec\theta\right)\sec\theta\tan\theta\,\mathrm{d}\theta \tag{5.178}$$

$$= A\int f\left(A\sqrt{\tan^2\theta}, A\sec\theta\right)\sec\theta\tan\theta\,\mathrm{d}\theta \tag{5.179}$$

$$= A\int f\left(A\tan\theta, A\sec\theta\right)\sec\theta\tan\theta\,\mathrm{d}\theta. \tag{5.180}$$

Integrals containing $\sqrt{A^2 + x^2}$

Let the integral be described by

$$\int f\left(\sqrt{A^2+x^2}, x\right) \mathrm{d}x. \tag{5.181}$$

We want to take advantage of $1 + \tan^2\theta = \sec^2\theta$, so substitute $x = A\tan\theta$. From table 5.1,

$$\frac{\mathrm{d}x}{\mathrm{d}\theta} = \frac{\mathrm{d}}{\mathrm{d}\theta}(A\tan\theta) \tag{5.182}$$

$$\mathrm{d}x = A\sec^2\theta. \tag{5.183}$$

Our integral then becomes

$$\begin{aligned}
&\int f\left(\sqrt{A^2+A^2\tan^2\theta}, A\tan\theta\right) A\sec^2\theta\,\mathrm{d}\theta \\
&= \int f\left(A\sqrt{1+\tan^2\theta}, A\tan\theta\right)\sec^2\theta\,\mathrm{d}\theta && (5.184)\\
&= \int f\left(A\sqrt{\sec^2\theta}, A\tan\theta\right)\sec^2\theta\,\mathrm{d}\theta && (5.185)\\
&= \int f\left(A\sec\theta, A\tan\theta\right)\sec^2\theta\,\mathrm{d}\theta. && (5.186)
\end{aligned}$$

5.6 Geometrical interpretation of trigonometric substitutions

CHAPTER 6

Merging calculus and dynamical systems

As said earlier, with the merge of calculus and trigonometry, all the cards are on the deck. And we can finally begin to analyze many interesting dynamical systems that is scattered all over the world.[1]

Since this is an exploratory chapter, I can't really provide a main *theme* for the chapter, except that it's a chapter that explores *differential equations*. So, I've decided to structure the chapter into examples and theory group. A section will start with an example of a dynamical system. Then, we'll try to describe the system using differential equations. Afterward, we'll solve the system, and analyze its physical and mathematical properties.

The chapter is planned out as follows:

1. Prelude: more advanced techniques of integration

[1]These dynamical systems are partially my excuses to teach various technique of integrations. So, be prepared for it.

2. Exponential growth and decay (Separated into two sections)
3. Logistic growths
4. Movement with drag forces
5. Realization of gravitational systems
6. Alternating currents

6.1 Prelude: more advanced techniques of integration

6.1.1 Rational quadratic integrals

There are many integrands of the form

$$\int \frac{Ax+B}{Cx^2+Dx+E}\,\mathrm{d}x \tag{6.1}$$

in this chapter. Normally I prefer to introduce them along the way, but it plays a very important role in this chapter that it deserves its own section.

Generally, the integral above can be simplified into a sum of integrals of the form

$$\int \frac{1}{x^2+bx+c}\,\mathrm{d}x, \tag{6.2}$$

which is way better. So, we'll deal with this reduced form instead. Because the denominator is quadratic, it's not a great idea to substitute $u = x^2 + bx + c$ straightaway, as $\mathrm{d}u = 2x + b$ but there's no x's in the numerator. It'd be better if we can turn the quadratic denominator into something else. From the section earlier, we know that trigonometric substitution can be used to integrate integrals of the form

$$\int \frac{1}{Ax^2+B}\,\mathrm{d}x. \tag{6.3}$$

It'd be very nice if we can remove the middle bx term. That's it! Completion of the square allows us to do just that!

$$x^2 + bx + c = (x)^2 + 2(x)\left(\frac{b}{2}\right) + \left(\left(\frac{b}{2}\right)^2 - \left(\frac{b}{2}\right)^2\right) + c \tag{6.4}$$

$$= \left(x^2 + 2x\frac{b}{2} + \frac{b^2}{4}\right) + \left(c - \frac{b^2}{4}\right) \tag{6.5}$$

$$= \left(x + \frac{b}{2}\right)^2 + \left(c - \frac{b^2}{4}\right) \tag{6.6}$$

Let's call $\frac{b}{2}$ as P and $\frac{b^2}{4}$ as Q^2 for shortness. Thus,

$$\int \frac{1}{x^2 + bx + c}\,\mathrm{d}x = \int \frac{1}{(x+P)^2 + Q^2}\,\mathrm{d}x \tag{6.7}$$

Then, we can do a simple substitution. Let $u = x + P$, then $\mathrm{d}u = \mathrm{d}x$; thus,

$$\int \frac{1}{(x+P)^2 + Q^2}\,\mathrm{d}x = \int \frac{1}{u^2 + Q^2}\,\mathrm{d}u \tag{6.8}$$

From here, there are actually two ways to integrate this: using the standard trigonometric substitution, or using partial fractions. Let's do both

Using trigonometric substitution

6.2 Exponential equations

6.2.1 Normal growth

A **normal growth** is an unconstrained growth: a type of growth that only happens when an organism has access to infinite resources. Such systems do not exist in reality, but there are some which can be approximated using this model, e.g., the population of bacteria in a Petri dish with plenty of food, the population of lilies in a well-regulated pond, or even the length of Jack's magic beanstalk. The differential equation is formed under the assumption that the growth rate is proportional to the current population size. Given a population size N which is a function of time t, and a constant α which controls the growth rate,

$$\frac{\mathrm{d}N(t)}{\mathrm{d}t} = \alpha N(t). \tag{6.9}$$

Given an initial condition $N(t=0) = n$, eq. (6.9) can be directly solved using separation of variables

$$\frac{\mathrm{d}N}{\mathrm{d}t} = \alpha N \tag{6.10}$$

$$\frac{1}{N}\,\mathrm{d}N = \alpha\,\mathrm{d}t \tag{6.11}$$

$$\int \frac{1}{N}\,\mathrm{d}N = \int \alpha\,\mathrm{d}t. \tag{6.12}$$

The L.H.S. is the integral of the reciprocal (section 3.7) which evaluates to $\ln(N)$, and the R.H.S. directly evaluates to αt by the power rule (section 3.3.2); therefore,

$$\ln(N) = \alpha t + C. \tag{6.13}$$

Substitute in the initial condition ($N(t=0) = n$) to find the integral constant C,

$$\ln(n) = \alpha(0) + C \tag{6.14}$$

$$\ln(n) = C; \tag{6.15}$$

therefore,

$$\ln(N) = \alpha t + \ln(n) \tag{6.16}$$

$$\ln(N) - \ln(n) = \alpha t \tag{6.17}$$

$$\ln\left(\frac{N}{n}\right) = \alpha t \tag{6.18}$$

$$N(t) = n\mathrm{e}^{\alpha t}. \tag{6.19}$$

This result is quite easy to interpret. The higher α is, the faster the growth. The more initial population there is, the steeper the growth is.

6.2.2 Decay systems: Newton's cooling law and radioactivity

6.2.3 Logistic growth: bacteria

The model of exponential functions in growth is ideal. Most of the time, growth can't be modeled using exponential functions, because the environment imposes a limit on how much population there can be. This limit can be from food shortage, lack of space, etc. Mathematically, this limit is called the **carrying capacity**. When the amount of living things reaches this carrying capacity, it stops to grow more, and the value stabilizes. How do we model these growth by using differential equations?

First, recall the normal growth equation (eq. (6.9), repeated here for ease of reading)

$$\frac{\mathrm{d}N}{\mathrm{d}t} = \alpha N$$

where N is the population, and α, the growth rate. The interpretation of this is: the growth rate is proportional to the amount of population that already exists in the system. If we want to make the growth rate slower when it's reaching the carrying capacity (ς), we can multiply in

$$1 - \frac{N}{\varsigma}. \tag{6.20}$$

This term represents how far away the system is from reaching the carrying capacity. If $N = \varsigma$, then $1 - \frac{N}{\varsigma} = 1 - \frac{\varsigma}{\varsigma} = 0$. If the system doesn't have any population yet, then $1 - \frac{N}{\varsigma} = 1 - \frac{0}{\varsigma} = 1$. By the condition that the population can't be more than the carrying capacity, $0 < N < \varsigma$; thus, eq. (6.20) is bounded between 0 and 1. Our equation then becomes

$$\frac{\mathrm{d}N}{\mathrm{d}t} = \alpha N\left(1 - \frac{N}{\varsigma}\right), \tag{6.21}$$

which is called the equation of **logistic growth**

Solving this equation is quite simple. First, expand the R.H.S. into

$$\frac{\mathrm{d}N}{\mathrm{d}t} = \frac{\varsigma N - N^2}{\varsigma} \tag{6.22}$$

and do a separation of variable

$$\frac{1}{\varsigma N - N^2}\,\mathrm{d}N = \varsigma\,\mathrm{d}t \tag{6.23}$$

$$\int \frac{1}{\varsigma N - N^2}\,\mathrm{d}N = \int \varsigma\,\mathrm{d}t \tag{6.24}$$

The R.H.S. directly evaluates to ςt. But the L.H.S. is harder. These quadratic integrands require the use of several techniques to solve. First, we have to rewrite the quadratic down below to be in the form $ax^2 + b$. From there, two possible approaches are trigonometric substitution or partial fraction decomposition. Since we've explored trigonometric substitution in the last chapter already, we'll do partial fraction decomposition in this chapter.

Completing the square

Firstly, notice that

$$\varsigma N - N^2 = -\Big((N)^2 - (2)(N)\Big(\frac{\varsigma}{2}\Big)\Big) \tag{6.25}$$

$$= -\left((N)^2 - (2)(N)\Big(\frac{\varsigma}{2}\Big) + \Big(\frac{\varsigma}{2}\Big)^2 - \Big(\frac{\varsigma}{2}\Big)^2\right) \tag{6.26}$$

$$= -\left(\Big(N - \frac{\varsigma}{2}\Big)^2 - \frac{\varsigma^2}{4}\right) \tag{6.27}$$

$$= \frac{\varsigma^2}{4} - \Big(N - \frac{\varsigma}{2}\Big)^2 \tag{6.28}$$

Partial fraction decomposition in integrals

6.3 Movement with drag forces

The second system that we're going to analyze is a system of a block moving with drag forces.

Drag is quite complex. It depends on many variables, e.g., air viscosity, air compressibility, object's shape, etc. So, there is no such thing as an "exact drag function." If there was, the amount of parameters you have to put in would probably be too much. Therefore, we assume that the drag function is continuous, and assume that it can be written in the terms of power series.

Most of the time, drag is a function of velocity. The faster the object is moving, the more drag. So we write

$$f_r(\mathbf{v}) = a_0 + a_1\mathbf{v} + a_2\mathbf{v}^2 + a_3\mathbf{v}^3 + \dots, \tag{6.29}$$

where f_r is the drag force. In a simplified model, we only consider the first three terms,

$$f_r(\mathbf{v}) \approx a_0 + a_1\mathbf{v} + a_2\mathbf{v}^2. \tag{6.30}$$

Firstly, a_0 has to be zero. Otherwise, the object will just accelerate constantly, which is no good. Thus, there can only be $a_1\mathbf{v} + a_2\mathbf{v}^2$ leftover, and Newton's second law reads

$$m\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\mathrm{d}x}{\mathrm{d}t}\right) = a_1\frac{\mathrm{d}x}{\mathrm{d}t} + a_2\left(\frac{\mathrm{d}x}{\mathrm{d}t}\right)^2.$$

Using fluxion notation,

$$\begin{aligned} m\frac{\mathrm{d}\dot{x}}{\mathrm{d}t} &= a_1\dot{x} + a_2\dot{x}^2 \\ \frac{\mathrm{d}\dot{x}}{\mathrm{d}t} &= \frac{a_1}{m}\dot{x} + \frac{a_2}{m}\dot{x}^2 \end{aligned}$$

For convenience, let $a_1 / m = P$ and $a_2 / m = Q$. The equation reads

$$\frac{\mathrm{d}\dot{x}}{\mathrm{d}t} = P\dot{x} + Q\dot{x}^2. \tag{6.31}$$

Obviously, both P and Q must be negative; otherwise, the object will just accelerate further if velocity increases. To pick out this negative sign, let's set $P = -p^2$ and $Q = -q^2$, where $p, q \in \mathbb{R}^+$. Therefore, our differential equation becomes

$$\frac{\mathrm{d}\dot{x}}{\mathrm{d}t} = -q^2\dot{x}^2 - p^2\dot{x}. \tag{6.32}$$

Set the initial condition, and we're ready to solve!

Let's set $x(t = 0) = 0$ and $v(t = 0) = V$. A simple separation of variables would do just fine

$$\frac{1}{-q^2\dot{x}^2 - p^2\dot{x}}\,\mathrm{d}x = \mathrm{d}t \tag{6.33}$$

$$\int \frac{1}{q^2\dot{x}^2 + p^2\dot{x}}\,\mathrm{d}x = -t + C. \tag{6.34}$$

This integral is quite a challenge, requiring the use of several techniques. First, we have to rewrite the the quadratic expression in the form $ax^2 + b$. From there, two possible approaches are trigonometric substitution or partial fraction decomposition. But since we've already explored trigonometric substitution in the previous chapter, I'll focus on using partial fraction decomposition for this problem. After this integral is solved, we're left with another differential equation of $\dot{x}$ and t, then we have to solve for x in terms of t. So there are actually two integrals that we have to do. It's going to be a long ride, so I shall break this problem into two parts.

6.3.1 First integral: completing the square and partial fraction decomposition

As the name suggests, we complete the square, then use a simple change of variables to solve the integral.

$$\int \frac{1}{q^2\dot{x}^2 + p^2\dot{x}} = \int \frac{1}{(q\dot{x})^2 + 2(\frac{p^2}{2q})(q\dot{x})}\,\mathrm{d}x \tag{6.35}$$

$$= \int \frac{1}{(q\dot{x})^2 + 2(\frac{p^2}{2q})(q\dot{x}) + (\frac{p^2}{2q})^2 - (\frac{p^2}{2q})^2}\,\mathrm{d}\dot{x} \tag{6.36}$$

$$= \int \frac{1}{(q\dot{x} + \frac{p^2}{2q})^2 - (\frac{p^2}{2q})^2} \,\mathrm{d}\dot{x} \tag{6.37}$$

Substitute $u = q\dot{x} + \frac{p^2}{2q}$, then

$$\frac{\mathrm{d}u}{\mathrm{d}\dot{x}} = \frac{\mathrm{d}}{\mathrm{d}\dot{x}}\left(q\dot{x} + \frac{p^2}{2q}\right) \tag{6.38}$$

$$\frac{\mathrm{d}u}{\mathrm{d}\dot{x}} = q \tag{6.39}$$

$$\frac{1}{q}\,\mathrm{d}u = \mathrm{d}\dot{x}. \tag{6.40}$$

Therefore, from eq. (6.37),

$$\int \frac{1}{q^2\dot{x}^2 + p^2\dot{x}} \,\mathrm{d}\dot{x} = \frac{1}{q}\int \frac{1}{u^2 - (\frac{p^2}{2q})^2} \,\mathrm{d}u \tag{6.41}$$

To go from here, we need to do partial fraction decomposition. I believe that you have learnt this in Algebra II already. If not, appendix C is devoted to that. Notice that the denominator can be decomposed into a product of linear factors

$$u^2 - \left(\frac{p^2}{2q}\right)^2 = \left(u + \frac{p^2}{2q}\right)\left(u - \frac{p^2}{2q}\right); \tag{6.42}$$

therefore, we set

$$\frac{1}{u^2 - (\frac{p^2}{2q})} = \frac{\alpha}{u + \frac{p^2}{2q}} + \frac{\beta}{u - \frac{p^2}{2q}} \tag{6.43}$$

and solve for α and β. First we multiply out the terms

$$\frac{1}{u^2 - (\frac{p^2}{2q})} = \frac{\alpha(u + \frac{p^2}{2q}) + \beta(u - \frac{p^2}{2q})}{(u - \frac{p^2}{2q})(u + \frac{p^2}{2q})} \tag{6.44}$$

$$1 = \alpha\left(u + \frac{p^2}{2q}\right) + \beta\left(u - \frac{p^2}{2q}\right) \tag{6.45}$$

$$(0)u + 1 = u\big(\alpha + \beta\big) + \frac{p^2}{2q}(\alpha - \beta). \tag{6.46}$$

By matching the leading coefficient, we get

$$\alpha + \beta = 0, \tag{6.47}$$

$$\frac{p^2}{2q}(\alpha - \beta) = 1. \tag{6.48}$$

Solving this is quite simple. Take eq. (6.48) and move $\frac{p^2}{2q}$ to the R.H.S. to get

$$\alpha - \beta = \frac{2q}{p^2}. \tag{6.49}$$

Then, subtract eq. (6.49) from eq. (6.47)

$$\alpha + \beta - \alpha + \beta = -\frac{2q}{p^2} \tag{6.50}$$

$$\beta = -\frac{q}{p^2}. \tag{6.51}$$

By eq. (6.47),

$$\alpha = \frac{q}{p^2}; \tag{6.52}$$

thus,

$$\frac{1}{u^2 - (\frac{p^2}{2q})^2} = \frac{q}{p^2}\frac{1}{u + \frac{p^2}{2q}} - \frac{q}{p^2}\frac{1}{u - \frac{p^2}{2q}} \tag{6.53}$$

$$= \frac{q}{p^2}\left(\frac{1}{u - \frac{p^2}{2q}} - \frac{1}{u + \frac{p^2}{2q}}\right). \tag{6.54}$$

Equation (6.41) then becomes

$$\frac{1}{q}\left(\frac{q}{p^2}\int \frac{1}{u - \frac{p^2}{2q}} - \frac{1}{u + \frac{p^2}{2q}}\,\mathrm{d}u\right) = \frac{1}{p^2}\int \frac{1}{u - \frac{p^2}{2q}} - \frac{1}{u + \frac{p^2}{2q}}\,\mathrm{d}u \tag{6.55}$$

$$= \frac{1}{p^2}\left(\int \frac{1}{u - \frac{p^2}{2q}}\,\mathrm{d}u - \int \frac{1}{u + \frac{p^2}{2q}}\,\mathrm{d}u\right)$$

Each of these integrals are in the form

$$\int \frac{1}{u + a}\,\mathrm{d}u, \tag{6.56}$$

which can be solved using a simple substitution. Let $w = u + a$, then $\mathrm{d}w = \mathrm{d}u$; thus,

$$\int \frac{1}{u + a}\,\mathrm{d}u = \int \frac{1}{w}\,\mathrm{d}w \tag{6.57}$$

Recall that this is just the integral of the reciprocal, discussed in sections 3.5 and 3.7. It is evaluated to $\ln(w)$. Because $w = u + a$,

$$\int \frac{1}{u+a}\,\mathrm{d}u = \ln(u+a); \tag{6.58}$$

thus,

$$\frac{1}{p^2}\left(\int \frac{1}{u-\frac{p^2}{2q}}\,\mathrm{d}u - \int \frac{1}{u+\frac{p^2}{2q}}\,\mathrm{d}u\right) = \frac{1}{p^2}\left(\ln\left(u-\frac{p^2}{2q}\right) - \ln\left(u+\frac{p^2}{2q}\right)\right). \tag{6.59}$$

Because $u = q\dot{x} + \frac{p^2}{2q}$, the integral evaluates to

$$\frac{1}{p^2}\left(\ln(q\dot{x}) - \ln\left(q\dot{x}+\frac{p^2}{q}\right)\right) \tag{6.60}$$

Equation (6.34) then becomes

$$-t + C = \frac{1}{p^2}\left(\ln(q\dot{x}) - \ln\left(q\dot{x}+\frac{p^2}{q}\right)\right). \tag{6.61}$$

Recall that our initial condition is $x(t=0) = 0$, and $v(t=0) = V$ ($\dot{x} = v$ by Newton's fluxion notation). Substitute those in to find C.

$$\begin{aligned}
-0 + C &= \frac{1}{p^2}\left(\ln(qV) - \ln\left(qV+\frac{p^2}{q}\right)\right) && (6.62)\\
C &= \frac{1}{p^2}\left(\ln(qV) - \ln\left(\frac{q^2V+p^2}{q}\right)\right) && (6.63)\\
&= \frac{1}{p^2}\left(\ln(q) + \ln(V) - \ln(q^2V+p^2) + \ln(q)\right) && (6.64)\\
&= \frac{1}{p^2}\left(\ln(V) - \ln(q^2V+p^2) + 2\ln(q)\right) && (6.65)
\end{aligned}$$

Substitute this C back into eq. (6.61) to get

$$\begin{aligned}
-t + \frac{1}{p^2}\left(\ln(V) - \ln(q^2V+p^2) + 2\ln(q)\right) &= \frac{1}{p^2}\left(\ln(q\dot{x}) - \ln\left(q\dot{x}+\frac{p^2}{q}\right)\right)\\
-p^2t + \ln(V) - \ln(q^2V+p^2) + 2\ln(q) &= \ln(q\dot{x}) - \ln\left(\frac{q^2\dot{x}+p^2}{q}\right)\\
-p^2t + \ln\left(\frac{V}{q^2V+p^2}\right) + 2\ln(q) &= \ln(q) + \ln(\dot{x})\\
&\quad - \ln(q^2\dot{x}+p^2) - \ln(q)
\end{aligned}$$

$$-p^2t + \ln\left(\frac{V}{q^2V + p^2}\right) = \ln(\dot{x}) - \ln(q^2\dot{x} + p^2) \tag{6.66}$$

$$-p^2t + \ln\left(\frac{V}{q^2V + p^2}\right) = \ln\left(\frac{\dot{x}}{q^2\dot{x} + p^2}\right). \tag{6.67}$$

For simplicity's sake, let $c \equiv \ln\left(\frac{V}{q^2V+p^2}\right)$. Our equation then becomes

$$-p^2t + c = \ln\left(\frac{\dot{x}}{q^2\dot{x} + p^2}\right). \tag{6.68}$$

6.3.2 Second integral: a simple substitution after the storm

I shall remind you again of what we're doing, because it's very easy to get lost. We are trying to solve the equation that describes a motion in one dimension with drag. Starting with the Newton's equation, we've isolated the variables and integrated it using the technique of completing the square and partial fractions. We're left with a first order differential equation (eq. (6.68)) which contains $\dot{x}$ and t. The next step that we're going to do is to turn $\dot{x}$ into $\frac{\mathrm{d}x}{\mathrm{d}t}$ and solve for x in terms of t.

Isolation of variables can be done by raising both sides to the power of e.

$$\mathrm{e}^{-p^2t+c} = \frac{\dot{x}}{q^2\dot{x} + p^2}. \tag{6.69}$$

Currently, $\dot{x}$ is on both the numerator and the denominator of the R.H.S., but preferrably we want it to be isolated either on the numerator only, or the denominator only. Thanksfully, there's a way to do just that: multiply the whole equation by q^2 to get

$$q^2\mathrm{e}^{-p^2t+c} = \frac{q^2\dot{x}}{q^2\dot{x} + p^2}, \tag{6.70}$$

and add $p^2 - p^2$ (zero) on the numerator

$$q^2\mathrm{e}^{-p^2t+c} = \frac{(q^2\dot{x} + p^2) - p^2}{q^2\dot{x} + p^2} \tag{6.71}$$

$$q^2 \mathrm{e}^{-p^2 t + c} = 1 - \frac{p^2}{q^2 \dot{x} + p^2}. \tag{6.72}$$

Some simple algebra will do it from here

$$1 - q^2 \mathrm{e}^{-p^2 t + c} = \frac{p^2}{q^2 \dot{x} + p^2} \tag{6.73}$$

$$q^2 \dot{x} + p^2 = \frac{p^2}{1 - q^2 \mathrm{e}^{-p^2 t + c}} \tag{6.74}$$

$$q^2 \frac{\mathrm{d}x}{\mathrm{d}t} = \frac{p^2}{1 - q^2 \mathrm{e}^{-p^2 t + c}} - p^2 \tag{6.75}$$

$$\int \mathrm{d}x = \frac{p^2}{q^2} \int \frac{1}{1 - q^2 \mathrm{e}^{-p^2 t + c}} \, \mathrm{d}t. \tag{6.76}$$

Don't freak out just yet. This integral is actually simpler than you think. It requires just a little substitution before the solution just pops right out of the page. Funnily enough, we can just substitute $u = q^2 \mathrm{e}^{-p^2 t + c}$, and we'll be done.

$$u = q^2 \mathrm{e}^{-p^2 t + c} \tag{6.77}$$

$$\frac{\mathrm{d}u}{\mathrm{d}t} = \frac{\mathrm{d}\mathrm{e}^{-p^2 t + c}}{\mathrm{d}t} \tag{6.78}$$

Using the chain rule, set $w = -p^2 t + c$

$$= \frac{\mathrm{d}\mathrm{e}^{w}}{\mathrm{d}w} \times \frac{\mathrm{d}w}{\mathrm{d}t} \tag{6.79}$$

$$\frac{\mathrm{d}u}{\mathrm{d}t} = -p^2 \mathrm{e}^{-p^2 t + c} \tag{6.80}$$

$$\mathrm{d}u = -p^2 \mathrm{e}^{-p^2 t + c} \, \mathrm{d}t. \tag{6.81}$$

6.3.3 Physical aspects of motion with drag

Motion with just linear drag

You don't really see linear drag in real life. It's mostly drag in moving liquid, e.g., a fish swimming in the water. Equation (6.32) simplifies

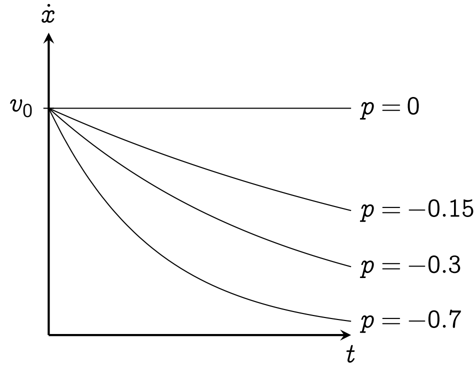


FIG. 6.1 | PLOT OF EQ. (6.83) WITH VARIOUS p.

to

$$\frac{\mathrm{d}\dot{x}}{\mathrm{d}t} = p\dot{x}.$$

I shall set v_0 as the initial velocity and x_0 as the initial position. There are two methods of solving this. First, by separating variables using analytical methods.

$$\int \frac{1}{\dot{x}}\,\mathrm{d}\dot{x} = p\int \mathrm{d}t$$

$$\ln(\dot{x}) + C = pt. \tag{6.82}$$

To find out what this integration constant should be, we have to use the initial condition $t = 0 \implies \dot{x} = v_0$.

$$\ln(v_0) + C = 0$$

$$C = -\ln(v_0).$$

Equation (6.82) then becomes

$$\ln(\dot{x}) - \ln(v_0) = pt$$

$$\dot{x} = v_0 \mathrm{e}^{pt}. \tag{6.83}$$

Figure 6.1 plots eq. (6.83) with various p. Notice that when $p = 0$, eq. (6.83) is just a straight line $\dot{x} = v_0$. The more negative p is, the faster it slows down, as illustrated. That is why sometimes, we call p the **dampening factor**. Also, v_0 here just scales the graph in the $\dot{x}$ direction.

To find $x(t)$, we rewrite eq. (6.83) as

$$\frac{\mathrm{d}x}{\mathrm{d}t} = v_0 \mathrm{e}^{pt}.$$

Then,

$$\mathrm{d}x = v_0 \mathrm{e}^{pt}\,\mathrm{d}t \tag{6.84}$$

$$x + C_1 = v_0 \int \mathrm{e}^{pt}\,\mathrm{d}t. \tag{6.85}$$

Here, I shall introduce an integration technique called **change of variables**, commonly known as u-substitution. We'll formally come back to this topic later in section 7.1. Basically, it's a way to convert integrals that we don't recognize into an easier integral. It's better if I just show the examples. We don't know the antiderivative of e^{pt} in eq. (6.85), however we know the antiderivative of e^{u}. So let's convert e^{pt} into that form. By letting a dummy variable $u = pt - v_0$, we have to convert $\mathrm{d}u$ into $\mathrm{d}t$ aswell.

$$\begin{aligned} u &= pt - v_0 \\ \frac{\mathrm{d}u}{\mathrm{d}t} &= \frac{\mathrm{d}}{\mathrm{d}t}(pt - v_0) \\ \frac{1}{p}\,\mathrm{d}u &= \mathrm{d}t. \end{aligned}$$

Then, substituting $\mathrm{d}t = \frac{1}{p}\,\mathrm{d}u$ and $u = pt$ into eq. (6.85), it reads

$$x + C_1 = v_0 \int \mathrm{e}^{u}\left(\frac{1}{p}\,\mathrm{d}u\right)$$

$$x + C_1 = \frac{v_0}{p}\mathrm{e}^{pt}. \tag{6.86}$$

We also have to take care of the C_1. Using $t = 0 \implies x = x_0$,

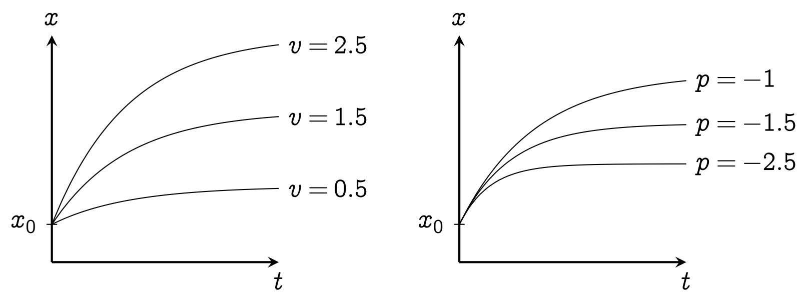
$$x_0 + C_1 = \frac{v_0}{p}\mathrm{e}^{p\cdot(0)}$$

$$C_1 = \frac{v_0}{p} - x_0.$$

Then, plug it into eq. (6.86). The equation becomes

$$\begin{aligned} x + \frac{v_0}{p} - x_0 &= \frac{v_0}{p}(\mathrm{e}^{pt}) \\ x &= x_0 + \frac{v_0}{p}(\mathrm{e}^{pt} - 1). \end{aligned} \tag{6.87}$$

I've plotted eq. (6.87) with varying v in fig. 6.2a and varying p in fig. 6.2b. The graph should match with what you expect intuitively: higher v will get you further, and the less the drag, the further you'll get.



(A) WITH VARYING v, SETTING $p = -1$ **(B)** WITH VARYING p, SETTING $v = 2$

FIG. 6.2 | PLOT OF EQ. (6.87), SETTING $x_0 = 0.5$.

Notice, this motion has a clear upper limit. *If* $p \in \mathbb{R}^-$, $\lim_{t\to\infty} e^{pt} = 0$. Taking the limit as $t \to 0$ on both sides of eq. (6.87), we get

$$\begin{aligned} \lim_{t\to\infty} x(t) &= \lim_{t\to\infty}\left(x_0 + \frac{v_0}{p}(e^{pt} - 1)\right). \\ &= x_0 + \frac{v_0}{p}\cancelto{0}{\lim_{t\to 0}(e^{pt})} - \frac{v_0}{p} = x_0 - \frac{v_0}{p}, \end{aligned}$$

which when $p \in \mathbb{R}^-$ and $v_0 \in \mathbb{R}^+$, the motion procedes forward, then gradually slows down and stops at $x_0 - \frac{v_0}{p}$ which is more than x_0.

Motion with just quadratic drag

This equation is even easier than linear drag, so I'd leave out some steps. **??** simplifies to

$$
\begin{aligned}
\frac{\mathrm{d}\dot{x}}{\mathrm{d}t} &= q\dot{x}^2 \\
\int \frac{1}{\dot{x}^2}\,\mathrm{d}\dot{x} &= q\int \mathrm{d}t^2 \\
-\frac{1}{\dot{x}} + C &= qt.
\end{aligned}
$$

Taking care of C: use $t = 0 \implies \dot{x} = v_0$.

$$
\begin{aligned}
-\frac{1}{v_0} + C &= q \cdot 0 \\
C &= \frac{1}{v_0}.
\end{aligned}
$$

Therefore,

$$
\begin{aligned}
-\frac{1}{\dot{x}} + \frac{1}{v_0} &= qt \\
\frac{\mathrm{d}t}{\mathrm{d}x} &= \frac{1 - qv_0 t}{v_0} \\
x &= v_0 \int \frac{1}{1 - qv_0 t}\,\mathrm{d}t.
\end{aligned} \tag{6.88}
$$

The structure of the integral is similar to $1/t$. Therefore, let $u = 1 - qv_0 t$. Then,

$$
\begin{aligned}
\frac{\mathrm{d}u}{\mathrm{d}u} &= \mathrm{d}u(1 - qv_0 t) \\
1 &= -qv_0 \frac{\mathrm{d}t}{\mathrm{d}u} \\
\mathrm{d}t &= -\frac{1}{qv_0}\,\mathrm{d}u.
\end{aligned}
$$

Substitute $u = 1 - qv_0 t$ and $\mathrm{d}t = -\frac{1}{qv_0}\,\mathrm{d}u$ into eq. (6.88):

$$
\begin{aligned}
x &= v_0 \int \frac{1}{u}\left(-\frac{1}{qv_0}\,\mathrm{d}u\right) \\
&= -\frac{1}{q}\int \frac{1}{u}\,\mathrm{d}u \\
x(t) &= -\frac{1}{q}\ln(1 - qv_0 t) + C_1.
\end{aligned} \tag{6.89}
$$

Taking care of C_1: use $t \to 0 \implies x(t) = x_0$.

$$
\begin{aligned}
x_0 &= -\frac{1}{q}\ln(1 - qv_0 \cdot (0)) + C_1 \\
x_0 &= C_1
\end{aligned}
$$

Plug this into eq. (6.89) to get the final answer:

$$
x(t) = -\frac{1}{q}\ln(1 - qv_0 t) + x_0.
$$

Surprisingly, quadratic drag does not have upper position bounds. A bit more thought would reveal that when $v < 0$, the quadratic drag $f_r(v) = a_2 v^2$ is smaller than $f_r(v) = a_1 v$. Thus, it should make sense that quadratic doesn't have bounds, but linear has an upper bound.

6.3.4 Damped harmonic motion

6.4 Realization of gravitational systems

I want to go back and revisit the falling ball that we've introduced in chapter 1, because it's a lie. The model $F = mg$ is a simplified one, and only works when the object isn't that high above the Earth's surface. The actual model of gravity that's given by Newton's law of universal gravitation:

$$F = -G\frac{Mm}{r^2}, \tag{6.90}$$

where G is the universal gravitational constant, M, the mass of the first object, and m, the mass of the second object. In this model, gravity is dependent on the position of the object. The further the object is, the weaker the gravity, given by the term r^2. The effect of the weakening gravity will be very prominent when the object is at a very great height. Newton's second law reads

$$F = -G\frac{Mm}{r^2} = m\ddot{r} \tag{6.91}$$

$$-G\frac{M}{r^2} = \ddot{r} \tag{6.92}$$

Various systems are encapsulated by this model, such as a rocket escaping Earth, or a meteor falling from great height (fig. 6.3). It's quite a general model, so let's solve eq. (6.92), then tinker around with initial conditions to see what happens.

Let's set $r(t=0) = R$, and $\dot{r}(t=0) = V$. The method of solving eq. (6.92) is very similar to the method of solving the harmonic oscillator (section 5.4). First, abuse the chain rule to reduce the second order derivative to a first order

$$\ddot{r} = \frac{\mathrm{d}^2 r}{\mathrm{d}t^2} = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\mathrm{d}r}{\mathrm{d}t}\right) = \frac{\mathrm{d}\dot{r}}{\mathrm{d}t} = \frac{\mathrm{d}\dot{r}}{\mathrm{d}r}\frac{\mathrm{d}r}{\mathrm{d}t} = \dot{r}\frac{\mathrm{d}\dot{r}}{\mathrm{d}r}; \tag{6.93}$$

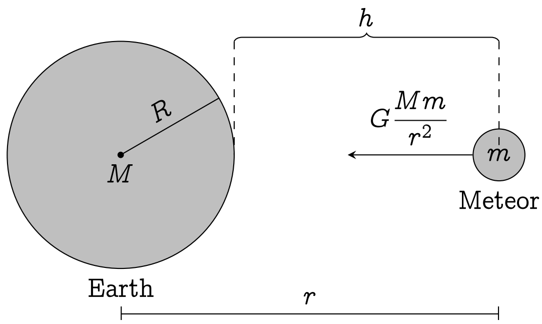


FIG. 6.3 | AN ILLUSTRATION OF A METEOR FALLING TO THE EARTH FROM HEIGHT h.

eq. (6.92) then turns into

$$-\frac{GM}{r^2} = \dot{r}\frac{\mathrm{d}\dot{r}}{\mathrm{d}r} \tag{6.94}$$

$$-GMr^{-2}\,\mathrm{d}r = \dot{r}\,\mathrm{d}r \tag{6.95}$$

$$-GM\int r^{-2}\,\mathrm{d}r = \int \dot{r}\,\mathrm{d}r \tag{6.96}$$

Using the reversed power rule (section 3.3.2),

$$-GM\frac{r^{-1}}{-1} = \frac{\dot{r}^2}{2} + C_1 \tag{6.97}$$

$$\frac{2GM}{r} - 2C_1 = \dot{r}^2. \tag{6.98}$$

To find C, substitute in the initial condition ($r(t=0) = R$, and $\dot{r}(t=0) = V$)

$$\frac{2GM}{R} - 2C_1 = V^2 \tag{6.99}$$

$$C_1 = \frac{GM}{R} - \frac{V^2}{2} \tag{6.100}$$

This is a bit long, so we'll just substitute C_1 in later. Now let's just continue from eq. (6.98). Take the square root of both sides and do another separation of variables.

$$\sqrt{\frac{2GM}{r} - 2C_1} = \dot{r} \tag{6.101}$$

$$\sqrt{\frac{2GM - 2C_1 r}{r}} = \frac{\mathrm{d}r}{\mathrm{d}t} \tag{6.102}$$

$$\int \mathrm{d}t = \int \sqrt{\frac{r}{2GM - 2C_1 r}}\,\mathrm{d}r \tag{6.103}$$

$$t + C_2 = \int \sqrt{\frac{r}{2GM - 2C_1 r}}\,\mathrm{d}r. \tag{6.104}$$

Now we have another integral to work with. It's quite a mess right now, but we can simplify it by substituting the leading constants with dummy variables. E.g., let $a = 2GM$ and $b = 2C_1$, our integral is in the form

$$\int \sqrt{\frac{r}{a - br}}\,\mathrm{d}r \tag{6.105}$$

Just a fair warning, this integral requires a bit of substitution, so try to stick around and not get lost here.

Firstly, having a linear term on the denominator of the square root is not favorable. So let's do a substitution to get rid of it. Let $u = a - br$; thus, $r = \frac{a-u}{b}$, and $-\frac{1}{b}\,\mathrm{d}u = \mathrm{d}r$. Thus, eq. (6.105) becomes

$$\int \frac{\sqrt{\frac{a-u}{b}}}{\sqrt{u}}\left(-\frac{1}{b}\,\mathrm{d}u\right) = -\frac{1}{b^{3/2}}\int \frac{\sqrt{a-u}}{\sqrt{u}}\,\mathrm{d}u \tag{6.106}$$

Another substitution would turn this integral into a more recognizable form. Let $\sqrt{u} = w$; thus, $u = w^2$, $a - u = a - w^2$, $\mathrm{d}u = 2w\,\mathrm{d}w$. Our integral becomes

$$-\frac{1}{b^{3/2}}\int \frac{\sqrt{a - w^2}}{w}(2w\,\mathrm{d}w) = -\frac{2}{b^{3/2}}\int \sqrt{a - w^2}\,\mathrm{d}w. \tag{6.107}$$

This integral can be easily solved by using a trigonometric substitution.

CHAPTER 7

Basic techniques of integration

Abstract

This chapter explores various techniques you can use to integrate stuffs. You still have to keep in mind that even though this chapter is very abstracted away, *integrals still have its geometric meaning: area under the graph.* This geometric interpretation is going to somewhat appear quite often throughout the chapter, so be aware of it.

Also, some integrals are definite: they have clear boundaries of integrations. You also have to keep track of what variable is bounded by that bound, otherwise you might end up evaluating incorrectly.

As an educator myself, I usually don't teach this topic but leave it as an exercise. Techniques of integrations come naturally with much experiences. Therefore, I list a table of integrations at the end of the chapter for the physicists out there.

7.1 Change of variables

Change of variables, or integration by substitution is a method to convert an unknown integrals into a more familiar one. Consider $\int x\mathrm{e}^(x^2)\,\mathrm{d}x$. This integral might seem daunting at first but, notice that the derivative of x^2 is exactly x. I shall introduce a new variable u where $u = x^2$. If we take the derivative w.r.t. x on both sides,

$$\frac{\mathrm{d}u}{\mathrm{d}x} = 2x$$

$$\mathrm{d}u = 2x\,\mathrm{d}x.$$

7.2 Integrals of inverse functions

7.3 Integrals of symmetric functions

7.3.1 Even and odd functions

7.3.2 Functions with axial and spherical symmetry

7.4 Integration by part

7.5 Recurrence relations and reduction formulas

7.6 Working with complex numbers and exponentials

7.7 Trigonometric substitution

7.8 Rational functions

7.8.1 Partial fraction decomposition

7.8.2 Quadratic under radicals

7.8.3 Ostrogradski method

7.9 Feynman's trick for integration

7.10 Polar integration

7.11 Gaussian integral

7.12 Cauchy's formula for repeated integrations

7.13 Numerical integration

CHAPTER 8

Advanced techniques of integration

Abstract

This is the chapter that explores more advanced techniques of integrations. You could go on with your life skipping this chapter and it'd still be fine. However, for the curious, there are some very interesting maths in here, so be sure to check this chapter out before going on to the next part of the book.

8.1 Rational functions

8.1.1 Quadratics denominator

An integral in the form

$$\int \frac{1}{ax^2 + bx + c}\,\mathrm{d}x \tag{8.1}$$

can be evaluated by completing the squares:

$$ax^2 + bx + c = a\left(x + \frac{b}{2a}\right)^2 + \left(c - \frac{b^2}{4a}\right) \tag{8.2}$$

8.1.2 Denominator quadratics under radicals

8.2 Mathematical interlude: trigonometric substitution

The trigonometric substitution technique earlier works for all the Pythagorean identity. The most used ones being

$$1 - \cos^2\theta = \sin^2\theta, \quad 1 + \tan^2\theta = \sec^2\theta \quad \text{and} \quad 1 - \sec^2\theta = \tan^2\theta.$$

They're used to take terms outside the square root, whether it'd be $\sqrt{a^2 - x^2}$, $\sqrt{a^2 + x^2}$, or $\sqrt{x^2 - a^2}$. Now let's go through these one by one.

Integrals with $\sqrt{a^2 - x^2}$: Let $x = a\sin\theta$; therefore, $\theta = \arcsin\left(\frac{x}{a}\right)$, and

$$\sqrt{a^2 - x^2} = a\sqrt{1 - \sin^2\theta} = a\sqrt{\cos^2\theta} = a\cos\theta. \tag{8.3}$$

The differential element $\mathrm{d}x$ transforms into

$$\frac{\mathrm{d}x}{\mathrm{d}\theta} = \frac{\mathrm{d}}{\mathrm{d}\theta}\big(a\sin\theta\big) \tag{8.4}$$

$$\mathrm{d}x = a\cos\theta\,\mathrm{d}\theta. \tag{8.5}$$

Example 8.2.1: $\int \frac{\sqrt{4-x^2}}{x^2}\,\mathrm{d}x$ This integral has a square root of the form $\sqrt{a^2 - x^2}$ where $a = 2$; thus, substitute in $x = 2\sin\theta$. Then,

$$\frac{\mathrm{d}x}{\mathrm{d}\theta} = \frac{\mathrm{d}}{\mathrm{d}\theta}\big(2\sin\theta\big) \tag{8.6}$$

$$\mathrm{d}x = 2\cos\theta\,\mathrm{d}\theta \tag{8.7}$$

The integral then becomes

$$\int \frac{\sqrt{4-4\cos^2\theta}}{4\sin^2\theta} 2\cos\theta \, \mathrm{d}\theta = \int \frac{2\sqrt{1-\cos^2\theta}}{4\sin^2\theta} 2\cos\theta \, \mathrm{d}\theta \tag{8.8}$$

$$= \int \frac{2\sin^2\theta}{4\sin^2\theta} 2\cos\theta \, \mathrm{d}\theta \tag{8.9}$$

$$= \int \cos\theta \, \mathrm{d}\theta = \sin\theta \tag{8.10}$$

Since $x = 2\sin\theta$, $\sin\theta = \frac{x}{2}$, our integral evaluates to $\frac{x}{2} + C$.

Integrals with $\sqrt{a^2+x^2}$: Let $x = a\tan\theta$; therefore, $\theta = \arctan\left(\frac{x}{a}\right)$, and

$$\sqrt{a^2+x^2} = a\sqrt{1+\tan^2\theta} = a\sqrt{\sec^2\theta} = a\sec\theta. \tag{8.11}$$

By **??**, the differential element $\mathrm{d}x$ transforms into

$$\frac{\mathrm{d}x}{\mathrm{d}\theta} = \frac{\mathrm{d}}{\mathrm{d}\theta}\left(a\sec\theta\right) \tag{8.12}$$

$$\mathrm{d}x = a\sec\theta\tan\theta. \tag{8.13}$$

8.2.1 Feynman's trick for integration

Formerly known as Leibniz's rule but popularized by Richard Feynman, is a very powerful integral technique that allows you to differentiate the function in the integral sign. Sadly, it only works for definite integrals. However, it's still a very nice trick to have just in case.

The idea is: integrals and derivatives can be swapped in

8.3 Integrals of other trigonometric functions

Part II

Analyzing differential equation

Part III

Signal analysis

Book II

Real analysis

CHAPTER 9

Invitation to real analysis

This part of the book is a bit out of touch with the other chapters. I named this whole series "Compact Calculus - Calculus for the practitioners," because I envision it to be a gentle and compact introduction to calculus that is used by most practitioners. One could argue that real analysis isn't for the practitioners, as it is considered to be one of the hardest pure mathematics course that an undergraduate can take. It's dedicated to rigorously analyzing the behaviors of the reals, redefining many of the hand-wavy definitions that we've used, and proving even the most trivial properties. Some proofs are so abstract that most have trouble wrapping their heads around them. So, why did I decide to write about real analysis?

After reading many analysis book[1], I've developed a deeper appreciation for the nature of mathematics. It's remarkable that from just a few axioms, mathematics has managed to prosper into the vast interconnecting webs of theories that we know today. So, I'd like to take my

[1] Even though I don't even have the point of doing so; I do physics, not mathematics.

appreciation and write them down on a book, where those who are interested can also appreciate the foundations of the reals like I did.

Part IV

Logic

CHAPTER 10

Mathematical logic

CHAPTER 11

Set theory

11.1 Sets and its definition

11.2 Operation on sets

Part V

Number systems and their arithmetrization

CHAPTER 12

The foundations: natural numbers

Since this book takes on a different approach to the Compact Basic Mathematics Book, I shall not go over how naturals are used, how the number line works, how objects can be represented as naturals, etc. They should be trivial for the students reading real analysis. So, I'd like to shift our attention to an arguably more important concept: the axiomatic construction ts of natural numbers.

One of the most crucial concept in this chapter is the induction axiom (axiom 12.1.6 and theorem 12.1.7), which will allow us to prove the majority of the results stated in this chapter. At the end, induction will lead our way to construct a proper mathematical system.

Most of the things stated here are trivial, but is crucial for maturing the skills of writing a good mathematical proof. Without the intention of developing this skill, this chapter is merely just a brain-decoration: practically useless to read. It prepares you for the later chapters which are very proof-heavy. But due to my style of writing, I can't, for the life of me, leave proofs of theorems as exercises. So, I'd have to excuse the reader here to refrain oneself from looking at the proofs before trying to write down one themselves.

12.1 Axiomatic formulation of natural numbers

The big question here is: *what does it mean to be a number*? Let's say if there's an alien species coming to us. How would you explain the concept of a number to them? You might point to your finger and say "This is the number one." The alien then says: "Ah, that finger is the number one". Clearly, this is incorrect. There's also other objects which can represent the number one. So you might point to your head and say that "I have only one head!" The alien then says: "Oh, the number one could be both your finger or your head." Again, the alien still misses the essence of what a number truly means.

The challenge arises because numbers are abstract concepts that aren't tied to any physical objects. There is no "the number one" in this world. So, it's practically impossible to tell the alien what a number is using a real world object. Instead, one must rely on abstractions, using purely mathematical concepts to tell the alien what a number actually is. Here, we shall only allow ourselves to use the following:

1. Logic
2. Sets and sets operations
3. Equality
4. Functions and mappings

We first start by defining what a set of naturals are:

Definition 12.1.1: *The set of natural numbers*

Let $\mathbb{N}$ be the set that contains all natural numbers.

There are many accepted construction of the natural numbers. But, the most well-accepted one is Peano's axioms. The first of which states that

Axiom 12.1.2: *Existence of zero as a natural number*

The number 0 is a natural number, i.e., $0 \in \mathbb{N}$

Then, we define a successor function $S(n)$, which takes in a natural number n and outputs the natural number that succeeds it. E.g., the successor of 0 is 1, and the successor of 1 is 2, etc. The successor function shall behave as follows:

Axiom 12.1.3: *The natural number is closed under the successor*

$$\forall(n \in \mathbb{N}) : S(n) \in \mathbb{N}.$$

Axiom 12.1.4: *The successor function is an injection*

$$\forall(n, m \in \mathbb{N}) : \big(S(n) = S(m)\big) \implies (n = m). \tag{12.1}$$

Axiom 12.1.5: *Zero is not the successor of any numbers*

$$\forall(n \in \mathbb{N}) : S(n) \neq 0. \tag{12.2}$$

Now let's see how these axioms create the natural numbers as we know it. Firstly, axiom 12.1.3 says that for all natural numbers, there is always a *next one*, which is given by the successor. Using axiom 12.1.2, we can directly form a chain of naturals from zero using the successor function:

$$0, S(0), S(S(0)), S(S(S(0))), \dots, \tag{12.3}$$

and by axiom 12.1.3, we can guarantee that all the elements in this chain will be a natural number. To make sure that the next natural number is different from the current natural number, axiom 12.1.4 is imposed, so cases like

$$S(n) = n \tag{12.4}$$

does not happen. Finally, we impose that zero is not the successor of anything by axiom 12.1.5

Traditionally, we assign names and symbols to each of the successor of zero:

$$\begin{aligned} S(0) &\equiv 1 \\ S(S(0)) = S(1) &\equiv 2 \\ S(S(S(0))) = S(2) &\equiv 3 \\ S(S(S(S(0)))) = S(3) &\equiv 4 \quad \dots \end{aligned} \tag{12.5}$$

and just like that, we have constructed the naturals. This sounds like a complete and functioning set of axioms. But, there are still some subtle details to it that we're missing. Let's say we have a chain of naturals:

$$S(0) = 1, S(1) = 2, S(2) = 3, \dots. \tag{12.6}$$

Let's add another two elements which is separate from the chain, a and b which has the following relation:

$$S(a) = b, S(b) = a. \tag{12.7}$$

It should be easy to convince yourself that all the axioms stated earlier (axioms 12.1.3 to 12.1.5) do permit a and b to exist as a part of the naturals. But this shouldn't happen as they're not in the chain of naturals that originated from zero. They would render $\mathbb{N}$ to be bigger than it should be. So, we have to add in another axiom that limit the natural to be the smallest set that satisfy all axioms prior (axioms 12.1.3 to 12.1.5): the induction axiom.

Axiom 12.1.6: *Induction axiom*

Consider any set T. If 0 is an element of the set, and that for every element n in T, $S(n)$ is also in T, then T must be isomorphic to $\mathbb{N}$. I.e.,

$$\forall T : \left((0 \in T) \wedge \left[\forall(n \in T) : S(n) \in T\right]\right) \implies T \cong \mathbb{N}. \tag{12.8}$$

This axiom assures that $\mathbb{N}$ is the smallest possible, and is analogous to the more well known "mathematical induction," which states that

Theorem 12.1.7: *Mathematical induction*

Let there be a statement $P(n)$ associated with all natural numbers n. If $P(0)$ is true, and $P(n)$ is true implies $P(S(n))$, then $P(n)$ is true for all natural numbers. I.e.,

$$P(0) \wedge \left[\forall(n \in \mathbb{N}) : P(n) \implies P(S(n))\right] \implies \left[\forall(m \in \mathbb{N}) : P(m)\right]. \tag{12.9}$$

Often times, we shall call $P(0)$ the **base case of induction**, and the step $P(n) \implies P(S(n))$, the **induction step**. Since we have to hypothesize first that $P(n)$ is true, then use that to prove that $P(n+1)$ must also be true, we also call the hypothesis that $P(n)$ is true, the **inductive hypothesis**.

For the rest of this textbook, induction should refer to theorem 12.1.7, as it's the easier version to work with. Still, we shall proof that axiom 12.1.6 and theorem 12.1.7 are equivalent.

Insert proof here

One important remark with theorem 12.1.7 is that one does not have to start from 0. However, that will be discussed after we've introduced the concept of order in section 12.6.

12.2 Set theoretical description of the naturals*

So far, we've tried to construct the naturals without actually referencing what they actually are. Anything that follows axioms 12.1.2 to 12.1.5 are said to behave exactly like the naturals; therefore, they are the naturals. Even though this abstract definition works, it might lack the intuitive tangibility one would expect for such a simple thing like the naturals. Therefore, I shall demonstrate how one may construct the naturals out of sets in this section.

The most accepted set-theoretical description of the natural numbers is the von-Neumann description which defines the following.

Definition 12.2.1: *Zero (von-Neumann description)*

Zero is represented by the empty set ($\emptyset$), i.e.,

$$0 \equiv \emptyset = \{\}.$$

Definition 12.2.2: *Successor function (von-Neumann description)*

The successor function of a natural number n is defined as the union between n and $\{n\}$, i.e.,

$$S(n) = n \cup \{n\}.$$

With the chain of naturals defined in eq. (12.5), let's try to construct the first natural number.

12.3 Addition on the naturals

Now that the naturals are constructed, we then construct a system of arithmetic from the ground up. Starting with addition.

Definition 12.3.1: *Addition on the naturals*

$$\forall(n, m \in \mathbb{N}) : n + S(m) = S(n + m). \tag{12.10}$$

Definition 12.3.2: *Zero and addition*

$$\forall(n \in \mathbb{N}) : n + 0 = n. \tag{12.11}$$

There has been nothing said about commutativity or associativity of addition yet; therefore, the order in which the successor function or zero is placed is very strict. Zero is defined solely as the **right identity element** of addition; therefore, we can't blindly conclude that $0 + n = n$. Def. 12.3.1 does not say about the successor function being placed on n instead of m either; therefore, statements like $S(n) + m = S(n + m)$ which seems trivial, still has to be rigorously proven. But before that, let's see some examples of how def. 12.3.1 and 12.3.2 comes together and capture the behavior of addition that we know.

Example 12.3.3: $2 + 2 = 4$ Given this chain of naturals

$$\begin{aligned}
S(0) &\equiv 1,\\
S(S(0)) = S(1) &\equiv 2,\\
S(S(S(0))) = S(2) &\equiv 3,\\
S(S(S(S(0)))) = S(3) &\equiv 4,\\
&\vdots
\end{aligned}$$

we have

$$2 + 2 = 2 + S(1) \tag{12.12}$$
$$= S(2 + 1). \quad \text{Def. 12.3.1} \tag{12.13}$$

and,

$$S(2 + 1) = S(2 + S(0)) \tag{12.14}$$
$$= S(S(2 + 0)) \quad \text{Def. 12.3.1} \tag{12.15}$$
$$= S(S(2)). \quad \text{Def. 12.3.2} \tag{12.16}$$

Since $2 = S(S(0))$,

$$S(S(2)) = S(S(S(S(0)))) = 4; \tag{12.17}$$

therefore, $2 + 2 = 4$.

Since we're building this from the ground up, it's going to be a very long process. So, let's set a roadmap. For all natural numbers n, m and k,

1. $0 + n = n$ (Lem. 12.3.4)
2. $S(n) + m = S(n + m)$ (Lem. 12.3.5)
3. $n + m = m + n$ (Lem. 12.3.6)
4. $(n + m) + k = n + (m + k)$ (Lem. 12.3.7)
5. $(m + n = k + n) \implies (m = k)$ (Lem. 12.3.8)
6. $n + 1 = S(n)$ (Corollary 12.3.9)

It's also worth knowing that there is no arithmetical rules yet. The only thing available for us are just the induction axiom and the rules of equality. So, both this section and the next section (section 12.4) shall serve as a very great way to practice mathematical induction.

Lemma 12.3.4: *Zero and addition from the left*

$$\forall(n \in \mathbb{N}) : 0 + n = n.$$

Proof. Let $P(n)$ be a statement that is true if $0 + n = n$.

Base case: Show that $P(0)$ is true. By def. 12.3.2,

$$0 + 0 = 0 + 0 = 0. \tag{12.18}$$

Inductive step: As an inductive hypothesis, assume that $P(n)$ is true, i.e., $0 + n = n$. Use this to prove that $P(S(n))$ is true, i.e., $0 + S(n) = S(n)$.

$$\begin{aligned} 0 + S(n) &= S(0 + n) && \text{Def. 12.3.1} && (12.19) \\ &= S(n), && \text{Inductive hypothesis} && (12.20) \end{aligned}$$

completing the induction □

Before proving that addition is commutative, we must first prove that def. 12.3.1 works from both sides:

Lemma 12.3.5: *Addition from the left*

$$\forall(n, m \in \mathbb{N}) : S(n) + m = S(n + m).$$

Proof. We induct on m while keeping n fixed; thus, let $P(m)$ be a statement that's true when $S(n) + m = S(n + m)$.

Base case: Show that $P(0)$ is true.

$$\begin{aligned} S(n) + 0 &= S(n) && \text{Def. 12.3.1} && (12.21) \\ &= S(n + 0). && \text{Def. 12.3.2} && (12.22) \end{aligned}$$

Inductive step: As an inductive hypothesis, assume that $P(m)$ is true, i.e., $S(n) + m = S(n + m)$. Use this to prove that $P(S(m))$ is true, i.e., $S(n) + S(m) = S(n + S(m))$.

$$\begin{aligned} S(n) + S(m) &= S(S(n) + m) && \text{Def. 12.3.1} && (12.23) \\ &= S(S(n + m)) && \text{Inductive hypothesis} && (12.24) \\ &= S(n + S(m)), && \text{Def. 12.3.1} && (12.25) \end{aligned}$$

completing the induction. □

We're now ready to prove that addition is **commutative**, i.e., changing the order in which the numbers are positioned does not affect the answer.

Lemma 12.3.6: *Commutativity of addition*

$$\forall(n, m \in \mathbb{N}) : n + m = m + n.$$

Proof. We induct on m while keeping n fixed; thus, let $P(m)$ be a statement that's true when $n + m = m + n$.

Base case: Show that $P(0)$ is true, i.e., $n + 0 = 0 + n$. By def. 12.3.2 and lem. 12.3.4,

$$n + 0 = 0 + n = n. \tag{12.26}$$

Inductive step: As an inductive hypothesis, assume that $P(m)$ is true, i.e., $n + m = m + n$. Use this to prove that $P(S(m))$ is true, i.e., $n + S(m) = S(m) + n$.

$$\begin{aligned} n + S(m) &= S(n + m) && \text{Def. 12.3.1} && (12.27) \\ &= S(m + n) && \text{Inductive hypothesis} && (12.28) \end{aligned}$$

$$= S(m) + n, \qquad \text{Lem. 12.3.5} \tag{12.29}$$

completing the induction. □

The next thing is that addition is **associative**, i.e., changing the order of groupings does not affect the result.

Lemma 12.3.7: *Associativity of addition on the naturals*

$$\forall (n, m, k \in \mathbb{N}) : (n + m) + k = n + (m + k).$$

Proof. We induct on k while keeping both n and m fixed; thus, let $P(k)$ be a statement that's true if $(n + m) + k = n + (m + k)$.

Base case: Show that $P(0)$ is true, i.e., $(n + m) + 0 = n + (m + 0)$.

$$(n + m) + 0 = n + m \qquad \text{Def. 12.3.2} \tag{12.30}$$

$$= n + (m + 0). \qquad \text{Def. 12.3.2} \tag{12.31}$$

Inductive step: As an inductive hypothesis, assume that $P(k)$ is true, i.e., $(n + m) + k = n + (m + k)$. Use this to prove that $P(S(k))$ is true, i.e., $(n + m) + S(k) = n + (m + S(k))$.

$$(n + m) + S(k) = S((n + m) + k) \qquad \text{Def. 12.3.1} \tag{12.32}$$

$$= S(n + (m + k)) \qquad \text{Inductive hypothesis} \tag{12.33}$$

$$= n + S(m + k) \qquad \text{Def. 12.3.1} \tag{12.34}$$

$$= n + (m + S(k)), \qquad \text{Def. 12.3.1} \tag{12.35}$$

completing the induction. □

Finally, we give ourselves the ability to cancel two sides of the equation.

Lemma 12.3.8: *Cancellation law of the addition on the naturals*

$$\forall(n, m, k \in \mathbb{N}) : (m + n = k + n) \implies (m = k).$$

Proof. We induct on n while keeping both m and k fixed; thus, let $P(n)$ be a statement that's true if $(m + n = k + n) \implies (m = k)$.

Base case: Show that $P(0)$ is true, i.e., $(m + 0 = k + 0) \implies (m = k)$.

$$m + 0 = k + 0 \tag{12.36}$$

$$m = k. \qquad \text{Lem. 12.3.4} \tag{12.37}$$

Inductive step: As an inductive hypothesis, assume that $P(n)$ is true, i.e., $(m + n = k + n) \implies (m = k)$. Use this to prove that $P(S(n))$ is true, i.e. $(m + S(n) = k + S(n)) \implies (m = k)$.

$$m + S(n) = k + S(n) \tag{12.38}$$

$$S(m + n) = S(k + n) \qquad \text{Lem. 12.3.5} \tag{12.39}$$

$$m + n = k + n \qquad \text{Axiom 12.1.4} \tag{12.40}$$

$$m = k. \qquad \text{Inductive hypothesis} \tag{12.41}$$

which completes the induction. □

To ease the usability of this system, we shall define the notion of *adding one*. Intuitively, the concept of *adding one* is analogous to the successor function. It is then left for us to establish the connection formally. Note that induction isn't used to prove the following corollary.

Corollary 12.3.9: *The successor function is analogous to addition of one*

$$S(0) \equiv 1, \quad \forall(n \in \mathbb{N}) : S(n) = n + 1.$$

Proof.

$$\forall(n \in \mathbb{N}) : S(n) = S(n+0) \qquad \text{Lem. 12.3.4} \tag{12.42}$$

$$= n + S(0) \qquad \text{Lem. 12.3.5} \tag{12.43}$$

$$= n + 1. \qquad \text{Definition of number 1} \qquad \square$$

Now that's we've proven that $n + 1 = S(n)$, it's very tempting to deprecate the successor function in favor of the plus sign. But when this is done in proof by induction, it can lead to some issues with clarity. Therefore, we shall keep it when we're using proof by induction, but use the plus one elsewhere in general arithmetic.

12.4 Multiplication

Similar to how addition is defined, it's also possible to define multiplication based on just two definitions. Informally, the first definition captures the fact that multiplication distributes over addition.

Definition 12.4.1: *Multiplication on the naturals*

$$\forall(n, m \in \mathbb{N}) : S(n) \times m = (n \times m) + m.$$

And, the second states that there exists an **absorbing element**, i.e., an element that when combined with any other element in the set yields the absorbing element itself.

Definition 12.4.2: *Absorbing element of multiplication*

$$\forall(n \in \mathbb{N}) : 0 \times n = 0.$$

We shall then prove the following: for all natural numbers n, m and k,

1. $n \times 0 = 0$ (Lem. 12.4.3)
2. $n \times S(m) = (n \times m) + n$ (Lem. 12.4.4)
3. $n \times m = m \times n$ (Lem. 12.4.5)
4. $n \times (m + k) = (n \times m) + (n \times k)$ (Lem. 12.4.6)
5. $(n \times m) \times k = n \times (m \times k)$ (Lem. 12.4.7)
6. $n \times 1 = n$ (Lem. 12.4.8)

In favor of conciseness, I shall quickly go through these and not spend much time on the induction process. Also, since addition is already built, I shall no longer reference the lemmas in this chapter. Let addition work normally as expected.

Lemma 12.4.3: *Zero is the absorbing element of multiplication*

$$\forall (n \in \mathbb{N}) : n \times 0 = 0 \times n = 0$$

Proof. Let $P(n)$ be true whenever $n \times 0 = 0$ is true.

Base case: This is similar to the proof of lem. 12.3.4

$$0 \times 0 = 0 \times 0 = 0. \tag{12.44}$$

Inductive step: Assume that $n \times 0 = 0$; prove that $S(n) \times 0 = 0$

$$S(n) \times 0 = (n \times 0) + 0 \tag{12.45}$$

$$= n \times 0 \qquad \text{Inductive hypothesis} \tag{12.46}$$

$$= 0, \tag{12.47}$$

completing the induction. □

Lemma 12.4.4: *The definition of multiplication works from both sides*

$$\forall(n, m \in \mathbb{N}) : n \times S(m) = (n \times m) + n$$

Proof. We induct on n while keeping m fixed.

Base case:

$$\begin{aligned}
0 \times S(m) &= 0 && \text{Lem. 12.4.3} && (12.48)\\
&= 0 + 0 && && (12.49)\\
&= (0 \times m) + 0. && \text{Lem. 12.4.3} && (12.50)
\end{aligned}$$

Inductive step: Assume that $n \times S(m) = (n \times m) + n$; prove that $S(n) \times S(m) = (S(n) \times m) + S(n)$.

$$\begin{aligned}
S(n) \times S(m) &= (n \times S(m)) + S(m) && \text{Def. 12.4.1} && (12.51)\\
&= ((n \times m) + n) + S(m) && \text{Inductive hypothesis} && (12.52)\\
&= (n \times m) + n + S(m)) && && (12.53)\\
&= (n \times m) + S(n) + m && && (12.54)\\
&= ((n \times m) + m + S(n) && && (12.55)\\
&= (S(n) \times m) + S(n), && \text{Def. 12.4.1} && (12.56)
\end{aligned}$$

completing the induction □

Lemma 12.4.5: *Commutativity of multiplication*

$$\forall(n, m \in \mathbb{N}) : n \times m = m \times n$$

Proof. We induct on n while keeping m fixed.

Base case: By def. 12.4.1 and lem. 12.4.4,

$$0 \times m = 0 = m \times 0. \qquad (12.57)$$

Inductive step: Assume that $n \times m = m \times n$; prove that $S(n) \times m = m \times S(n)$.

$$S(n) \times m = (n \times m) + m \qquad \text{Def. 12.4.1} \tag{12.58}$$
$$= (m \times n) + m \qquad \text{Inductive hypothesis} \tag{12.59}$$
$$= m \times S(n), \qquad \text{Lem. 12.4.4} \tag{12.60}$$

completing the induction. ☐

Lemma 12.4.6: *Distributivity of multiplication over addition*

$$\forall (n, m, k \in \mathbb{N}) : n \times (m + k) = (n \times m) + (n \times k)$$

Proof. We induct on k while keeping n and m fixed.

Base case:

$$n \times (m + 0) = n \times m \tag{12.61}$$
$$= n \times m + 0 \tag{12.62}$$
$$= (n \times m) + (n \times 0). \qquad \text{Def. 12.4.2} \tag{12.63}$$

Inductive step: Assume that $n \times (m + k) = (n \times m) + (n \times k)$; prove that $n \times (m + S(k)) = (n \times m) + (n \times S(k))$

$$n \times (m + S(k)) = n \times S(m + k) \tag{12.64}$$
$$= (n \times (m + k)) + n \qquad \text{Lem. 12.4.4} \tag{12.65}$$
$$= (n \times m) + (n \times k) + n \tag{12.66}$$
$$= (n \times m) + (n \times S(k)), \qquad \text{Lem. 12.4.4} \tag{12.67}$$

completing the induction. ☐

Lemma 12.4.7: *Associativity of multiplication*

$$\forall(n, m, k \in \mathbb{N}) : (n \times m) \times k = n \times (m \times k).$$

Proof. We induct on k while keeping n and m fixed.

Base case:

$$(n \times m) \times 0 = 0 \qquad \text{Def. 12.4.2} \tag{12.68}$$

$$= n \times 0 \tag{12.69}$$

$$= n \times (m \times 0). \qquad \text{Def. 12.4.2} \tag{12.70}$$

Inductive step: Assume that $(n \times m) \times k = n \times (m \times k)$; prove that $(n \times m) \times S(k) = n \times (m \times S(k))$.

$$(n \times m) \times S(k) = (n \times m) \times k + (n \times m) \qquad \text{Def. 12.4.1} \tag{12.71}$$

$$= n \times (m \times k) + (n \times m) \qquad \text{Inductive hypothesis} \tag{12.72}$$

$$= n \times ((m \times k) + m) \qquad \text{Lem. 12.4.6} \tag{12.73}$$

$$= n \times (m \times S(k)), \qquad \text{Lem. 12.4.4} \tag{12.74}$$

completing the induction. □

Structurally, it would make more sense if I just introduce the cancellation law for multiplication straightaway. However, it requires a more general concept of induction, as the base case starts at one instead of zero. Therefore, I shall postpone it until we've fully discussed the concepts of order in section 12.6. For those who want to skip through and read it straightawawy, it's at lem. 12.6.5.

Next, we prove that one is the identity of multiplication

Lemma 12.4.8: *Identity of multiplication*

$$\forall(n \in \mathbb{N}) : n \times 1 = n.$$

Proof. **Base case:** $0 \times 1 = 0$

Inductive step: Assume that $n \times 1 = n$, prove that $S(n) \times 1 = S(n)$.

$$\begin{aligned} S(n) \times 1 &= (n \times 1) + 1 && \text{Def. 12.4.1} && (12.75)\\ &= n + 1 && \text{Induction hypothesis} && (12.76)\\ &= S(n), && \text{Corollary 12.3.9} && (12.77) \end{aligned}$$

completing the induction. □

12.5 Positivity of natural numbers

This section concerns the property of **positive numbers** and the ordering of naturals.

Definition 12.5.1: *Positive natural numbers*

$$(n \text{ is a positive natural}) \iff \left[(n \neq 0 \in \mathbb{N})\right].$$

To be fair, we wouldn't be using this definition much because we could just say $n \neq 0$, and it's more compact notationally.

With the definition of positive natural numbers, we've essentially split the naturals into two categories: zero and positive.

Corollary 12.5.2: *Dichotomy of natural numbers*

For all natural numbers n, either n is zero, or n is positive.

Proof. If n is zero, then by def. 12.5.1, it isn't positive. If it is positive, then by def. 12.5.1, it isn't zero. $\square$

The next corollary is a simple deduction of axiom 12.1.5 which says that there are no naturals in which its successor is zero.

Corollary 12.5.3: *Positive numbers are a successor of other numbers*

$$\forall(n \neq 0 \in \mathbb{N})\exists(m \in \mathbb{N}) : n = S(m)$$

For the rest of this section, we shall develop how positivity interacts with the arithmetical operations that we've defined. Firstly, addition.

Lemma 12.5.4: *The sum of a positive number and a natural is positive*

$$\forall(n \neq 0 \in \mathbb{N})\forall(m \in \mathbb{N}) : n + m \neq 0.$$

Note that n is restricted to be positive, but m isn't.

Proof. From corollary 12.5.3,

$$(n \neq 0 \in \mathbb{N}) \implies \left[\exists(k \in \mathbb{N}) : n = S(k)\right]; \tag{12.78}$$

therefore,

$$n + m = S(k) + m \tag{12.79}$$
$$= S(k + m). \tag{12.80}$$

From axiom 12.1.5, $S(k + m)$ can't be zero, meaning that $n + m$ isn't zero either; hence, $n + m$ is positive. $\square$

The converse statement is also true

Corollary 12.5.5

$$\forall(n, m \in \mathbb{N}) : (n + m = 0) \implies ((n = 0) \land (m = 0)).$$

This is the first time that a proof by contradiction is used. To prove a statement by contradiction, we begin by assuming that its negation is true. Next, we show that the assumption that the negation is true leads to a contradiction, implying that the negation cannot possibly hold; therefore, the original statement must be true. Since the corollary above is a three part statement, let's declare some propositions to ease the proof.

Proof. Let p represents $n + m = 0$, q represents $n = 0$, and r represents $m = 0$. The original statement can then be written as $p \implies (q \land r)$. The negation of this statement is then

$$\neg(p \implies (q \land r))$$

$$\equiv \neg(p \lor \neg(q \land r)) \qquad \text{Definition of conditional} \tag{12.81}$$

$$\equiv \neg p \land (q \land r) \qquad \text{de Morgan's law} \tag{12.82}$$

$$\equiv \neg(n + m = 0) \land (n = 0) \land (m = 0) \tag{12.83}$$

$$\equiv (n + m \neq 0) \land (n = 0) \land (m = 0) \tag{12.84}$$

Since the negation of *for all* ($\forall$) is *there exists* ($\exists$), the negated statement can be written as

$$\exists(n, m \in \mathbb{N}) : (n + m \neq 0) \land (n = 0) \land (m = 0) \tag{12.85}$$

To show that this is definitely false, let's assume that $(n = 0) \land (m = 0)$ is true. The sum $n + m$ must be equal to zero, which creates a contradiction because in order for eq. (12.85) to be true, $n + m$ must not be equal to zero. Therefore, the original proposition

$$\forall(n, m \in \mathbb{N}) : (n + m = 0) \implies ((n = 0) \land (m = 0)) \tag{12.86}$$

must be true. □

We next deal with the interaction between positivity and multiplication.

Lemma 12.5.6: *The product of two positive numbers is positive*

$$\forall(n, m \in \mathbb{N}) : \big((n \neq 0) \wedge (m \neq 0)\big) \implies (n \times m \neq 0).$$

Proof. Let p represents $n \neq 0$, q represents $m \neq 0$, and $n \times m \neq 0$. The original statement can be written as $(p \wedge q) \implies r$. The negation of this statement is then

$$\begin{aligned}
&\neg\big((p \wedge q) \implies r\big) && \\
&\equiv \neg\big((p \wedge q) \vee \neg r\big) && \text{Definition of conditional} \\
&\equiv \neg(p \wedge q) \wedge r && \text{de Morgan's law} \quad (12.87) \\
&\equiv (\neg p \vee \neg q) \wedge r && \text{de Morgan's law} \quad (12.88) \\
&\equiv \big(\neg(n \neq 0) \vee \neg(m \neq 0)\big) \wedge (n \times m \neq 0) && (12.89) \\
&\equiv \big((n = 0) \vee (m = 0)\big) \wedge (n \times m \neq 0). && (12.90)
\end{aligned}$$

Since the negation of *for all* ($\forall$) is *there exists* ($\exists$), the negated statement can be written as

$$\exists(n, m \in \mathbb{N}) : \big((n = 0) \vee (m = 0)\big) \wedge (n \times m \neq 0) \tag{12.91}$$

To show that this is definitely false, let's assume that $(n = 0) \vee (m = 0)$ is true, meaning that either $n = 0$ or $m = 0$. Either ways, from def. 12.4.2 and lem. 12.4.3, the product $n \times m$ must be zero, which creates a contradiction. In order for eq. (12.91) to be true, $n \times m$ must not be equal to zero. Therefore, the original proposition

$$\forall(n, m \in \mathbb{N}) : \big((n \neq 0) \wedge (m \neq 0)\big) \implies (n \times m \neq 0) \tag{12.92}$$

must be true. □

12.6 Ordering of the natural numbers

The ultimate goal of this section is to show that the natural numbers has a well-ordering property, i.e., every non-empty subset of the naturals have a smallest element. The statement of well-ordering property is actually equivalent to the induction axiom, and one can be used to prove the other. To do that, we first have to discuss the concept of order.

Definition 12.6.1: *Orders on the naturals*

For all naturals n and m, n is **strictly greater than** m and m is **strictly less than** n, if there exists a positive number a such that $n = m + a$. This relation is denoted as $n > m$ and $m < n$ respectively.

$$\forall(n, m \in \mathbb{N}) : \\ \big[(n > m) \wedge (m < n)\big] \iff \big[\exists(a \neq 0 \in \mathbb{N}) : n = m + a\big].$$

If a isn't restricted to be positive, we say that n is **greater than or equal to** m and m is **less than or equal to** n, denoted as $n \geq m$ and $m \leq n$ respectively.

$$\forall(n, m \in \mathbb{N}) : \big[(n \geq m) \wedge (m \leq n)\big] \iff \big[\exists(a \in \mathbb{N}) : n = m + a\big].$$

In text, we usually write *greater than* and *less than* instead of the *strictly greater than* and *strictly less than* in favor of concision.

There's also another commonly accepted definition of non-strict ordering that's equivalent to def. 12.6.1, i.e.,

$$\forall(n, m \in \mathbb{N}) : \left[(n \geq m) \wedge (m \leq n)\right] \\ \iff \left[\left(\exists(a \neq 0 \in \mathbb{N}) : n = m + a\right) \wedge (n \neq m)\right].$$

It's trivial to prove that this definition and the one of def. 12.6.1 is actually equivalent, and is left as a side-exercise for the reader.

We then begin to prove the basic properties of order. Since these proofs are pretty short, I've contained all of them in one proposition.

Proposition 12.6.2: *Basic properties of order on the naturals* For all natural numbers n, m and k,

1. $n \geq n$ (Reflexivity)
2. $\left[(n \geq m) \wedge (m \geq k)\right] \implies (n \geq k)$ (Transitivity)
3. $\left[(n \geq m) \wedge (m \geq n)\right] \implies (n = m)$ (Symmetric)
4. $(n \geq m) \iff (n + k \geq m + k)$
5. $(n > m) \iff (n \geq S(m))$

Proof of item 1 of prop. 12.6.2. From def. 12.6.1, we must show that there exists a natural a which for all naturals n, $n = n + a$. Obviously, $a = 0$, proving that $n \geq n$ for all n. □

Proof of item 2 of prop. 12.6.2. From def. 12.6.1,

$$n \geq m \iff \left[\exists(a \in \mathbb{N}) : n = m + a\right], \tag{12.93}$$

$$m \geq k \iff \left[\exists(b \in \mathbb{N}) : m = k + b\right]. \tag{12.94}$$

Substituting $m = k + b$ into $n = m + a$ yields

$$n = k + (a + b). \tag{12.95}$$

Let $(a + b) = c$, we get $n = k + c$; thus by def. 12.6.1, $n \geq k$. □

Proof of item 3 of prop. 12.6.2. From def. 12.6.1,

$$n \geq m \iff \left[\exists(a \in \mathbb{N}) : n = m + a\right], \tag{12.96}$$

$$m \geq n \iff \left[\exists(b \in \mathbb{N}) : m = n + b\right]. \tag{12.97}$$

Combining $n = m + a$ and $m = n + b$ yields

$$n + m = m + n + a + b \tag{12.98}$$

$$0 = a + b. \qquad \text{Lem. 12.3.8} \tag{12.99}$$

By corollary 12.5.5, $a = 0$ and $b = 0$ meaning that $n = m + 0$ and $m = n + 0$; thus, $n = m$. ☐

Proof of item 4 of prop. 12.6.2. Since this is a biconditional, we must prove both the forwards and backwards direction, i.e.,

$$\forall(n, m, k \in \mathbb{N}) : \left[(n \geq m) \implies (n + k \geq m + k)\right] \wedge \left[(n + k \geq m + k) \implies (n \geq m)\right]. \tag{12.100}$$

Either ways, we can start with def. 12.6.1

$$(n \geq m) \iff \left[\exists(a \in \mathbb{N}) : n = m + a\right]. \tag{12.101}$$

To prove the forward direction, we can add $k = k$ to $n = m + a$ to get

$$n + k = m + k + a. \tag{12.102}$$

By def. 12.6.1, $n + k \geq m + k$. And for the backwards direction, just use the cancellation law for addition on the naturals (lem. 12.3.8). ☐

Proof of item 5 of prop. 12.6.2. Similarly, this is also a biconditional. Let's start with the forward direction, i.e.,

$$\forall(n, m \in \mathbb{N}) : (n > m) \implies (S(n) \geq m). \tag{12.103}$$

By def. 12.6.1,

$$(n > m) \iff \big[\exists(a \in \mathbb{N}) : (a \neq 0) \land (n = m + a)\big], \tag{12.104}$$

where a is positive. Because positive numbers are a successor of other naturals (corollary 12.5.3), there exists a natural b in which $a = S(b)$; therefore,

$$n = m + S(b) \tag{12.105}$$

$$= S(m) + b. \tag{12.106}$$

Since b isn't constrained to be positive, $n \geq S(m)$. To prove the backwards direction, i.e.,

$$\forall(n, m \in \mathbb{N}) : \big(S(n) \geq m\big) \implies (n > m), \tag{12.107}$$

we first notice that $n \geq S(m)$ means there exists some natural a in which $n = S(m) + a$. Rewriting this gives $n = m + S(a)$. By axiom 12.1.5, $S(a) \neq 0$, meaning $S(a)$ is positive. Thus, $n \geq m$ by def. 12.6.1. □

The next thing that I'd like to focus on is the trichotomy for orders.

> **Lemma 12.6.3:** *Order trichotomy* For all natural numbers n and m, only one of these statements are true: $n = m$, $n > m$, or $n < m$.

Trichotomy means a *division into three category*. If something is trichotomous, then that thing can be in only one of the category at a time. These families of trichotomy statement also appear in the construction of the integers, rationals and reals, where the numbers are distinctly categorized into negatives, zero, and positive. In the naturals, order trichotomy is used to develop a stronger form of mathematical induction called the *strong induction*, and also used to prove the cancellation law for multiplication. It can also be used to prove various statements on the reals.

The logical form of a trichotomous statement is given as follows. Let there be an object x, and three disjoint sets A, B, and C. Let p represents $x \in A$, q represents $x \in B$, and r represents $x \in C$. If x is trichotomous, then

$$\left(p \wedge \neg q \wedge \neg r\right) \vee \left(\neg p \wedge q \wedge \neg r\right) \vee \left(\neg p \wedge \neg q \wedge r\right). \tag{12.108}$$

An equivalent form of this statement is

$$\left(p \vee q \vee r\right) \wedge \neg\left((p \wedge q) \vee (q \wedge r) \vee (p \wedge r)\right) \tag{12.109}$$

which is easier to both interpret and prove. The first part $(p \vee q \vee r)$ directly translates to "*at least one of the statement is true.*" The second part $(p \wedge q) \vee (q \wedge r) \vee (p \wedge r)$ translates to "*at least one of the statement is true.*" Together, they ensure that one of the statements are guaranteed to be true. With this simplified logical form, we can then prove the order trichotomy of the naturals.

Proof. Let there be two naturals n and m. Firstly, we prove that no two statements can hold simultaneously.

1. Proving that $n < m$ and $n = m$ can't be true simultaneously: If $n < m$, then by def. 12.6.1, $n \neq m$
2. Proving that $n > m$ and $n = m$ can't be true simultaneously: If $n > m$, then by def. 12.6.1, $n \neq m$.
3. Proving that $n < m$ and $n > m$ can't be true simultaneously: For the sake of contradiction, assume that $n < m$ and $n > m$. Then by

prop. 12.6.2, $n = m$ which by the two statement that's proven earlier, is a contradiction.

And secondly, we prove that at least one of the statement is true. We induct on n while keeping m fixed. Let $P(n)$ be the statement $(n < m) \vee (n > m) \vee (n = m)$.

Base case: When $n = 0$, $n \leq m$ for all naturals m; thus, either $n = m$ or $n < m$.

Inductive step: Assume that $P(n)$ is true, check that $P(S(n))$ is also true.

1. If $n < m$, then by prop. 12.6.2, $S(n) \leq m$, i.e., either $S(n) < m$ or $S(n) = m$. Therefore, $P(S(n))$ is true because there is at least one true statement.
2. If $n = m$, then by prop. 12.6.2, $S(n) > m$: $P(S(n))$ is true.
3. If $n > m$, then by prop. 12.6.2, $S(n) > m$: $P(S(n))$ is true.

Therefore, $P(n)$ must hold true for all natural numbers, i.e., either $n = m$, $n < m$, or $n > m$. Combined with the earlier proof that no two statements can hold simultaneously, we've shown that the order trichotomy is true for all natural numbers n and m. $\square$

Then, we can develop a stronger form of induction which is equivalent to theorem 12.1.7:

Theorem 12.6.4: *Strong mathematical induction*

Let there be a statement $P(n)$ associated with all natural numbers. Let there be two naturals that's less than m, m_0 where $m > m_0$. If the following proposition is true: $P(n)$ is true for all $m_0 \leq n < m$

implies $P(m)$ is true, then $P(n)$ is true for all natural numbers that's more than m_0. I.e.,

$$\left[\{\forall(n \in \mathbb{N} : m_0 \leq n \leq m \in \mathbb{N}) : P(n)\} \implies P\big(S(m)\big)\right] \implies [\forall(k \geq m_0) : P(k)]. \tag{12.110}$$

Before completing the arithmetical system on the naturals by proving the cancellation law for multiplication, we first prove that theorem 12.1.7 and theorem 12.6.4 are equivalent.

Proof. Let $Q(n)$ be a statement that's true when $P(n)$ is true for all $m_0 \leq m < n$. □

Proof by strong induction also includes the base case and the inductive step, just like the standard mathematical induction. First, we show that $P(m_0)$ is true. Then, we assume that $P(m_0)$ all the way to $P(m)$ is true for some natural m as our induction hypothesis, and show that $P(S(m))$ is true, closing the induction.

Finally, we can complete the arithmetical systems on the naturals by proving the cancellation law for multiplication

Lemma 12.6.5: *Cancellation law for multiplication*

$$\forall(n, m \in \mathbb{N})\forall(c \neq 0 \in \mathbb{N}) : \big(n \times c = m \times c\big) \implies (n = m).$$

Proof. Using prove by contradiction, we assume that the negation of the statement is true, i.e., let

$$\exists(n, m \in \mathbb{N})\exists(c \neq 0 \in \mathbb{N}) : (n \times c \neq m \times c) \wedge (n = m) \tag{12.111}$$

We assume that $n \times c \neq m \times c$. By the trichotomy of order, either $n \times c < m \times c$, $n \times c = m \times c$, or $n \times c > m \times c$. $\square$

CHAPTER 13

The integers

All numbers that are an integer will be marked with an asterisk. E.g., 2 is in $\mathbb{N}$ but $2^{\ddagger}$ is in $\mathbb{Z}$

13.1 An informal introduction to the integers

In this section, I shall not refrain myself from using the subtraction sign. The subtraction sign here is used to better illustrate what we shall be doing in this chapter. In the following sections, we shall formally construct the integers from the ideas that are illustrated here.

The purpose of defining the integers is to define what subtraction and negative numbers are. In a sense, the integers are an extension of the naturals: we can use the naturals and the operations we have to define what an integer is. Although, with the use of some equivalence relations.

Ultimately, an integer is difference between two naturals, i.e.,

$$a^{\ddagger} = n - m. \tag{13.1}$$

Naturally, we could define an integer as an ordered pair (n, m), and say that this ordered pair represent the integer $n - m$. Although, we haven't defined what subtraction is yet. We'll see how we can deal with that later. However, this definition alone wouldn't suffice, because now, one integer can be represented by many pairs of naturals, e.g., $(5, 3)$ and $(4, 2)$ both represent $2^\ddagger$. So, we need an equivalence relation to tell us whether a pair of naturals can represent the same integer or not.

Consider two integers (n, m) and (k, l). These two integers are equivalent *iff*

$$n - m = k - l \tag{13.2}$$

From the assumed knowledge of arithmetic, we can move the terms around to get rid of the subtraction:

$$n + l = k + m \tag{13.3}$$

Therefore, an equivalence relation can be defined by

$$(n, m) \sim (k, l) \iff n + l = k + m. \tag{13.4}$$

Addition and multiplication of integers are also defined with the definition that's inspired by an assumed knowledge of arithmetic. I.e., if $a^\ddagger = (n, m)$ and $b^\ddagger = (k, l)$, then

$$a^\ddagger + b^\ddagger = (n, m) + (k, l) \tag{13.5}$$

$$= (n - m) + (k - l) \tag{13.6}$$

$$= (n + k) - (m + l) \tag{13.7}$$

$$= (n + k, m + l), \tag{13.8}$$

and

$$a^\ddagger b^\ddagger = (n, m) \times (k, l) \tag{13.9}$$

$$= (n-m)(k-l) \tag{13.10}$$

$$= nk - nl - mk + ml \tag{13.11}$$

$$= (nk+ml) - (nl+mk) \tag{13.12}$$

$$= (nk+ml, nl+mk). \tag{13.13}$$

To define negative numbers, we just have to revert the order of the pair. If $(n,m) = n - m = a^{\ddagger}$, then $(m,n) = m - n = -a$. Then, subtraction is simply defined by $a^{\ddagger} - b^{\ddagger} = a^{\ddagger} + (-b^{\ddagger})$

13.2 Construction of the integers

Definition 13.2.1: *Integers*

An integer is an ordered pair in the form (n,m) where $n,m \in \mathbb{N}$. If $a^{\ddagger}$ is an integer, then $\exists(n,m \in \mathbb{N}) : a^{\ddagger} = (n,m)$.

Then, we construct the set of all integers

Definition 13.2.2: *Set of all integers ($\mathbb{Z}$)*

The set $\mathbb{Z}$ contains all integers, i.e.,

$$\mathbb{Z} = \left\{(n,m) : n,m \in \mathbb{N}\right\} \tag{13.14}$$

The equivalence relation is then stated as follows:

Definition 13.2.3: *Equivalence relation for integers*

Given two integers $a^{\ddagger} = (n,m)$ and $b^{\ddagger} = (k,l)$, where $n,m,k,l \in \mathbb{N}$. If these two integers are equivalent, then we write $a^{\ddagger} \sim b^{\ddagger}$, or $(n,m) \sim (k,l)$. And,

$$a^{\ddagger} \sim b^{\ddagger} \iff n + l = m = k. \tag{13.15}$$

Because we want to later define arithmetic on the integers, we must verify that the equivalence relation for integers also satisfy the three equality axioms, i.e.,

1. Reflexivity $(\forall(a^\ddagger \in \mathbb{Z}) : a^\ddagger \sim a^\ddagger)$
2. Symmetry $(\forall(a^\ddagger, b^\ddagger \in \mathbb{Z}) : a^\ddagger \sim b^\ddagger \iff b^\ddagger \sim a^\ddagger)$
3. Transitivity $(\forall(a^\ddagger, b^\ddagger, c^\ddagger \in \mathbb{Z} : (a^\ddagger \sim b^\ddagger) \wedge (b^\ddagger \sim c^\ddagger) \implies a^\ddagger \sim c^\ddagger)$

Once these three are verified, we can actually replace the equivalence sign $(\sim)$ with the equality sign $(=)$.

Corollary 13.2.4: *Equivalence of integers is reflexive*

$$\forall(a^\ddagger \in \mathbb{Z}) : a^\ddagger \sim a^\ddagger. \tag{13.16}$$

Proof. Let $\exists(n, m \in \mathbb{N})$ where $(n, m) = a^\ddagger$. Since $n + m = n + m$, $(n, m) \sim (n, m)$; thus, $a^\ddagger \sim a^\ddagger$. □

Corollary 13.2.5: *Equivalence of integers is symmetric*

$$\forall(a^\ddagger, b^\ddagger \in \mathbb{Z}) : a^\ddagger \sim b^\ddagger \iff b^\ddagger \sim a^\ddagger \tag{13.17}$$

Proof. Since the backwards direction can be obtained by switching a and b, we shall only prove the forward direction.

Let $\exists(n, m, k, l \in \mathbb{N})$ where $(n, m) = a^\ddagger$ and $(k, l) = b^\ddagger$. Since $a^\ddagger \sim b^\ddagger$, $(n, m) \sim (k, l)$; thus, $n + l = m + k$. By the symmetry of equality, $m + k = n + l$. Because addition is commutative, $k + m = n + l$. Therefore, we get $(k, l) \sim (n, m)$, i.e., $b^\ddagger \sim a^\ddagger$. □

Corollary 13.2.6: *Equivalence of integers is transitive*

$$\forall(a^\ddagger, b^\ddagger, c^\ddagger \in \mathbb{Z} : (a^\ddagger \sim b^\ddagger) \wedge (b^\ddagger \sim c^\ddagger) \implies a^\ddagger \sim c^\ddagger \tag{13.18}$$

Proof. Let $\exists(n, m, k, l, p, q \in \mathbb{N})$ where $(n, m) = a^\ddagger$, $(k, l) = b^\ddagger$, and $(p, q) = c^\ddagger$.

$$a^\ddagger \sim b^\ddagger \iff (n, m) \sim (k, l) \iff n + l = m + k, \tag{13.19}$$

$$b^\ddagger \sim c^\ddagger \iff (k, l) \sim (p, q) \iff k + p = l + q. \tag{13.20}$$

Combining these two equations together we get

$$n + l + k + p = m + k + l + q. \tag{13.21}$$

By the cancellation law on addition of naturals,

$$n + p = m + q, \tag{13.22}$$

which is equivalent to saying $(n, m) \sim (p, q)$, i.e., $a^\ddagger \sim c^\ddagger$. □

By corollaries 13.2.4 to 13.2.6, we can now use the equal sign instead of the equivalence sign to represent the equality of integers. But as said earlier, there are still one thing that is still not verified: the substitution axiom, i.e., if $f(n)$ be an arithmetical function of the integers, then

$$\big(f(n) = m\big) \wedge \big(n = n'\big) \implies f(n') = m. \tag{13.23}$$

13.3 Arithmetic on the integers

Definition 13.3.1: *Addition of integers*

Let there be two integers $a^\ddagger$ and $b^\ddagger$ where $\exists(n, m, k, l \in \mathbb{N})$ such that $(n, m) = a^\ddagger$ and $(k, l) = b^\ddagger$. The sum of $a^\ddagger$ and $b^\ddagger$, denoted $a^\ddagger + b^\ddagger$, is defined as:

$$a^\ddagger + b^\ddagger = (n + k, m + l). \tag{13.24}$$

Definition 13.3.2: *Product of integers*

Let there be two integers $a^\ddagger$ and $b^\ddagger$ where $\exists(n, m, k, l \in \mathbb{N})$ where $(n, m) = a^\ddagger$ and $(k, l) = b^\ddagger$. Then, the product of $a^\ddagger$ and $b^\ddagger$, $a^\ddagger + b^\ddagger$ is defined as:

$$a^\ddagger \times b^\ddagger = a^\ddagger b^\ddagger = (nk + ml, nl + mk). \tag{13.25}$$

Here, we prove that both addition and multiplication of integers obey the substitution axiom

Corollary 13.3.3: *Addition and multiplication of integers obey the substitution axiom* Let n, m, n', m', k, l be natural numbers. If $(n, m) = (n', m')$, then

$$(n, m) + (k, l) = (n', m') + (k, l), \tag{13.26}$$

$$(n, m) \times (k, l) = (n', m') \times (k, l) \tag{13.27}$$

Proof. The relation $(n, m) = (n', m')$ means $n + m' = m + n'$.

Firstly, we prove that addition obeys the substitution axiom. Evaluate both sides separately

$$(n, m) + (k, l) = (n + k, m + l) \tag{13.28}$$

$$(n', m') + (k, l) = (n' + k, m' + l) \tag{13.29}$$

Which, for an equivalence test, we check whether $(n + k, m + l)$ is equal to $(n' + k, m' + l)$ or not.

$$(n + k, m + l) \stackrel{?}{=} (n' + k, m' + l) \tag{13.30}$$

$$n + k + m' + l \stackrel{?}{=} m + l + n' + k. \tag{13.31}$$

Using cancellation of addition on natural numbers, we get

$$n + m' \stackrel{?}{=} m + n'. \tag{13.32}$$

From our assumption that $(n, m) = (n', m')$, $n + m'$ does indeed equal $m + n'$. Therefore, our statement is true: addition obeys the substitution axiom.

For multiplication,

$$(n, m) \times (k, l) = (nk + ml, nl + mk) \tag{13.33}$$

$$(n', m') \times (k, l) = (n'k + m'l, n'l + m'k). \tag{13.34}$$

Then, we check whether $(nk+ml, nl+mk)$ is equal to $(n'k+m'l, n'l+m'k)$ or not.

$$(nk + ml, nl + mk) \stackrel{?}{=} (n'k + m'l, n'l + m'k) \tag{13.35}$$

$$nk + ml + n'l + m'k \stackrel{?}{=} nl + mk + n'k + m'l \tag{13.36}$$

$$k(n + m') + l(m + n') \stackrel{?}{=} l(n + m') + k(m + n'). \tag{13.37}$$

Since $n + m' = m + n'$, the two sides must be equal. Therefore, our statement is true: multiplication obeys the substitution axiom. □

CHAPTER 14

The rationals

CHAPTER 15

Real numbers and sequences

15.1 Informal invitation to sequences

I'm sure that you are probably already familiar with the concept of **sequence**. $1, 2, 3, 4, 5, \dots$ is a sequence of numbers which are increasing by one, $2, 4, 8, 16, 32, \dots$ is a sequence of numbers which are doubling, etc. If the sequence is increasing by d each time, it's an arithmetic sequence. If it's multiplied by r each time, it's a geometric sequence. It's a basic concept that most have already encountered in high school algebra. So, why do we have to analyze it here again?

The answer is two-fold. Firstly, we use sequences to actually construct the reals. Secondly, sequences are used to define limit of functions, which are the building blocks of derivatives. We shall answer the problem of constructing the reals in this chapter, and the limit of functions in later chapters.

The real numbers are constructed differently than all other numbers that we've constructed. We can't just introduce another operation and some equivalence class. You might say that we can just construct rational exponentiation, and we can get the reals. Yes, there are infinitely many real numbers that's obtainable by construction of new operations, such as square root or logarithm; however, there are many real numbers which cannot be constructed by algebraic operations. Those numbers are called **non-algebraic numbers** or **transcendental numbers**. Even more so, there are infinitely many of them. So our construction of the reals must be able to capture all these cases. How? The answer is through the analysis of sequences.

Firstly, let's define what a sequence is

Definition 15.1.1: *Sequence*

A **sequence** a_n of a set $\mathbb{F}$ is a mapping a from $\mathbb{N}$ to $\mathbb{F}$.

Notation 15.1.2

A sequence of a set $\mathbb{F}$ is written as $(a_n)_{n=i}^{f}$ where the starting point $i \in \mathbb{N}$ and the ending point is $f \in \mathbb{N}$. If $f = \infty$, then $(a_n)_{n=i}^{\infty}$ is an infinite sequence which starts at i.

For example, a finite sequence of rationals, $(a_n)_{n=2}^{7}$ defined as $a_n = \frac{1}{n}$ forms the set

$$\left\{\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \frac{1}{6}, \frac{1}{7}\right\}.$$

A finite sequence of rationals, $(b_n)_{n=5}^{7}$, defined as $b_n = n^2$ forms the set

$$\{25, 36, 49, 64, 81, 100\}.$$

An infinite sequence of rationals $(c_n)_{n=3}^{\infty}$ defined as $c_n = n + 2$ starting at $n = 3$ is the set

$$\{5, 6, 7, 8, 9, 10, \ldots\}.$$

Cauchy constructed the real numbers by saying that the real numbers are the *limit* of these *converging rational sequences*. To illustrate what *limits* and *converging sequences* are, let's look at some examples. Consider the sequence

$$3, 3.1, 3.14, 3.141, 3.1415, 3.14159, \ldots .$$

This sequence, although filled with rational numbers, are *gradually approaching* the real number π, which is not rational. Consider another sequence

$$1, 1.4, 1.41, 1.4142, 1.41421, \ldots . \tag{15.1}$$

This sequence is a sequence of rationals that *gradually approaches* $\sqrt{2}$. These numbers here, π and $\sqrt{2}$, though not in the set of rationals, are the number that these sequence *gradually approaches*. In math lingo, if a sequence *gradually approaches* some number L, then that sequence is a **convergent sequence** and L is said to be the **limit** of that sequence, i.e., the sequence converges to L. Here, you can clearly see how the reals can be possibly defined to be the limit of said converging sequence. However, we quickly run into a problem.

As you'll see later, the problem with convergence is that one has to know the limit beforehand before showing that it actually *converges*. E.g., we have to know what the exact value of π in order to show that the sequence

$$3, 3.1, 3.14, 3.141, 3.1415, 3.14159, \ldots$$

is actually a convergence sequence in the first place. I.e., we want to construct the reals by using the limit of a rational sequence, but we need to know what the reals are in order to show that the sequence actually converges. This is circular reasoning, and is undesirable in mathematics. To break out of this reasoning loop, we must show that a sequence of rational converges without knowing its limit in the first place.

Cauchy's constructed the reals by showing that all convergent sequences are Cauchy sequence. Informally,

> *A Cauchy sequence is a sequence whose elements become arbitrarily close to each other as the sequence progresses.*

After which, the algebraic operations on the reals will then be defined as the operation on each term of the sequence. E.g., if the real x is defined as the limit of the sequence $(a_n)_{n=i}^{\infty}$ and the real y as the limit of the sequence $(b_n)_{n=i}^{\infty}$. Their sum can then be represented as the limit of $(a_n + b_n)_{n=i}^{\infty}$, their product as the limit of $(a_n b_n)_{n=i}^{\infty}$, and their ratio as the limit of $\left(\frac{a_n}{b_n}\right)_{n=i}^{\infty}$, etc.

From now on, I'd like to focus on sequence of rationals because we want to construct the reals from the rationals. So, every sequence from now on is assumed to be a rational sequence. When the reals ($\mathbb{R}$) are finally constructed, one can replace all $\mathbb{Q}$ with $\mathbb{R}$.

15.2 Cauchy sequences

Definition 15.2.1: *Cauchy sequences*

If a rational sequence $(a_n)_{n=m}^{\infty}$ is Cauchy, then

$$\forall(\varepsilon > 0 \in \mathbb{Q})\exists(N > m \in \mathbb{N})\forall(j, k \geq N \in \mathbb{N}) : |a_j - a_k| \leq \varepsilon. \quad (15.2)$$

I understand that this definition looks very unfriendly, so let's break it down and see how each element come together to capture concept of a sequence whose elements get closer together as the sequence progresses. In my humble opinion, the approach taken by Terence Tao in his *Analysis I* book [4] is one of the most intuitive; therefore, I shall follow his teachings and deconstruct the definition of Cauchy sequences into two bite-sized ones: ε-steadiness and eventual ε-steadiness.

The first definition defines what it means for *two elements to be close to each other.*

Definition 15.2.2: *ε-closeness*

Two numbers a and b are ε-close if $|a - b| \leq \varepsilon$

This definition measures the distance on the number line. If the distance between a and b is less than ε, we say that a and b are ε-close.

Definition 15.2.3: *ε-steadiness*

A rational sequence $(a_n)_{n=i}^{f}$ is ε-steady if for all j, k between i and f, a_j and a_k are ε-close.

One interpretation of a sequence that's ε-steady is that the range of the sequence ($\max\left((a_n)_{n=i}^{f}\right) - \min\left((a_n)_{n=i}^{f}\right)$) is restricted to be less than or equal to ε:

$$\max\left((a_n)_{n=i}^{f}\right) - \min\left((a_n)_{n=i}^{f}\right) \leq \varepsilon. \tag{15.3}$$

Because of this, the distance between each term in an ε-steady sequence is always less than or equal to ε.

This definition also works for infinite sequence. However, the concept of maxima and minima is currently a bit loose. Sometimes, the maxima and minima doesn't even exist, so it'd be wiser for now to think of an ε-steady sequence as a sequence that the distance between each and every term is less than ε. Now, let's look at some examples and non-examples.

Example 15.2.4: *ε-steady sequences*

1. The sequence $(a_n)_{n=0}^{\infty}$: $3, 3.1, 3.14, 3.141, 3.1415, 3.14159, \ldots$, is 0.1-steady but not 0.01-steady, because

$$|a_0 - a_1| = |3 - 3.1| = 0.1, \quad \text{which is more than} \quad 0.01.$$

2. The sequence $(b_n)_{n=0}^{\infty}$: $-1, 0, 1, 0, -1, 0, \ldots$ is 2-steady, but not 1-steady because

$$|b_0 - b_2| = |-1 - 1| = 2, \quad \text{which is more than} \quad 1.$$

3. The sequence $(c_n)_{n=4}^{9}$: $2, 3, 1, 5, 4, 6$ is 10-steady, but not 4-steady because

$$|c_6 - c_9| = |6 - 1| = 5, \quad \text{which is more than} \quad 4.$$

Before moving on, I'd like to discuss the image of ε-steady sequence. Consider a sequence $(a_n)_{n=0}^{\infty}$ shown in fig. 15.1. Let's draw a rectangle which infinite extends along the x-axis but has the height of ε. The sequence is ε-steady *iff* you can find a position of this rectangle such that every term in the sequence lies in that rectangle. Here, $(a_n)_{n=0}^{\infty}$ is ε-steady because there's a rectangle configuration in which every point of the sequence lies in the rectangle shown in the right most figure. The left most and middle figure doesn't count because there are some points, drawn in red, which is outside the rectangle.

The concept of a sequence being ε-steady is quite simple, but it doesn't capture the limiting behavior of a sequence. Since we want to construct the reals from the *limit* of a *converging sequence*, we shouldn't really care about the initial terms. Therefore, we improve on the definition of

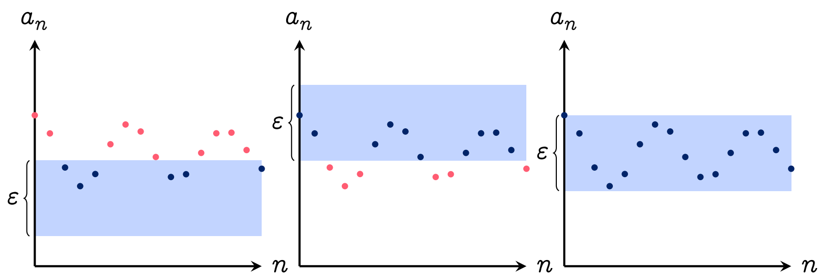


FIG. 15.1 | EXAMPLES OF RECTANGLE CONFIGURATIONS TO CHECK THAT A SEQUENCE IS ε-STEADY.

ε-steadiness to eventual ε-steadiness: a definition that captures what it means for *the elements of a sequence to get close together as the sequence progresses*.

Definition 15.2.5: *Eventual ε-steadiness*

A rational sequence $(a_n)_{n=i}^{f}$ is eventually ε-steady if there exists some natural $N \geq i$ in which the sequence $(a_n)_{n=N}^{f}$ is ε-steady.

The intuition for this is that, if it's possible to find some number N in which the removal of the first N terms of the sequence makes the sequence ε-steady, then the sequence is eventually ε-steady. Let's look at some examples and non-examples.

Example 15.2.6: *Eventually ε-steady sequences*

1. The sequence $(a_n)_{n=1}^{\infty}$: $1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots,$ is not $\frac{1}{2}$-steady because

$$|a_0 - a_2| = \left|1 - \frac{1}{3}\right| = \frac{2}{3}, \quad \text{which is more than} \quad \frac{1}{2};$$

however, the sequence is eventually $\frac{1}{2}$-steady because the removal of the first term yields a sequence that is $\frac{1}{2}$-steady:

$$\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \ldots,$$

2. The sequence $(b_n)_{n=0}^{\infty}$: $1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \dots$, is not $\frac{1}{4}$-steady because

$$|a_2 - a_4| = \left|\frac{1}{2} - \frac{1}{8}\right| = \frac{3}{8}, \quad \text{which is more than} \quad \frac{1}{4};$$

however, the sequence is eventually $\frac{1}{4}$-steady because the removal of the first two terms yields a sequence that is $\frac{1}{4}$-steady:

$$\frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \frac{1}{32}, \dots .$$

3. The sequence $(c_n)_{n=0}^{\infty}$: $69, 2, 2, 2, 2, \dots$ is not 3-steady because for all $j > 0$,

$$|a_0 - a_j| = |69 - 2| = 67, \quad \text{which is more than} \quad 3;$$

however, the sequence is eventually 3-steady because the removal of the first term yields a sequence that is 3-steady:

$$2, 2, 2, 2, 2, \dots .$$

And now, we're ready to define what Cauchy sequences are in simple terms:

Corollary 15.2.7: *Cauchy sequences* A Cauchy sequence is an infinite sequence which is eventually ε-steady for all $\varepsilon > 0 \in \mathbb{Q}$. I.e., for a sequence $(a_n)_{n=m}^{\infty}$ to be Cauchy,

$$\forall(\varepsilon > 0 \in \mathbb{Q})\exists(N > m \in \mathbb{N})\forall(j, k \geq N \in \mathbb{N}) : |a_j - a_k| \leq \varepsilon$$

A natural question to ask from here would be, *how to prove that a sequence is Cauchy?* It is quite hard, but the core concept is to show that for every ε, there exists a corresponding N which makes the sequence

$(a_n)_{n=N}^{\infty}$, ε-steady. This would be one of the first proof that must have a preparation stage before the actual proof. We do some arithmetic first to decide what N might be in terms of ε, and use that to show that for all $j, k \geq N$, the sequence is ε-steady.

Example 15.2.8: *Show that $(a_n)_{n=0}^{\infty}$ is Cauchy* We prepare the proof by first fixing ε and find N in terms of ε. For all $n \geq N$, we demand that

$$|a_j - a_k| \leq \varepsilon \tag{15.4}$$
$$\left|\frac{1}{j} - \frac{1}{k}\right| \leq \varepsilon \tag{15.5}$$
$$\left|\frac{1}{j}\right| + \left|\frac{1}{k}\right| \leq \varepsilon. \qquad \text{Triangle inequality} \tag{15.6}$$

If we demand both $\left|\frac{1}{j}\right|$ and $\left|\frac{1}{k}\right|$ to be less than $\frac{\varepsilon}{2}$, then we would get $|a_j - a_k| \leq \varepsilon$ as desired; thus, let

$$\left|\frac{1}{j}\right|, \left|\frac{1}{k}\right| \leq \frac{\varepsilon}{2} \tag{15.7}$$

Since $j, k \geq N$, we have $\frac{1}{j}, \frac{1}{k} \leq \frac{1}{N}$. Combining this with the equation above gives

$$\frac{1}{N} \leq \frac{\varepsilon}{2} \tag{15.8}$$
$$N \geq \frac{2}{\varepsilon}. \tag{15.9}$$

Here, we've shown that $N \geq \frac{2}{\varepsilon}$ causes $|a_j - a_k|$ where $j, k \geq N$ to be less than ε for every ε; thus, the sequence is Cauchy. Now onto the formal proof.

Proof.

$$|a_j - a_k| = \left|\frac{1}{j} - \frac{1}{k}\right| \tag{15.10}$$

$$\left|\frac{1}{j}-\frac{1}{k}\right| \leq \left|\frac{1}{j}\right|+\left|\frac{1}{k}\right| \qquad \text{Triangle's inequality} \tag{15.11}$$

$$\left|\frac{1}{j}\right|+\left|\frac{1}{k}\right| \leq \frac{1}{N}+\frac{1}{N} \tag{15.12}$$

Choose $N \geq \frac{2}{\varepsilon}$, then

$$\frac{1}{N}+\frac{1}{N} \geq \frac{\varepsilon}{2}+\frac{\varepsilon}{2} \tag{15.13}$$

$$\frac{\varepsilon}{2}+\frac{\varepsilon}{2} = \varepsilon. \tag{15.14}$$

Therefore, $N \geq \frac{2}{\varepsilon}$ causes $|a_j - a_k|$ to be less than ε for all $j, k \geq N$.

Thus, the sequence is Cauchy. □

Before constructing the reals, I'd like to discuss the notion of boundedness which is later used to prove that multiplication on the reals is well-defined

Definition 15.2.9: *Boundedness of sequences*

A sequence $(a_n)_{n=i}^{f}$ is bounded by a rational $M > 0$ if for all $n \in \mathbb{N}$ that's between i and f, $|a_n| \leq M$.

Corollary 15.2.10: *Finite sequences are bounded* Every finite sequence $(a_n)_{n=i}^{f}$ is bounded

Proof. We shall use strong induction to prove this.

Base case: Trivially, the sequence a_1 is bounded by $M = |a_1|$.

Inductive step: Assume that the sequence $(a_n)_{n=i}^{N}$ where $i < N < f$ is bounded by some rational M. Then, the sequence $(a_n)_{n=i}^{S(N)}$ must also be bounded by the rational $M + |a_{S(N)}|$; therefore, any finite sequence is bounded, closing the induction. □

The next thing is to establish a connection between Cauchy sequences and the concept of boundedness.

Proposition 15.2.11: *Every Cauchy sequence is bounded* Every Cauchy sequence $(a_n)_{n=i}^{\infty}$ is bounded.

Proof. Since the sequence is Cauchy, for every ε, there exists a natural N such that for all $j, k \geq N$, $|a_j - a_k| \leq \varepsilon$. The sequence can then be split into two parts:

$$a_1, a_2, \dots, a_{N-1}, \quad \text{and} \quad a_N, a_{N+1}, a_{N+2}, \dots .$$

By corollary 15.2.10, the first part of the sequence is bounded by some rational M. Since the distance between every term in the second part is less than ε. If the smallest term in the second part of the sequence is K, then the entire part must be bounded by $K + \varepsilon$; therefore, the whole sequence is bounded by $M + K + \varepsilon$. □

15.3 Construction of the reals

Now we are ready to construct the reals. Similar to how we've constructed other objects, we do not care what the object actually is, but rather, how it interacts with others. Therefore, let's define a function that takes in a Cauchy sequence and spits out a real number:

Definition 15.3.1: *Real numbers*

A real number is of the form $\mathfrak{L}(a_n)$ where $(a_n)_{n=m}^{\infty}$ is a Cauchy sequence.

Similar to how integers are defined, there are many Cauchy sequences which represent the same real number. I.e.,

$$1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \ldots, \quad \text{and} \quad 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \ldots, \text{[1]} \tag{15.15}$$

both of which seems to approach the number 0. Therefore, we must define the concept of equivalence sequences:

Definition 15.3.2: *Equivalent sequences*

Two sequences, $(a_n)_{n=m_a}^{\infty}$ and $(b_n)_{n=m_b}^{\infty}$ are equivalent *iff* for all rational ε, there exists a natural $N \geq \max(m_a, m_b)$ in which for all $n \geq N$, a_n and b_n are ε-close, i.e.,

$$\forall(\varepsilon > 0 \in \mathbb{Q})\exists(N \geq \max(m_a, m_b) \in \mathbb{N})\forall(n \geq N \in \mathbb{N}) : \\ |a_n - b_n| \leq \varepsilon. \tag{15.16}$$

$(a_n)_{n=m_a}^{\infty}$ and $(b_n)_{n=m_b}^{\infty}$ represent the same number, i.e., $\mathfrak{L}(a_n) = \mathfrak{L}(b_n)$ *iff* they are both Cauchy and equivalent

Notation 15.3.3: *Equivalent sequences*

If $(a_n)_{n=m_a}^{\infty}$ and $(b_n)_{n=m_b}^{\infty}$ are equivalent, we write

$$(a_n)_{n=m_a}^{\infty} \sim (b_n)_{n=m_b}^{\infty}.$$

The first thing to notice is that the notion of equivalent (def. 15.3.2) doesn't care about the starting point of the sequences. We analyze the equivalency for $n \geq \max(m_a, m_b)$ only, because we only care about the behavior of the sequence as it *progresses*. Similar to how we've defined Cauchy sequences, we can also break def. 15.3.2 into two parts: ε-close,

[1]It is left as an exercise for the reader to prove that these two sequences are Cauchy.

and eventual ε-close. I shall go over them very briefly as it is analogous to how Cauchy sequences (def. 15.2.1) are defined.

Two sequences $(a_n)_{n=m_a}^{\infty}$ and $(b_n)_{n=m_b}^{\infty}$ are ε-close *iff*

$$|a_n - b_n| \leq \varepsilon \tag{15.17}$$

These sequences are eventually ε-close if there exists a number $N > \max(m_a, m_b)$ in which $(a_n)_{n=N}^{\infty}$ and $(b_n)_{n=N}^{\infty}$ are ε-close. Finally, these sequences are equivalence if it is eventually ε-close for all rational ε.

The second thing to notice is that def. 15.3.2 doesn't care about whether the sequences are Cauchy or not. E.g.,

$$1, 2, 3, 4, 5, 6, \ldots, \quad \text{and} \quad 3, 3, 3, 4, 5, 6, \ldots, \tag{15.18}$$

are equivalent but not Cauchy.[2] But if two sequences are equivalent and one of them is Cauchy, we can guarantee that the other one must also be Cauchy.

Corollary 15.3.4 Let $(a_n)_{n=m_a}^{\infty} \sim (b_n)_{n=m_b}^{\infty}$. If one of the sequence is Cauchy, then the other must also be. If one of the sequence isn't Cauchy, then the other is also not Cauchy.

Proof. By def. 15.3.2, $(a_n)_{n=m_a}^{\infty} \sim (b_n)_{n=m_b}^{\infty}$ means

$$\forall(\varepsilon > 0 \in \mathbb{Q})\exists\big(N \geq \max(m_a, m_b) \in \mathbb{N}\big)\forall(n \geq N \in \mathbb{N}) : |a_n - b_n| \leq \varepsilon. \tag{15.19}$$

Let's assume that $(a_n)_{n=m_a}^{\infty}$ is Cauchy, i.e.,

$$\forall(\varepsilon > 0 \in \mathbb{Q})\exists\big(N \geq \max(m_a, m_b) \in \mathbb{N}\big)\forall(j, k \geq N \in \mathbb{N}) : |a_j - a_k| \leq \varepsilon. \tag{15.20}$$

[2] It is left as an exercise for the reader to prove that these two sequence aren't Cauchy, but are still equivalent.

We then want to prove that $(b_n)_{n=m_b}^{\infty}$ is also Cauchy, i.e.,

$$\forall(\delta > 0 \in \mathbb{Q})\exists\big(N \geq \max(m_a, m_b) \in \mathbb{N}\big)\forall(j, k \geq N \in \mathbb{N}) : |b_j - b_k| \leq \delta.$$

This may seem a bit outrageous, but notice that

$$|b_j - b_k| = |b_j - a_j + a_j - a_k + a_k - b_k|. \tag{15.21}$$

By the triangle's inequality, we get

$$|b_j - a_j + a_j - a_k + a_k - b_k| \leq |b_j - a_j| + |a_j - a_k| + |a_k - b_k|. \tag{15.22}$$

From eq. (15.20) and section 15.3,

$$|b_j - a_j| + |a_j - a_k| + |a_k - b_k| \leq 3\varepsilon, \tag{15.23}$$

or $|b_j - b_k| \leq 3\varepsilon$. If $\delta = 3\varepsilon$, then $|b_j - b_k| \leq \delta$ as desired. □

15.4 Addition and multiplication on the reals

As said earlier, we want to define arithmetic on the reals as operations on Cauchy sequences. Since the definition of division has much nuance, we shall focus on addition, subtraction, and multiplication for now. Similar to how addition and multiplication on lower number systems, we shall define the operations, then prove that they are well-defined.

Definition 15.4.1: *Addition on the reals*

Let there be two Cauchy sequences $(a_n)_{n=m_a}^{\infty}$ which represents the real $x = \mathfrak{L}(a_n)$, and $(b_n)_{n=m_b}^{\infty}$ which represents the real $y = \mathfrak{L}(b_n)$. The sum of x and y, denoted $x + y$, is given by the real number $\mathfrak{L}(a_n + b_n)$ represented by the sequence $(a_n + b_n)_{n=\max(m_a, m_b)}^{\infty}$, i.e.,

$$x + y = \mathfrak{L}(a_n) + \mathfrak{L}(b_n) = \mathfrak{L}(a_n + b_n). \tag{15.24}$$

To actually prove that addition on the reals is well-defined, we have to make sure that the sum of two Cauchy sequences are Cauchy first.

Lemma 15.4.2: *The sum of two Cauchy sequences are Cauchy* Let there be two Cauchy sequences $(a_n)_{n=m_a}^{\infty}$ and $(b_n)_{n=m_b}^{\infty}$. Then, $(a_n + b_n)_{n=\max(m_a,m_b)}^{\infty}$ is also a Cauchy sequence.

Proof. $(a_n)_{n=m_a}^{\infty}$ and $(b_n)_{n=m_b}^{\infty}$ are Cauchy means that for all rational ε,

$$\exists(A > m_a \in \mathbb{N})\forall(j,k \geq A \in \mathbb{N}) : |a_j - a_k| \leq \varepsilon, \tag{15.25}$$

$$\exists(B > m_b \in \mathbb{N})\forall(j,k \geq A \in \mathbb{N}) : |b_j - b_k| \leq \varepsilon. \tag{15.26}$$

Let $C = \max(A, B)$; therefore,

$$\forall(j,k \geq C \in \mathbb{N}) : \big(|a_j - a_k| \leq \varepsilon\big) \wedge \big(|b_j - b_k| \leq \varepsilon\big). \tag{15.27}$$

We want to show that $(a_n + b_n)_{n=\max(m_a,m_b)}^{\infty}$ is Cauchy, i.e.,

$$\forall(\delta > 0 \in \mathbb{Q})\exists\big(N > \max(m_a, m_b)\big)\forall(j,k \geq C \in \mathbb{N}) : \\ \Big|(a_j + b_j) - (a_k + b_k)\Big| \leq \delta \tag{15.28}$$

From eq. (15.27), we get that for all $j, k > C$,

$$|a_j - a_k| + |b_j - b_k| \leq 2\varepsilon \tag{15.29}$$

By the triangle's inequality, $|a_j - a_k + b_j - b_k| \leq |a_j - a_k| + |b_j - b_k|$; thus,

$$\Big|a_j - a_k + b_j - b_k\Big| \leq 2\varepsilon \tag{15.30}$$

$$\Big|(a_j + b_j) - (a_k + b_k)\Big| \leq 2\varepsilon. \tag{15.31}$$

Let $\delta = 2\varepsilon$, and we get $\Big|(a_j + b_j) - (a_k + b_k)\Big| \leq \delta$ as desired. □

The direct consequence of lem. 15.4.2 is that the product of two Cauchy sequence is also bounded because Cauchy sequences are bounded. Then, we prove that addition on the reals is well-defined.

Proposition 15.4.3: *Addition on the reals is well-defined* Let there be two Cauchy sequences in which $(a_n)_{n=m_a}^{\infty} \sim (a'_n)_{n=m_{a'}}^{\infty}$, and another Cauchy sequence $(b_n)_{n=m_b}^{\infty}$. If $x = \mathfrak{L}(a_n)$, $x' = \mathfrak{L}(a'_n)$, and $y = \mathfrak{L}(b_n)$ then

$$x + y = x' + y \tag{15.32}$$

One other way to write this proposition is that sums of equivalent Cauchy sequences are equivalent.

Proof. Let $m = \max(m_a, m_{a'}, m_b)$, then $(a_n)_{n=m}^{\infty} \sim (a'_n)_{n=m}^{\infty}$ can be written as

$$\exists(N \in \mathbb{N})\forall(n \geq N \in \mathbb{N}) : |a_n - a'_n| \leq \varepsilon. \tag{15.33}$$

We want to show that $x + y = x' + y$, i.e., $(a_n + b_n)_{n=m}^{\infty} \sim (a'_n + b_n)_{n=m}^{\infty}$. By lem. 15.4.2, these two sequences must be Cauchy, so the final result is guaranteed to be a real number. This proof uses the same trick as the one before: adding and subtracting the same term. From eq. (15.33),

$$|a_n - a'_n| \leq \varepsilon \tag{15.34}$$

$$|a_n - b_n + b_n - a'_n| \leq \varepsilon \tag{15.35}$$

$$|(a_n - b_n) - (a'_n + b_n)| \leq \varepsilon, \tag{15.36}$$

meaning that $(a_n + b_n)_{n=m}^{\infty} \sim (a'_n + b'_n)_{n=m}^{\infty}$; thus,

$$\mathfrak{L}(a_n + b_n) = \mathfrak{L}(a'_n + b_n). \tag{15.37}$$

By def. 15.4.1,

$$\mathfrak{L}(a_n) + \mathfrak{L}(b_n) = \mathfrak{L}(a'_n) + \mathfrak{L}(b_n) \tag{15.38}$$

$$x + y = x' + y$$

□

Because I do not want to write separate proof for addition and subtraction, I shall embed the concept of negative numbers in the reals and define $x - y = x + (-y)$. This way, we only have to introduce the negation of the real numbers, and prove that it is well-defined instead of defining subtraction as its own operation, which would be more cumbersome.

Definition 15.4.4: *Negation of a real number*

Given a Cauchy sequence $(a_n)_{n=m}^{\infty}$ in which $x = \mathfrak{L}(a_n)$, we define

$$-x = \mathfrak{L}(-a_n). \tag{15.39}$$

Proposition 15.4.5: *Negation of a real is well-defined* If two Cauchy sequences $(a_n)_{n=m_a}^{\infty} \sim (a'_n)_{n=m_{a'}}^{\infty}$, and $x = \mathfrak{L}(a_n)$ and $x' = \mathfrak{L}(a'_n)$. Then,

$$-x = -x' \tag{15.40}$$

Another way to say this is that the negation of two equivalent Cauchy sequences are equivalent.

Proof. $(a_n)_{n=m_a}^{\infty} \sim (a'_n)_{n=m_{a'}}^{\infty}$ means for all rational ε,

$$(N \geq \max(m_a, m_b) \in \mathbb{N})\forall(n \geq N \in \mathbb{N}) : |a_n - a'_n| \leq \varepsilon. \tag{15.41}$$

From the property of absolute values, $|a_n - a'_n| = |(-a'_n) - (-a_n)|$; therefore, $(-a_n)_{n=m_a}^{\infty} \sim (-a'_n)_{n=m_{a'}}^{\infty}$ and $x = -x'$ as desired. □

Subtraction is then defined as addition of a negated real number.

Definition 15.4.6: *Subtraction on the reals*

If $x = \mathfrak{L}(a_n)$ and $y = \mathfrak{L}(b_n)$, then the difference between x and y is

$$x - y = x + (-y) = \mathfrak{L}(a_n) + \mathfrak{L}(-b_n).$$

Since both negation and addition is well-defined, and negation is written using addition, subtraction must also be well-defined on the reals. We then move on to define multiplication.

Definition 15.4.7: *Multiplication on the reals*

Let there be two Cauchy sequences, $(a_n)_{n=m_a}^{\infty}$ and $(b_n)_{n=m_b}^{\infty}$, in which $x = \mathfrak{L}(a_n)$ and $y = \mathfrak{L}(b_n)$. The product of x and y, denoted xy, is given by the real number $\mathfrak{L}(a_n b_n)$ represented by the sequence $(a_n b_n)_{n=\max(m_a, m_b)}^{\infty}$, i.e.,

$$xy = \mathfrak{L}(a_n)\mathfrak{L}(b_n) = \mathfrak{L}(a_n b_n). \tag{15.42}$$

Similarly, we must also prove that the product of two Cauchy sequences are Cauchy:

Lemma 15.4.8: *The product of two Cauchy sequences are Cauchy* Let there be two Cauchy sequences $(a_n)_{n=m_a}^{\infty}$ and $(b_n)_{n=m_b}^{\infty}$. Then, $(a_n b_n)_{n=\max(m_a, m_b)}^{\infty}$ is also a Cauchy sequence.

Proof. $(a_n)_{n=m_a}^{\infty}$ and $(b_n)_{n=m_b}^{\infty}$ are Cauchy means that for all rational ε,

$$\exists(A > m_a \in \mathbb{N})\forall(j, k \geq A \in \mathbb{N}) : |a_j - a_k| \leq \varepsilon, \tag{15.43}$$

$$\exists(B > m_b \in \mathbb{N})\forall(j, k \geq A \in \mathbb{N}) : |b_j - b_k| \leq \varepsilon. \tag{15.44}$$

Let $C = \max(A, B)$; therefore,

$$\forall(j, k \geq C \in \mathbb{N}) : \left(|a_j - a_k| \leq \varepsilon\right) \wedge \left(|b_j - b_k| \leq \varepsilon\right). \tag{15.45}$$

We want to show that $(a_n b_n)_{n=\max(m_a, m_b)}^{\infty}$ is Cauchy, i.e.,

$$\forall(\delta > 0)\exists(N \geq C \in \mathbb{N})\forall(j, k \geq N \in \mathbb{N}) : |a_j b_j - a_k b_k| \leq \varepsilon. \tag{15.46}$$

Notice that

$$|a_jb_j - a_kb_k| = |a_jb_j - a_kb_j + a_kb_j - a_kb_k| \tag{15.47}$$

$$= |b_j(a_j - a_k) + a_k(b_j - b_k)| \tag{15.48}$$

$$|b_j(a_j - a_k) + a_k(b_j - b_k)| \leq |b_j||a_j - a_k| + |a_k||b_j - b_k| \tag{15.49}$$

$$\tag{15.50}$$

From eq. (15.45),

$$|b_j||a_j - a_k| + |a_k||b_j - b_k| \leq \varepsilon\big(|b_j| + |a_k|\big). \tag{15.51}$$

It is quite tempting to let $\delta = \varepsilon\big(|b_j| + |a_k|\big)$, but it'd be better if the R.H.S. is dependent on the property of the sequence as a whole and not restricted to individual terms. Here is where boundedness comes in. From prop. 15.2.11, every Cauchy sequences are bounded. So, let's make U and V be the bound of $(a_n)_{n=m_a}^{\infty}$ and $(b_n)_{n=m_b}^{\infty}$ respectively, i.e.,

$$\forall(n \geq C \in \mathbb{N}) : \big(|a_n| \leq U\big) \wedge \big(|b_n| \leq V\big); \tag{15.52}$$

thus, $\varepsilon\big(|b_j| + |a_k|\big) \leq \varepsilon(U + V)$. If $\delta = \varepsilon(U + V)$, we get $|a_jb_j - a_kb_k| \leq \delta$ as desired. □

The direct consequence of lem. 15.4.8 is, the product of two Cauchy sequence is also bounded. Then, we prove that multiplication on the reals is well-defined.

Proposition 15.4.9: *Multiplication on the reals is well-defined* Let there be two Cauchy sequences in which $(a_n)_{n=m_a}^{\infty} \sim (a'_n)_{n=m_{a'}}^{\infty}$, and another Cauchy sequence $(b_n)_{n=m_b}^{\infty}$. If $x = \mathfrak{L}(a_n)$, $x' = \mathfrak{L}(a'_n)$, and $y = \mathfrak{L}(b_n)$ then

$$xy = x'y \tag{15.53}$$

One other way to write this proposition is that the product of equivalent Cauchy sequences are equivalent.

Proof. Let $C = \max(m_a, m_{a'}, m_b)$, then $\left(a_n\right)_{n=m_a}^{\infty} \sim \left(a'_n\right)_{n=m_{a'}}^{\infty}$ means that for all rational ε,

$$\exists(N \geq M \in \mathbb{N})\forall(n \geq N \in \mathbb{N}) : |a_n - a'_n| \leq \varepsilon. \tag{15.54}$$

Because all the sequences are Cauchy,

$$\begin{aligned}&\forall(\varepsilon > 0 \in \mathbb{Q})\exists(N \geq C)\forall(j, k \geq N \in \mathbb{N}) : \\ &\qquad \left(\left|b_j - b_k\right| \leq \varepsilon.\right) \wedge \left(\left|a_j - a_k\right| \leq \varepsilon\right) \wedge \left(\left|a'_j - a'_k\right| \leq \varepsilon\right).\end{aligned} \tag{15.55}$$

We want to show that $\left(a_n b_n\right)_{n=C}^{\infty} \sim \left(a'_n b_n\right)_{n=C}^{\infty}$, i.e.,

$$\forall(\delta > 0 \in \mathbb{Q})\exists(M \geq C)\forall(m \geq M \in \mathbb{N}) : |a_m b_m - a'_m b_m| \leq \delta. \tag{15.56}$$

First,

$$|a_m b_m - a'_m b_m| = |b_m||a_m - a'_m| \tag{15.57}$$

- From $\left(a_n\right)_{n=m_a}^{\infty} \sim \left(a'_n\right)_{n=m_{a'}}^{\infty}$, $|a_m - a'_m| \leq \delta$.
- Because $\left(b_n\right)_{m=C}^{\infty}$ is Cauchy, it is bounded by some rational B; thus, $\forall(m \leq C) : |b_m| \leq B$.

Combining these together we get

$$|b_m||a_m - a'_m| \leq B\varepsilon. \tag{15.58}$$

If $\delta = B\varepsilon$, then $|a_m b_m - a'_m b_m| \leq \delta$ as desired. □

15.5 Reciprocation and division on the reals

Since division is just multiplication by the reciprocal, we shall only focus on reciprocation for now.

The reciprocal of $(a_n)_{n=m}^{\infty}$ is defined *iff*,

1. The sequence $(a_n)_{n=m}^{\infty}$ is *not* equivalent to the zero sequence $(0)_{n=m}^{\infty}$: any sequences which are equivalent to the zero sequence represents zero, and the reciprocal of zero is not defined.

2. None of the terms in the sequence are zero. If one of the term in the sequence is zero, say $a_j = 0$, then $\frac{1}{a_j}$ is ∞. Since ∞ isn't a rational number, the sequence $\left(\frac{1}{a_n}\right)_{n=m}^{\infty}$ is not a rational sequence, thus by def. 15.3.1, it is not a real number.

These two restrictions forces us to define what sequence is allowed to be reciprocated. If a sequence doesn't follow these criteria, then its reciprocation is either not a Cauchy sequence of rationals; thus, can't be used to represent real numbers. Let's see some examples.

Example 15.5.1: *Sequences which cannot be reciprocated*

1. Consider the sequence $(a_n)_{n=m_a}^{\infty}$

$$1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \dots \tag{15.59}$$

The reciprocation of this sequence, $\left(\frac{1}{a_n}\right)_{n=m_a}^{\infty}$ is the sequence

$$1, 2, 3, 4, 5, \dots, \tag{15.60}$$

which is obviously not Cauchy. This is because the original sequence is actually equivalent to the zero sequence $(0)_{n=m}^{\infty}$.

2. Consider the sequence $(b_n)_{n=m_b}^{\infty}$:

$$0, 0.9, 0.99, 0.999, 0.9999, \dots \tag{15.61}$$

The reciprocation of this sequence, $\left(\frac{1}{b_n}\right)_{n=m_b}^{\infty}$ is the sequence

$$\infty, 1.11111\dots, 1.01010\dots, 1.00100\dots, 1.00010\dots, \dots. \tag{15.62}$$

Even though this looks like a Cauchy sequence that's somehow approaching the number one, the sequence itself isn't Cauchy because ∞ is not a number. This is because there are 0 in the original sequence.

This problem leads us to define a type of sequence that can be used for reciprocation: *sequences bounded away from zero*.

Definition 15.5.2: *Sequences bounded away from zero*

A sequence $(a_n)_{n=m}^{\infty}$ is bounded away from zero *iff* there exists some rational c in which for all natural $n \geq m$, $|a_n| \geq c$, i.e.,

$$\exists(c > 0 \in \mathbb{Q})\forall(n \geq m \in \mathbb{N}) : |a_n| \geq c. \tag{15.63}$$

This definition is actually a very strong definition which requires the sequence to be strictly non-zero. A natural question to ask is why we say that there exists some rational $c > 0$ in which $|a_n| \geq c$ instead of just saying $|a_n| > 0$? Let's look at some examples.

Example 15.5.3: *Sequences bounded away from zero vs. Non-zero sequences* Consider the sequence $(a_n)_{n=m}^{\infty}$:

$$1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \dots. \tag{15.64}$$

Although each term of the sequence is non-zero, i.e., $|a_n| > 0$ for all $n > m \in \mathbb{N}$, it is not bounded away from zero. You can find a term that's ε-close to zero for every ε. By def. 15.5.2, there doesn't exist any rational c in which all $|a_n| \geq c$ because for all c, there is always a term, $a_{\frac{1}{c}+1}$ that's smaller than c. E.g., if $c = \frac{1}{10}$, then $a_{\frac{1}{\frac{1}{10}}+1} = a_{11} = \frac{1}{11}$ is smaller than $\frac{1}{10}$.

An immediate consequence is that if a sequence is bounded away from zero, it is not equivalent to the zero sequence. This is because the sequence isn't c-close to zero.

15.6 Ordering the reals

15.7 Exponentiation of a real to a rational number

15.8 The supremum and least upper bound

15.9 Boundedness of sequences

15.10 Convergence

You may need to grab a cup of tea and think of what this definition actually means. It's important for you to understand it fully. Let it sink in for a while. Well, I'd tell you that this definition is quite a humongous one. So, it'd be better if I break it down into two more digestible chunks: ε-boundedness and eventual ε-boundedness, in order to ease the understanding.[3]

[3]These two definitions are not standard in the literature.

Definition 15.10.1: *(L, ε)-boundedness*

A sequence of rationals $(a_n)_{n=i}^{f}$ is (L, ε)-bounded if for all terms in the sequence,

$$|a_n - L| \leq \varepsilon.$$

Expanding the absolute value yields

$$-\varepsilon \leq a_n - L \leq \varepsilon \tag{15.65}$$

$$L - \varepsilon \leq a_n \leq L + \varepsilon, \tag{15.66}$$

i.e., all a_n lies inside $L - \varepsilon$ and $L + \varepsilon$. Let's consider a sequence of numbers $(a_n)_{n=0}^{\infty}$ illustrated.

Example 15.10.2: *(L, ε)-bounded sequences*

1. The sequence $-1, 1, -1, 1, -1, \ldots$ is $(0, 1)$-bounded, but not $(1, 1)$-bounded because -1 doesn't lie between $1 - 1 = 0$ and $1 + 1 = 2$.
2. The sequence $0.9, 0.99, 0.999, 0.9999, 0.99999, \ldots$ is $(0.2, 1)$-bounded, but not $(0.001, 1)$-bounded because 0.9 and 0.99 doesn't lie between $1 + 0.01 = 1.01$ and $1 - 0.01 = 0.99$.
3. The sequence $1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \ldots$ is $(0.5, 0.5)$-bounded, but not $(0.5, 0.25)$-bounded because every term that isn't $\frac{1}{2}$, $\frac{1}{3}$ and $\frac{1}{4}$, doesn't lie between $0.5 + 0.25 = 0.75$ and $0.5 - 0.25 = 0.25$.

Definition 15.10.3: *Eventual (L, ε)-boundedness*

A sequence of rationals $(a_n)_{n=i}^{f}$ is eventually (L, ε)-bounded if there exists a number m that lies between i and f, such that $(a_n)_{n=m}^{f}$ is

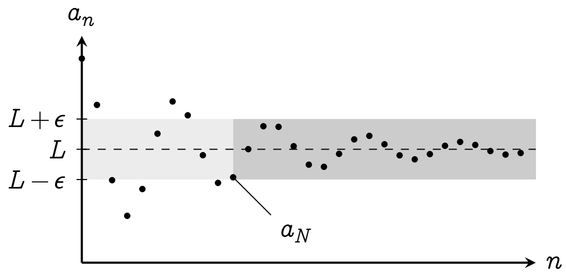


Fig. 15.2 | Convergence of sequences

(L, ε)-bounded.

The explanation: Consider any arbitrary sequence of rationals $(a_n)_{n=0}^{\infty}$ where the limit of a_n is L, shown in fig. 15.2. From the definition, there exists some N in which for all n that follows, $|a_n - L| \leq \varepsilon$. Expanding the absolute value yields

$$-\varepsilon \leq a_n - L \leq +\varepsilon \tag{15.67}$$

$$L - \varepsilon \leq a_n \leq L + \varepsilon. \tag{15.68}$$

I.e., for all $n \geq N$, a_n is between $L - \varepsilon$ and $L + \varepsilon$. Illustrated in fig. 15.2, it means that there exists some point a_N in which all the terms that follows it lies in the gray rectangle.

Part VI

Sequences and series

CHAPTER 16

Sequences

It's now time to study the sequence of reals in full. All the definitions about sequences in the chapter earlier that includes $\mathbb{Q}$ can now be replaced with $\mathbb{R}$.

16.1 Boundedness and convergence

Definition 16.1.1: *Convergence of sequences*

A sequence $(a_n)_{n=m}^{\infty}$ converges to a rational L *iff* for all rational $\varepsilon > 0$, there exists a natural N such that for all natural $n \geq N$, $|a_n - L| \leq \varepsilon$. When this happens, L is called the limit of a_n, denoted $\lim_{n\to\infty}(a_n)$. Formally, this can be written as

$$\lim_{n\to\infty}(a_n) = L \iff \forall(\varepsilon > 0 \in \mathbb{Q})\exists(N \in \mathbb{N})\forall(n \geq N \in \mathbb{N}) : |a_n - L| \leq \varepsilon \quad (16.1)$$

Book III

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